Curvature-Calibrated Exploration for Differentiable-Model Bandits

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Abstract

Curvature surrogates in differentiable-model bandits change exploration widths, but greater fidelity to full curvature need not improve online decisions. We use the full generalized Gauss–Newton (GGN)/Fisher operator as a reference for analysis and falsification, computing its widths with curvature–vector products and conjugate gradients (CG) without materializing a $d \times d$ matrix. For Gaussian rewards and squared loss, a transfer theorem separates linearization, centering, feature and operator transfer, CG error, and realized width complexity. Predictable Welford and residual statistics instantiate its nonlinear certificates with $O(d)$ additional state. A closed sublinear rate is conditional: a supplied fixed r-dimensional tangent subspace, a bounded near-linear path, and the stated width, optimizer, and CG premises yield $O(r\sqrt{T}\log T) + O(T/\sqrt{W})$. A separate growing-window result gives a conditional $T^{1-q/2}$ regret versus $T^{1+3q/2}$ sample-CVP tradeoff under persistent excitation. Experiments are a reference-operator falsification study, not a performance demonstration. A linear Gram witness isolates an off-diagonal action change, while a 3,600-run scaling grid, a mechanism map, and a matched contextual benchmark reject a uniform fidelity-based regret ordering in the tested regimes. The grid has a clean low-rank slice but optimizer and path-event failures at larger ranks; its fitted slopes are diagnostics. Diagonal and last-layer policies often have lower regret.

Preliminary work. Under review by AISTATS 2026. Do not distribute.

1 Introduction

Contextual bandits are a standard framework for sequential decision-making under partial feedback, and increasingly the reward predictor is a large differentiable parametric model—a pretrained network for text, vision, or multimodal data composed with a reward head (Zhou et al., 2020; Riquelme et al., 2018). The central algorithmic question is how to build an exploration bonus that faithfully reflects epistemic uncertainty over such a model's predictions. Our transfer reduction allows generic differentiable parametric models, but the closed sublinear rate below requires a bounded, low-rank tangent path; foundation models are a motivation, not a covered unrestricted-training regime.

In the *white-box* setting—gradient and curvature–vector-product access to the model—the canonical bonus is the GGN/Fisher-linearized predictive variance

$$s_t^2(a) \;=\; \mathbf{g}_t(a)^\top \mathbf{C}_t^{-1}\, \mathbf{g}_t(a), \tag{1}$$

where $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ is the gradient feature (Jacobian of the mean reward) and $\mathbf{C}_t$ is the damped Gauss–Newton (GGN) curvature at the current estimate $\boldsymbol{\theta}_t$. Exact full-history GGN/Fisher widths are costly at large d or t: dense materialization uses $O(d^2)$ state, while matrix-free CG replays history at every iteration. NeuralUCB retains the full neural-gradient Gram in its theoretical algorithm but diagonalizes it in experiments (Zhou et al., 2020); neural-linear and scalable Laplace methods instead use last-layer or structured curvature approximations (Riquelme et al., 2018; Ritter et al., 2018; Daxberger et al., 2021). Outside online bandits, implicit Jacobian/Fisher products and CG have already been used to compute full-network linearized predictive variances without forming the Gram matrix (Maddox et al., 2021). Each curvature surrogate changes the *spectrum* of $\mathbf{C}_t$ and hence the value of $s_t^2(a)$. The spectral change does not by itself imply a change in the selected action, and therefore in regret; whether it does is an empirical question, which we make measurable and study directly.

We use full curvature as a reference operator in a falsification study. Common-trajectory diagnostics isolate surrogate-induced width distortion, while independently executed policies test whether those distortions survive adaptive feedback. The target is the uniform claim that greater fidelity to full GGN orders regret: one controlled reversal rejects that claim, whereas agreement does not establish superiority. Because executed policies follow different histories, their regret contrasts are descriptive rather than causal operator effects. We adapt the established matrix-free predictive-variance primitive to online contextual exploration: curvature–vector products (CVPs), available through automatic differentiation, let us solve $\mathbf{C}_t\,\mathbf{u} = \mathbf{g}_t(a)$ by conjugate gradients (CG) *without forming* $\mathbf{C}_t$: only $O(d)$ *additional curvature-state* vectors (a constant number of CG vectors) are maintained and no $d \times d$ curvature matrix or per-sample gradient vector is stored—though exact full-history curvature–vector products still require retaining or replaying the historical raw examples to recompute their Jacobian products, and model parameters, optimizer state, activations, replay data, and any frozen-feature parameter checkpoints are not counted in this $O(d)$ curvature state. Because the damped GGN is symmetric positive definite, CG converges in exact arithmetic. All CG convergence and residual-to-energy statements below use exact arithmetic unless a verified rounding-error enclosure is supplied. Building on this, we propose **Curvature-Calibrated UCB (CC-UCB)**, a bandit algorithm whose exploration bonus is the predictive variance under the *uncompressed* GGN/Fisher metric, accessed entirely through implicit CG solves.

The analysis tracks predictable certificate requirements for each approximation source and constructs them for the nonlinear path terms. The Gaussian/squared-loss theorem (Theorem 1) handles a general predictable SPD curvature sequence with a one-sided transfer factor, a per-round CG factor, and a realized width complexity $\Lambda_T^{\mathcal{C}}$. Its dynamic determinant identity separates endpoint growth from determinant-destroying refreshes. More importantly, the previously external nonlinear inputs are computable: Welford parameter moments and observed collection residuals upper-bound feature transfer and centering, while an analytic Taylor schedule supplies the confidence radius (Lemma 7 and Corollary 2). These schedules are conservative and need known smoothness constants and a certified optimizer residual. For a supplied fixed-rank tangent path satisfying the stated width-dependent smoothness, residual, and optimizer envelopes, we close information gain and give an explicit $O(r\sqrt{T}\log T) + O(T/\sqrt{W})$ rate; a persistently excited growing window gives a computation–regret tradeoff (Corollary 9 and Theorem 3). Neither result certifies unrestricted fine-tuning.

Contributions. **(1)** A rank-closed conditional rate for fixed-subspace adaptation with exact current GGN, with bounded-path and width assumptions explicit (Corollary 9). **(2)** A certified growing-window rate exposing the $T^{1-q/2}$ regret versus $T^{1+3q/2}$ sample-CVP tradeoff (Theorem 3). **(3)** Predictable $O(d)$-state nonlinear transfer and centering certificates (Lemma 7 and Corollary 2). **(4)** Matrix-free full-GGN/Fisher widths via residual-checked CG/CVP solves, plus an off-diagonal linear-Gram witness relevant to the GGN metric (Proposition 26). **(5)** Executed nonlinear, mechanism, baseline, and systems audits that separate online outcomes from post-hoc diagnostics (Section 5).

2 Problem Setup

We present the problem setup for the *Gaussian/squared-loss* instantiation, which is the setting analyzed in Theorem 1. An extension to exponential-family likelihoods is discussed in Appendix H.

Protocol. At each round $t = 1, \dots, T$ the learner observes a context $\mathbf{x}_t \in \mathcal{X}$ and a finite action set $\mathcal{A}_t$. It selects $a_t \in \mathcal{A}_t$ and observes a stochastic reward

$$r_t \;=\; \mu^*(\mathbf{x}_t, a_t) + \eta_t, \qquad \eta_t \text{ is } \sigma\text{-sub-Gaussian}, \quad (2)$$

where $\mu^* : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$ is the unknown mean reward. The goal is to minimize cumulative regret $R_T = \sum_{t=1}^{T}\big[\mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)\big]$, where $a_t^* = \arg\max_{a \in \mathcal{A}_t} \mu^*(\mathbf{x}_t, a)$. We assume $|\mathcal{A}_t| \le A_{\max}$ deterministically for all t.

Reward model and parameterization convention. We fix a pretrained initialization $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ and parameterize the model by the *displacement* $\boldsymbol{\theta} \in \mathbb{R}^d$ from it: the deployed predictor is $\mu_{\boldsymbol{\theta}_0+\boldsymbol{\theta}}(\mathbf{x}, a)$, and we abbreviate $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) := \mu_{\boldsymbol{\theta}_0+\boldsymbol{\theta}}(\mathbf{x}, a)$ throughout. The initial displacement is $\boldsymbol{\theta}_1 = \mathbf{0}$ (the deployed model starts at the pretrained weights $\boldsymbol{\theta}_0$). With this convention the ridge penalty below is a penalty on the *displacement from the pretrained weights*, not a zero-centered prior on the raw weights, and the realizability radius S bounds $\|\boldsymbol{\theta}^\star\| = \|\boldsymbol{\theta}^\star_{\mathrm{orig}} - \boldsymbol{\theta}_0\|$, the distance of the true parameter from the pretrained initialization. Define the regularized loss at round t:

$$\mathcal{L}_t(\boldsymbol{\theta}) \;:=\; \frac{1}{2\sigma^2}\sum_{s=1}^{t-1}\big(r_s - \mu_{\boldsymbol{\theta}}(\mathbf{x}_s, a_s)\big)^2 \;+\; \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2. \quad (3)$$

We write $\boldsymbol{\theta}_t$ for the parameter returned by the warm-started optimizer on $\mathcal{L}_t$. In convex settings (e.g. a frozen backbone with a linear head), under exact optimization this coincides with the global minimizer; in nonconvex settings $\boldsymbol{\theta}_t$ may be a local minimizer or approximate stationary point. The gap between $\boldsymbol{\theta}_t$

and an idealized global minimizer is absorbed by the action-scaled centering certificate (Assumption 4).

Gradient features and curvature matrices. Define the *gradient feature* (Jacobian of the mean reward) at the current parameter estimate:

$$\mathbf{g}_t(\mathbf{x}, a) \;:=\; \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \;\in\; \mathbb{R}^d. \tag{4}$$

The algorithm uses the *relinearized curvature matrix*

$$\mathbf{C}_t \;=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1} \mathbf{g}_t(\mathbf{x}_s, a_s)\,\mathbf{g}_t(\mathbf{x}_s, a_s)^\top, \tag{5}$$

the *damped GGN curvature* for the Gaussian/squared-loss model at the current parameter $\boldsymbol{\theta}_t$. (For this likelihood, the GGN coincides with the true Fisher information; see the remark below.) While $\mathbf{C}_t$ is the natural object for computing predictive variances at $\boldsymbol{\theta}_t$, it re-evaluates all past Jacobians at $\boldsymbol{\theta}_t$ and therefore does *not* admit a sequential rank-one update—all summands change when $\boldsymbol{\theta}_t$ changes.

For the regret analysis we introduce the *frozen-feature Gram matrix*:

$$\bar{\mathbf{C}}_t \;:=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s)\,\mathbf{g}_s(\mathbf{x}_s, a_s)^\top, \tag{6}$$

where each $\mathbf{g}_s(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$ is evaluated at the parameter $\boldsymbol{\theta}_s$ that was current when the sample was collected. This matrix satisfies the recursive update

$$\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top, \tag{7}$$

and, critically, each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is measurable with respect to the pre-reward sigma-algebra $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ of (68), which is the filtration used in the self-normalized bound. Both the additive update (7) and this measurability are structural requirements of the self-normalized martingale bound and the elliptic potential lemma invoked in Section 4. The algorithm computes UCB widths using the algorithmic curvature $\mathcal{C}_t$ (via CG; the exact case is $\mathcal{C}_t = \mathbf{C}_t$), but the confidence analysis is conducted in terms of $\bar{\mathbf{C}}_t$; the one-sided optimism-transfer factor of Assumption 5, certified by Lemma 11, provides the bridge.

3 Method: Curvature-Calibrated Exploration

3.1 CC-UCB Algorithm

For a predictable SPD oracle $\mathcal{C}_t \succeq \lambda\mathbf{I}$, set $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ for the original nonlinear center. Algorithm 1 is the analyzed full-enumeration policy. The same fixed curvature operator is reused across separate per-action CG solves; a solution for one action is not reused as the solution for another action.

Algorithm 1 Curvature-Calibrated UCB (CC-UCB)

Observe context $\mathbf{x}_t$ and finite action set $\mathcal{A}_t$.
Construct a fixed predictable SPD oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$ with $\mathcal{C}_t \succeq \lambda\mathbf{I}$.
for each $a \in \mathcal{A}_t$ **do**
  Compute $m_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.
  If $\mathbf{g}_t(a) \neq \mathbf{0}$, solve $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$ separately by CG until the certified tolerance holds; otherwise set $\tilde{\mathbf{u}} = \mathbf{0}$.
  Set $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}$.
end for
Select and play $a_t \in \arg\max_{a\in\mathcal{A}_t}\{m_t(a) + \omega_t\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}\sqrt{\tilde{s}_t^2(a)}\}$.
Observe r_t, append $(\mathbf{x}_t, a_t, r_t)$, and update $\boldsymbol{\theta}_{t+1}$.

The oracle is accessed only through matrix–vector products. At each round the agent computes, for every candidate action a, the gradient feature $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and the *curvature-calibrated variance proxy*, the generic algorithmic width,

$$s_{\mathcal{C},t}^2(a) \;=\; \mathbf{g}_t(a)^\top\mathcal{C}_t^{-1}\,\mathbf{g}_t(a). \tag{8}$$

Under the GGN/Fisher-linearized Gaussian approximation with covariance $\mathcal{C}_t^{-1}$, the quantity $s_{\mathcal{C},t}^2(a)$ is the metric-dependent predictive variance of the linearized reward at $(\mathbf{x}_t, a)$. The *exact* full-history instantiation of CC-UCB takes $\mathcal{C}_t = \mathbf{C}_t$, the relinearized curvature (5); weighted replay, subsampling, sliding windows, periodic refreshes, and randomized sketches are special implementations of $\mathcal{C}_t$ (§A.3), covered by the theory only when the corresponding one-sided transfer factor is certified.

Because $\mathcal{C}_t \succ 0$ by construction, conjugate gradients (CG) is guaranteed to converge on the system $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$, provided the operator is held *fixed* across all iterations of the solve (the randomized sketch/subsample is drawn into Ω_t before action selection; §2). For each action with $\mathbf{g}_t(a) \neq \mathbf{0}$ we run CG to the per-action relative energy-norm error

$$\varepsilon_{t,a} \;:=\; \frac{\|\mathbf{u}_{t,a} - \tilde{\mathbf{u}}_{t,a}\|_{\mathcal{C}_t}}{\|\mathbf{u}_{t,a}\|_{\mathcal{C}_t}}, \qquad \mathbf{u}_{t,a} := \mathcal{C}_t^{-1}\mathbf{g}_t(a), \tag{9}$$

yielding the computable proxy $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}_{t,a}$; if $\mathbf{g}_t(a) = \mathbf{0}$ we set $\tilde{s}_t^2(a) = 0$ and skip the solve (leaving the residual ratio $\|r_I\|/\|\mathbf{g}\|$ undefined), and by convention put $\varepsilon_{t,a} := 0$ in this case, so that $\varepsilon_{t,a}$ is defined for every $a \in \mathcal{A}_t$ and (57) holds trivially there. Lemma 17 shows $\tilde{s}_t^2(a)$ approximates $s_{\mathcal{C},t}^2(a)$ within a multiplicative $(1 \pm \varepsilon_{t,a})$ factor. To ensure the UCB bonus *upper-bounds* the confidence width $\omega_t\,\bar{s}_t(a)$ despite the approximation, the algorithm inflates the width by $\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}$ (line 8), where $\bar{\varepsilon}_t \in (0,1)$ is a *per-round* known tolerance enforced uniformly over all actions,

$$\max_{a\in\mathcal{A}_t} \varepsilon_{t,a} \;\le\; \bar{\varepsilon}_t \;<\; 1, \tag{10}$$

and $u_t(a) \ge 1$ is the one-sided transfer factor (Assumption 5).

4 Theoretical Analysis

We state a regret bound for CC-UCB (Algorithm 1, with full action enumeration $K{=}|\mathcal{A}_t|$) under the Gaussian/squared-loss setup of Section 2 and Assumptions 1 to 5. The analysis does not constitute a full exponential-family or generalized linear bandit proof; the extension in Appendix H is a curvature-definition extension only. The analysis handles a *general predictable SPD algorithmic curvature sequence* $\mathcal{C}_t$ (37). Confidence is proved for the frozen-feature Gram $\bar{\mathbf{C}}_t$ (Eq. 6), which has the sequential-update and predictability properties required by the self-normalized martingale bound; the *one-sided* optimism-transfer factor $u_t(a)$ (Assumption 5) bridges from $\bar{\mathbf{C}}_t$ to $\mathcal{C}_t$ in the exploration direction only (no lower drift sandwich); the CG approximation enters through the per-round factor α_t; and the nonmonotonicity of $\mathcal{C}_t$ is priced pathwise by the dynamic log-determinant potential (Lemma 14).

The theorem is a modular transfer result. The path statistics below instantiate its nonlinear inputs from pre-action quantities; smallness still requires a restricted parameter path and is not claimed for unrestricted fine-tuning.

Core conditions and links. Assume conditionally σ-sub-Gaussian noise, $\|\mathbf{g}_t(\mathbf{x}_s,a)\| \le G$, realizability by $\|\boldsymbol{\theta}^*\| \le S$, and a local expansion with round-t remainder at most $\varepsilon_{\mathrm{lin}}(t)$. For the original center assume the action-scaled discrepancy is at most $\bar{\psi}_t\bar{s}_t(a)$. The operator $\mathcal{C}_t$ is predictable, SPD, satisfies $\mathcal{C}_t \succeq \lambda\mathbf{I}$ and $\mathcal{C}_1 = \lambda\mathbf{I}$, and remains fixed throughout every CG solve. Define

$$\bar{s}_t^2(a) = \mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a), \quad s_{\mathcal{C},t}^2(a) = \mathbf{g}_t(a)^\top\mathcal{C}_t^{-1}\mathbf{g}_t(a),$$
$$\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}_t(a). \tag{11}$$

The analysis requires only the one-sided transfer and uniform CG relations

$$\begin{gathered}\bar{s}_t^2(a) \le u_t(a)s_{\mathcal{C},t}^2(a),\\ (1-\bar{\varepsilon}_t)s_{\mathcal{C},t}^2(a) \le \tilde{s}_t^2(a) \le (1+\bar{\varepsilon}_t)s_{\mathcal{C},t}^2(a),\end{gathered} \tag{12}$$

for every candidate action, where $u_t(a) \ge 1$, $\bar{\varepsilon}_t < 1$, and $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$. Sufficient feature-drift and approximate-operator constructions are given in Lemma 11 and Theorem 24; no lower spectral transfer is needed.

Dynamic width complexity. The realized quantity used below is

$$\Lambda_T^{\mathcal{C}} := \sum_{t=1}^{T}\log\bigl(1+\sigma^{-2}s_{\mathcal{C},t}^2(a_t)\bigr)\,. \tag{13}$$

Let $\mathcal{C}_t^+ = \mathcal{C}_t + \sigma^{-2}\mathbf{g}_t(a_t)\mathbf{g}_t(a_t)^\top$ and $\Xi_t = (\mathcal{C}_t^+)^{-1/2}(\mathcal{C}_{t+1}-\mathcal{C}_t^+)(\mathcal{C}_t^+)^{-1/2}$. The pathwise determinant identity and its width consequence are

$$\begin{gathered}\Lambda_T^{\mathcal{C}} = \log\frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} - \sum_{t=1}^{T}\log\det(\mathbf{I}+\Xi_t),\\ \sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \le (\sigma^2 + G^2/\lambda)\Lambda_T^{\mathcal{C}}.\end{gathered} \tag{14}$$

Thus relinearization, windowing, and refreshes are charged through the actual widths and determinant-destroying operator changes; the identity does not imply that this complexity is small. A rank-r_t contraction with normalized spectral loss at most $\nu_t < 1$ contributes at most $r_t\log(1/(1-\nu_t))$ (Lemma 15). Complete statements and proofs are in Appendices A.5 and D.

4.1 Regret Bound

Let $\mathcal{E}_{\mathrm{conf}}$ denote the simultaneous self-normalized confidence event and let $\mathcal{E}_{\mathrm{cert}}$ collect the predictable linearization, centering, transfer, condition-number, CG, and randomized-operator certificates. The appendix defines each component and its failure allocation; write their total failure probability as δ_{cert}.

Theorem 1 (Dynamic transfer regret bound for CC-UCB (Gaussian/squared-loss))**.** *Under the conditions above and Assumptions 1 to 5, suppose $|\mathcal{A}_t| \le A_{\max}$, the optimal action is present, and Algorithm 1 enumerates all actions using a predictable schedule $\bar{\beta}_t \ge \beta_t$ and uniform per-round CG tolerance. Let $E_T = \sum_{t=1}^{T}\varepsilon_{\mathrm{lin}}(t)$, $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, and $S_T = \sum_{t=1}^{T}\alpha_t^2 u_t(a_t)\omega_t^2$. On $\mathcal{E}_{\mathrm{conf}}\cap\mathcal{E}_{\mathrm{cert}}$,*

$$R_T \;\le\; 2\sqrt{(\sigma^2+G^2/\lambda)\;\Lambda_T^{\mathcal{C}}\;S_T} \;+\; 2\,E_T, \tag{15}$$

with probability at least $1-\delta-\delta_{\mathrm{cert}}$. Deterministic or almost-sure certificates contribute zero to δ_{cert}. For the corrected center, the same statement holds without Assumption 4 after replacing ω_t by $\bar{\beta}_t$.

In particular, the bound is sublinear whenever $\Lambda_T^{\mathcal{C}}S_T = o(T^2)$ and $E_T = o(T)$.

The exact-curvature instantiation is $\mathcal{C}_t = \mathbf{C}_t$ with $u_t = (1 + \bar{\chi}_t)^2$ from Lemma 11; approximate operators are covered through Theorem 24. The proof is in Appendix D.

Proof sketch. Self-normalized confidence plus the centering and linearization terms gives $|\mu^*(a) - \mu_{\boldsymbol{\theta}_t}(a)| \leq \omega_t \bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t)$. The one-sided relation and CG lower bound make the computed bonus optimistic, so instantaneous regret is at most $2w_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t)$. The CG upper bound gives $w_t(a_t) \leq \alpha_t \omega_t \sqrt{u_t(a_t)} s_{\mathcal{C},t}(a_t)$. Summing, applying Cauchy–Schwarz only once, and using (14) yields (15); hence time-varying factors remain inside S_T rather than being replaced by horizon maxima.

The corrected center replaces ω_t by $\bar{\beta}_t$ and removes the centering certificate, but requires exact or certified frozen-feature access; the formal statement and implementation discussion are in Corollary 5.

4.2 Online nonlinear certificates

At the start of round t, let $n_t = t - 1$, and let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ be maintained by a vector Welford update. With collection residuals $c_s = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) - r_s$, define

$$Q_t = J_{<t} + n_t \|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2, \qquad R_t^{\mathrm{col}} = \sum_{s<t} c_s^2. \quad (16)$$

Both are pre-action quantities and $Q_t = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$ exactly. If prediction gradients are L_g-Lipschitz on the certified region, set

$$\bar{\chi}_t = \frac{L_g \sqrt{Q_t}}{\sigma \sqrt{\lambda}}, \qquad u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2, \quad (17)$$

where $\kappa_{+,t} = 1$ for exact full curvature and for an unrescaled current-parameter subset, and otherwise is a supplied one-sided operator certificate. If $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\| \leq \zeta_t$, an observable centering schedule is

$$\bar{\psi}_t = \frac{\zeta_t + \bar{M}_t}{\sqrt{\lambda}},$$
$$\bar{M}_t = \sigma^{-2} \Big[L_g \sqrt{R_t^{\mathrm{col}} Q_t} + \frac{3}{2} G L_g Q_t + \frac{1}{2} L_g^2 Q_t^{3/2} \Big]. \quad (18)$$

For a known radius-R trust region, the final numerator term improves to $L_g^2 R Q_t$. Finally, for an L_μ-Lipschitz prediction gradient define $\bar{\epsilon}_t^{\mathrm{lin}} = \frac{L_\mu}{2}(S + \|\boldsymbol{\theta}_t\|)^2$, $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\mathrm{lin}})^2$, and

$$\bar{\beta}_t = \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} + \sqrt{\lambda} S + \sigma^{-1} \sqrt{\bar{F}_t},$$
$$\hat{\gamma}_{t-1} = \sum_{s<t} \log(1 + q_s), \qquad q_s = \frac{u_s(a_s) \tilde{s}_s^2(a_s)}{\sigma^2 (1 - \bar{\varepsilon}_s)}. \quad (19)$$

Corollary 2 (Operational CC-UCB)**.** *Under the structural hypotheses of Theorem 1, with its certificate inputs instantiated by valid smoothness constants, the optimizer residual bound ζ_t, uniform CG certificates, and a valid $\kappa_{+,t}$,* (16)–(19) *are predictable and satisfy every nonlinear input of the theorem. Thus Theorem 1 holds with E_T replaced by $\bar{E}_T = \sum_{t \leq T} \bar{\epsilon}_t^{\mathrm{lin}}$ and the observable CG upper bound on $\Lambda_T^{\mathcal{C}}$ from Corollary 16.*

Proof. Lemma 7 proves $\bar{\chi}_t \geq \chi_t$, $\bar{\psi}_t \geq \psi_t$, $\bar{F}_t \geq F_t$, and $\bar{E}_T \geq E_T$. The one-sided operator relation and Corollary 21 give u_t and $\hat{\gamma}_{t-1} \geq \gamma_{t-1}$, hence $\bar{\beta}_t \geq \beta_t$. All quantities use the summary through round $t-1$ and the already formed $\boldsymbol{\theta}_t$; uniform candidate-action CG accuracy is checked before selection. Substitution in Theorem 1 and Corollary 16 proves the claim. ☐

The proof, filtration-order pseudocode, and a scalar-tanh specialization with $G \leq B_\phi$ and $L_\mu = L_g \leq 4B_\phi^2/(3\sqrt{3})$ are in Lemma 7 and Corollary 10.

For a width parameter W, suppose $\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \leq R$, $L_\mu \leq c_\mu/\sqrt{W}$, $L_g \leq c_g/\sqrt{W}$, $\zeta_t \leq \zeta_0/\sqrt{W}$, and $R_t^{\mathrm{col}} \leq C_R t P_T$ for a supplied $P_T \geq 1$. Then

$$\bar{E}_T \leq \frac{2c_\mu R^2 T}{\sqrt{W}}, \qquad \bar{F}_{T+1} \leq \frac{4c_\mu^2 R^4 T}{W},$$
$$\bar{\chi}_t \leq \frac{2c_g R\sqrt{t-1}}{\sigma\sqrt{\lambda W}}. \quad (20)$$

and Corollary 8 gives an explicit trust-region bound on $\bar{\psi}_t$. If a supplied fixed active subspace gives rank bound $r \geq 1$, define

$$\Gamma_{T,r} = r_T \log\left(1 + \frac{TG^2}{r_T \lambda \sigma^2}\right), \qquad r_T = \min\{r, T\}. \quad (21)$$

and set $\Gamma_{0,r} := 0$ for the round-one confidence schedule. The bound r must be known before selection; a terminal realized rank is only post-hoc. Then $\gamma_T \leq \Gamma_{T,r}$ and Corollary 9 gives the fully explicit exact-current-GGN bound

$$R_T \leq 2\alpha_I \sqrt{\rho_+/\rho_-}\, \Omega_{T,r} \sqrt{(\sigma^2 + G^2/\lambda) T \Gamma_{T,r}} + 2\bar{E}_T, \quad (22)$$

where $x_T = 2c_g R\sqrt{T-1}/(\sigma\sqrt{\lambda W})$, $\rho_\pm = 1 \pm (2x_T + x_T^2)$, and $\Omega_{T,r}$ is displayed with every $T, W, r, P_T, \lambda, \sigma$ dependence in the appendix. At fixed r and problem constants, $W = \Omega(T^2 P_T)$ makes the path inflation bounded and gives $R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W})$. This is a low-rank tangent-space or parameter-efficient adaptation regime, not generic full-network training.

Theorem 3 (Growing-window computation–regret tradeoff)**.** *Adopt the hypotheses, events, and valid predictable schedules of Theorem 1, let $m_t = \min\{t-1, \lceil t^q \rceil\}$ for $q \in (0, 1]$, and let $\mathcal{C}_t$ be the unrescaled* *current-parameter Gram over the last m_t observations. Thus Proposition 23 gives $\kappa_{+,t} = 1$ and $u_t = (1 + \bar{\chi}_t)^2$. Suppose candidate gradients lie in a fixed subspace U. For a supplied burn-in index $\tau_w \in \{1, \dots, T + 1\}$, assume the predictable window-excitation condition*

$$\mathcal{C}_t \succeq \lambda \mathbf{I} + \kappa_w m_t P_U$$

holds for a known $\kappa_w > 0$ and every $t \geq \tau_w$. Then, on the same event of probability at least $1 - \delta - \delta_{\mathrm{cert}}$ as Theorem 1,

$$\begin{aligned}
\Lambda_T^{\mathcal{C}} &\leq L_q(T; \tau_w), \\
L_q(T; \tau_w) &:= (\tau_w - 1) \log\left(1 + \frac{G^2}{\lambda \sigma^2}\right) \\
&\quad + \frac{G^2}{\sigma^2} \sum_{t=\tau_w}^{T} \frac{1}{\lambda + \kappa_w m_t}, \qquad (23) \\
L_q(T; \tau_w) &= \begin{cases} O(\tau_w + T^{1-q}), & q < 1, \\ O(\tau_w + \log T), & q = 1. \end{cases}
\end{aligned}$$

For $\tau_w = 1$, more explicitly, $L_q(T; 1) \leq (G^2/\sigma^2)[\lambda^{-1} + \kappa_w^{-1} h_q(T)]$, where, for $T \geq 2$, $h_q(T) = 1 + ((T - 1)^{1-q} - 1)/(1 - q)$ for $q < 1$ and $h_1(T) = 1 + \log(T - 1)$, while $h_q(1) := 0$ in both cases. If $H_T = \max_{t \leq T} \alpha_t \sqrt{u_t(a_t)}\, \omega_t$, then

$$R_T \leq 2H_T \sqrt{(\sigma^2 + G^2/\lambda) T L_q(T; \tau_w)} + 2E_T. \qquad (24)$$

For fixed τ_w, $H_T = \widetilde{O}(1)$ gives $R_T = \widetilde{O}(T^{1-q/2}) + 2E_T$. Certified worst-case sample-CVP cost is $O(KT^{1+3q/2} \log(1/\bar{\varepsilon}))$ when a fixed uniform target $\bar{\varepsilon} \in (0, 1)$ is enforced.

Proof. Before τ_w, $\mathcal{C}_t \succeq \lambda \mathbf{I}$ bounds each logarithm in $\Lambda_T^{\mathcal{C}}$ by $\log(1 + G^2/(\lambda\sigma^2))$. Thereafter, for $g_t(a) \in U$, inversion of the excitation inequality gives $s_{\mathcal{C},t}^2(a) \leq G^2/(\lambda + \kappa_w m_t)$; summing $\log(1 + x) \leq x$ proves (23). Since $S_T \leq T H_T^2$, substitution in Theorem 1 proves (24). The condition-number bound $\kappa(\mathcal{C}_t) \leq 1 + m_t G^2/(\lambda\sigma^2)$ makes CG use $O(\sqrt{m_t} \log(1/\bar{\varepsilon}))$ iterations, each replaying m_t samples; summing $K m_t^{3/2}$ proves the cost claim. □

Under the hypotheses of Corollary 9, if the window policy uses the predictable rank envelope $\Gamma_{t-1,r}$ instead of the realized $\hat{\gamma}_{t-1}$ in its confidence radius, then $H_T \leq \alpha_I (1 + x_T) \Omega_{T,r}^{\mathrm{win}}$, where $\Omega_{T,r}^{\mathrm{win}}$ is the explicit $\Omega_{T,r}$ in Corollary 9 with $A_{\inf} = 1$. Hence (24) contains no uncontrolled information-gain placeholder. The excitation premise is restrictive and standard in persistently excited identification; contextual diversity alone does not imply it. For $\dim U = r > 1$ one must choose τ_w after a rank-r burn-in, as the formal early charge records. The predictable rank schedule in Corollary 9 makes H_T polylogarithmic under the same bounded-path conditions. For fixed r, fixed τ_w, and fixed problem constants, the Pareto exponents $(q, \mathrm{regret}, \mathrm{cost})$ are $(1/2, 3/4, 7/4)$, $(2/3, 2/3, 2)$, and $(1, 1/2, 5/2)$.

5 Experiments

All regret rows are independently executed policies. Common random numbers are used only where pairing is valid. Exact teacher and dense replay quantities are post-action audits; common-trajectory operator comparisons never report causal regret. Tuning and evaluation seeds are disjoint. The derived full-grid aggregate retains seed-level checkpoints and complete hash provenance; release scope and missing raw tiers are stated in Appendix J.

The grid contains three ambient dimensions, three active ranks, eight methods, and 50 fixed seeds. For the column-orthonormal embedding E_d,

$$(E_d g)^\top (\lambda I_d + E_d A_t E_d^\top)^{-1} (E_d g) = g^\top (\lambda I_r + A_t)^{-1} g.$$

Thus non-diagonal widths are ambient-invariant in exact arithmetic. Empirically, the maximum cross-d regret range is zero; the dynamic-width range is 5.7×10^{-14} for non-diagonal methods and 2.2501×10^3 for the embedding-dependent diagonal method. Ambient-CVP runtime is excluded.

The policy predeclares $\zeta_t = 10^{-6}$ and stops after at most 40 optimizer iterations; it does not enforce that envelope. A failed field therefore means the executed radius does not instantiate the theorem on that round, so the path is an uncertified diagnostic. Five-point slopes and fit residuals are in Table 4.

Full CG is a one-step equivalence test: each right-hand side is an active coordinate and the Gram is diagonal in that basis. Width errors are about 10^{-15}, yet near-tie perturbations can change an action and the subsequent adaptive trajectory. The rotated-SPD audit in Appendix C, not this scaling row, tests nontrivial Krylov convergence.

5.1 Predictable nonlinear execution

We first use the genuinely nonlinear Gaussian model $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $d = 6$, four bounded actions, $T = 100$, $\|\theta^*\| \leq 0.35$, and a radius-0.5 projected optimizer. The analytic constants in Corollary 10, exact full-objective gradient norm, the path statistics in (16), and the analytic condition bound $1 + (t - 1)G^2/(\lambda\sigma^2)$ are computed before action selection. No teacher, future reward, exact χ_t, exact ψ_t, or exact Taylor error enters the policy. A fixed protocol is evaluated on 50 seeds (160–209); an eight-cell

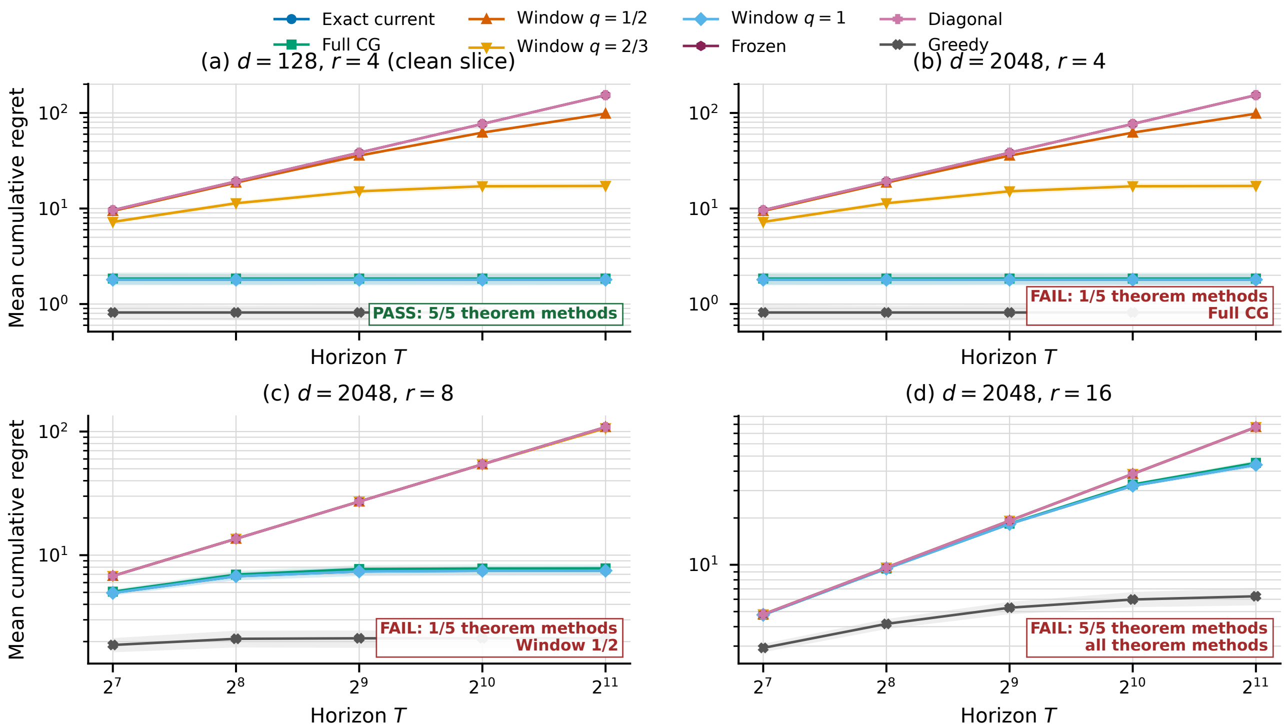

Figure 1: Finite-horizon theorem-scaling diagnostic. The first panel is the event-clean $d = 128, r = 4$ reference; the remaining panels use $d = 2048$. Curves are 50-seed means and bands are pointwise percentile-bootstrap 95% intervals at five preregistered prefixes. Panel labels report the conjunction of float64 theorem-event fields for exact, CG, and the three window methods. Greedy is uncertified. No exponent comparison tests Theorem 3 unless every premise and audit field for that method/rank cell passes.

one-factor sensitivity grid is confined to ten tuning seeds.

Every original- and corrected-center trajectory passes every recorded transfer, centering, linearization, information, CG, confidence, optimism, and regret event: zero observed failures over 5,000 rounds per center. Mean regret is 7.86 $[7.66, 8.07]$ for the original center and 5.69 $[5.46, 5.92]$ for the corrected center. The schedules are deliberately conservative: at the final round, original χ_t averages 0.0216 versus $\bar{\chi}_t = 4.89$, and ψ_t averages 0.0520 versus $\bar{\psi}_t = 128.1$. The mean observable RHS/regret ratios are 6,956.7 and 1,328.9, respectively, so the bounds are numerically vacuous. Because the residual and dense audit checks are ordinary float64 point computations, these executions are classified *post-hoc theorem-event verified with fully predictable schedules*, not verified-enclosure theorem certified.

5.2 Reference-operator falsification and matched baselines

The phase map crosses active-spectrum condition, rotation relative to the decision direction, gap, nuisance strength, effective rank, and damping in eight balanced cells. Each of seven operators runs 30 independent policies; the exact-full path is also replayed for per-action width ratios, ratio coefficient of variation, Spearman/Kendall ranks, top-score disagreement before and after scalar rescaling, decision-margin distortion, eigenspace alignment, global and action-set transfer, effective rank, and condition number. Exact full CG matches dense full regret in all cells. Diagonal has lower regret in all eight cells, and block diagonal in six (two unresolved); full has lower regret than window and stale refresh in all eight, and than rank-3 Lanczos in seven (one unresolved). Thus fidelity alone does not order regret. Scalar-rescaled disagreement is often much smaller than raw score disagreement, consistent with the invariance in Proposition 25; the remaining action changes are anisotropic.

The separately executed off-diagonal linear-Gram construction makes that anisotropy concrete (Figure 3 and table 6). Exact full curvature and residual-checked full CG coincide in the analytic witness, whereas raw diagonalization changes the selected action and accumulates regret. No uniform full-curvature ranking follows: the witness is existential, the noisy extensions are un-

Cell (d/r)	Exact	CG	$q = 1/2$	$q = 2/3$	$q = 1$	Frozen	Diag.	Greedy
128/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/8	7.50	7.82	108.61	106.44	7.50	108.61	108.61	2.13
2048/16	43.95	45.24	76.80	76.80	43.95	76.80	76.80	6.27

	Theorem-event status and failed round–event categories			
Method	$d = 128, r = 4$	$d = 2048, r = 4$	$d = 2048, r = 8$	$d = 2048, r = 16$
Exact current	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]
Full CG	PASS	FAIL [O=1]	PASS	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 1/2$	PASS	PASS	FAIL [O=1]	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 2/3$	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]
Window $q = 1$	PASS	PASS	PASS	FAIL [O=49, P_e=46, P_λ=45]

PASS means that all recorded float64 theorem-event fields passed for that method and cell.
Failure counts are round–event fields: O is optimizer residual, P_e is ψ-excitation, and P_λ is ψ-lambda.

Table 1: Preregistered scaling summary. Panel (a) reports terminal mean regret for every executed method. Panel (b) decomposes failed float64 theorem-event fields by method: O is the prescribed optimizer-residual envelope, P_e is the path bound using the excitation floor, and P_λ is the path bound using damping. Pass means zero observed failures for the displayed fields, not a verified numerical enclosure. Any nonzero entry precludes interpreting that cell's slope as a test of the corresponding conditional theorem.

certified, and the broader phase map above contains cells where diagonal curvature has lower regret.

For a separate balanced nonlinear contextual benchmark, all five arms are optimal on the exact 16-context support. Ten tuning seeds choose each method's ridge and bonus from the same prespecified grid; 30 fresh seeds share contexts and reward noise. The local baselines include LinUCB, linear Thompson sampling, a Bayesian frozen-representation NeuralLinear policy, local full-network NeuralUCB/NeuralTS, diagonal full-network UCB, frozen last-layer UCB, greedy, and two context-free policies. These local implementations are identified in the artifact and are not claimed to reproduce every published training detail. At $T = 200$, CC-UCB full GGN-CG has regret 21.94 $[20.64, 23.23]$; diagonal has 16.11 $[14.62, 17.61]$, LinUCB 10.58 $[9.82, 11.33]$, NeuralLinear 10.01 $[9.33, 10.68]$, and frozen last-layer UCB 7.44 $[6.90, 7.99]$. Context-free UCB1 and Thompson sampling have 90.35 and 89.40, so the contextual sanity check passes, but the experiment is negative evidence for uniform full-curvature superiority.

5.3 Audits and systems scope

All 280 bounded $d = 53$ linear traces pass the float64 theorem checks, but unenclosed spectral estimates leave CG, diagonal, subsample, and Lanczos rows post-hoc verified. The dense pre-action oracles test semantics, not scalable certification; RHS/regret at $T = 1000$ ranges from 260.2 to 13,456.0. The supplementary 45-parameter stress test retains its failed schedules.

The synthetic CPU implementation records a 1.8× row-batched/separate timing ratio at $d = 128, n = 512, K = 10$ and reaches $d = 8192$ without dense allocation. These unmatched, contention-affected primitives provide resource accounting, not a speed comparison. The defining autodiff JVP/VJP benchmark was not run because the declared Buck PyTorch dependency does not configure; no trained-model or accelerator performance is inferred.

Real-data baseline check. The Covertype study (Appendix C.1) fails its basic baseline check: at every horizon, deployable UCB1 and Beta–Bernoulli Thompson sampling beat every contextual method. It is not external validation of curvature-aware exploration.

6 Conclusion

CC-UCB supplies a matrix-free full-GGN/Fisher reference with explicit transfer, CG, and width terms. Its closed rate is conditional fixed-subspace adaptation, not unrestricted neural training. Empirically, the operator is a reference for testing width and action-order preservation: the tanh bound is vacuous, the scaling grid exposes failed envelopes, and full curvature is not uniformly preferred. Covertype fails its baseline check.

Limitations. Theory assumes Gaussian squared loss, enumeration, realizability, known smoothness, and a certified optimizer residual. Approximate operators, warm starts, and screening need separate certificates. Float64 checks are not verified enclosures; bounds are vacuous; actual autodiff timing is absent; and adaptive paths confound non-causal operator comparisons. No universal full-curvature advantage is claimed.

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A Deferred main-text derivations

This appendix collects the formal statements, proofs, and derivations deferred from the compact main text; all cross-referenced results are stated here.

For the sequential frozen-feature Gram, the usual information gain is

$$\gamma_T := \log \frac{\det(\bar{\mathbf{C}}_{T+1})}{\det(\lambda \mathbf{I})} = \sum_{t=1}^{T} \log\bigl(1 + \sigma^{-2} \bar{s}_t^2(a_t)\bigr) . \tag{25}$$

Lemma 4 (Rank and statistical-dimension information closure)**.** *For $A \succeq 0$, $r \geq \operatorname{rank}(A)$, and $q_A = \|A\|_{\mathrm{op}}/\lambda$,*

$$\log\det(\mathbf{I} + A/\lambda) \leq \min\left\{ r \log\left(1 + \frac{\operatorname{tr} A}{r\lambda}\right), k(q_A) d_{\mathrm{eff},\lambda}(A) \right\}, \tag{26}$$

where $d_{\mathrm{eff},\lambda}(A) = \operatorname{tr}[A(A+\lambda\mathbf{I})^{-1}]$, $k(0) = 1$, and $k(q) = (1+q)\log(1+q)/q$ for $q > 0$; the rank term is defined as zero when $r = 0$ (which forces $A = 0$). Consequently, if $\operatorname{rank}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq r$, then $\gamma_T \leq \Gamma_{T,r}$ from (21).

Proof. Writing $x_i = \lambda_i(A)/\lambda$, concavity of $\log(1+x)$ after padding the spectrum with zeros gives the rank–trace term. Also

$$\log(1+x) = \frac{(1+x)\log(1+x)}{x} \frac{x}{1+x}.$$

The first factor is increasing because its derivative is $(x - \log(1+x))/x^2 \geq 0$; bounding it at q_A and summing proves the statistical-dimension term. For the frozen Gram, $\operatorname{tr}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq TG^2/\sigma^2$, which gives (21) (with $r_T = \min\{r, T\}$). □

The second term is useful only with a predictable or deterministic envelope on $d_{\mathrm{eff},\lambda}$; a realized value is post-hoc. It is not the empirical trace effective rank $\operatorname{tr}(A)/\|A\|_{\mathrm{op}}$, which does not in general control log determinant.

The dynamic-potential and CG-observable consequences of Theorem 1 are

$$\begin{aligned} R_T &\leq 2\sqrt{(\sigma^2 + G^2/\lambda)\Gamma_T^{\mathrm{dyn}} S_T} + 2E_T, \\ R_T &\leq 2\sqrt{(\sigma^2 + G^2/\lambda)\widehat{\Lambda^{\mathcal{C}}}_T S_T} + 2E_T. \end{aligned} \tag{27}$$

Corollary 5 (Corrected-center variant, no centering certificate)**.** *Replace the predictive center in Algorithm 1 by*

$$m_t^{\mathrm{corr}}(a) := \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a) + \mathbf{g}_t(a)^\top (\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}_t) \tag{28}$$

and use $\omega_t = \bar{\beta}_t$. Under the hypotheses of Theorem 1 except Assumption 4, the bounds (15)–(27) hold.

Proof. Local linearization gives $\mu^*(\mathbf{x}_t, a) - m_t^{\mathrm{corr}}(a) = \mathbf{g}_t(a)^\top(\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) + \rho_t$. Lemma 19 bounds the first term by $\beta_t \bar{s}_t(a)$ and $|\rho_t| \leq \varepsilon_{\mathrm{lin}}(t)$, which is the first step of the regret proof with $\omega_t = \bar{\beta}_t$; all later steps are unchanged. □

Remark (Cost of the corrected center)**.** *Exact evaluation requires a solve against $\bar{\mathbf{C}}_t$. A constant number of $O(d)$ CG vectors is insufficient by itself: one additionally needs stored historical frozen gradients, raw replay data with the historical parameter checkpoints needed to reconstruct them, a frozen-feature matrix–vector oracle, an explicit low-dimensional representation, or a certified sketch. Maintaining $\bar{\mathbf{C}}_t$ explicitly costs $O(d^2)$ memory. Approximate frozen-feature access introduces an error that must be returned to the centering certificate unless separately controlled.*

A.1 Assumptions

Assumption 1 (Sub-Gaussian noise)**.** *The noise η_t is conditionally σ-sub-Gaussian given the context and action: $\mathbb{E}[\exp(s\eta_t) \mid \mathcal{G}_t] \leq \exp(s^2\sigma^2/2)$ for all $s \in \mathbb{R}$, where $\mathcal{G}_t := \mathcal{H}_t \vee \sigma(a_t)$ is the pre-reward sigma-algebra of (68), available immediately after the action is selected and before the reward is revealed.*

Assumption 2 (Bounded gradient features)**.** *For a constant $G > 0$,*

$$\sup_{t \le T} \sup_{s \le t} \sup_{a} \|\nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a)\| \;\le\; G.$$

This bounds both the current-round features $\mathbf{g}_t(\mathbf{x}_t, a)$ and the replayed features $\mathbf{g}_t(\mathbf{x}_s, a_s)$ appearing in the curvature matrix (5)*, as well as the frozen features $\mathbf{g}_s(\mathbf{x}_s, a_s)$ in* (6)*.*

Assumption 3 (Local linearization)**.** *There is a known convex parameter region Θ containing $\boldsymbol{\theta}^*$ and every realized $\boldsymbol{\theta}_t$, on which $\nabla_\theta \mu_\theta(\mathbf{x}, a)$ is L_μ-Lipschitz uniformly over candidate $(\mathbf{x}, a)$. Thus*

$$\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) \;=\; \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) + \mathbf{g}_t(\mathbf{x}, a)^\top (\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \rho_t(\mathbf{x}, a, \boldsymbol{\theta}), \tag{29}$$

where the remainder satisfies $|\rho_t(\mathbf{x}, a, \boldsymbol{\theta})| \le \frac{L_\mu}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_t\|^2$ for $\boldsymbol{\theta} \in \Theta$. Evaluating at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ gives a time-varying linearization error

$$\varepsilon_{\text{lin}}(t) \;:=\; \sup_{a \in \mathcal{A}_t} |\rho_t(\mathbf{x}_t, a, \boldsymbol{\theta}^*)| \;\le\; \frac{L_\mu}{2}\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\|^2. \tag{30}$$

For Theorem 1 to give a sublinear guarantee, we require the cumulative linearization error $E_T := \sum_{t=1}^T \varepsilon_{\text{lin}}(t) = o(T)$; see Theorem 1 for the precise dependence. Concretely, $E_T = O(1)$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ (since $\varepsilon_{\text{lin}}(t) = O(1/t^2)$ and $\sum t^{-2} < \infty$; achievable in convex or strongly convex regimes), and $E_T = O(\sqrt{T})$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$. In practice this is enforced by restricting fine-tuning depth, using trust-region updates, or freezing the backbone (which renders the last-layer model convex or near-linear with small L_μ). These rate examples are illustrative only. The present paper does not prove convergence of the nonconvex optimization path $\{\boldsymbol{\theta}_t\}$ to $\boldsymbol{\theta}^*$; a sublinear guarantee from this bound therefore requires this term to be controlled by an external stability or convergence argument.

Assumption 4 (Action-scaled centering certificate)**.** *Lemma 6 below proves the geometric bound $\left|\mathbf{g}_t(\mathbf{x}_t, a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})\right| \le \bar{s}_t(a)\, \psi_t$ for every $a \in \mathcal{A}_t$, where*

$$\psi_t \;:=\; \frac{\zeta_t}{\sqrt{\lambda}} \;+\; \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}, \tag{31}$$

$\zeta_t \ge 0$ is an $\mathcal{H}_t$-measurable optimizer gradient-residual bound ($\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \le \zeta_t$) and M_t is the gradient mismatch vector defined there. We assume there is an $\mathcal{H}_t$-measurable upper bound $\bar{\psi}_t \ge \psi_t$ (the centering certificate*) such that, for every action $a \in \mathcal{A}_t$,*

$$\left|\mathbf{g}_t(\mathbf{x}_t, a)^\top \big(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}}\big)\right| \;\le\; \bar{\psi}_t\, \bar{s}_t(a). \tag{32}$$

The discrepancy is thus action-scaled*: it multiplies the confidence width $\bar{s}_t(a)$ rather than entering additively. The certificate absorbs the same three sources as before—optimizer residual ζ_t, replayed-Jacobian drift inside M_t, and higher-order nonlinear terms—but now inside a single per-round scalar $\bar{\psi}_t$. In the convex-in-parameters regime (frozen backbone with a linear reward head), under exact optimization $\hat{\boldsymbol{\theta}}_t^{\text{lin}} = \boldsymbol{\theta}_t$ and $\bar{\psi}_t = 0$. The corrected-center variant of Corollary 5 removes $\bar{\psi}_t$ entirely, at the cost of an explicit frozen-feature solve.*

Remark (Status of Assumption 4)**.** *Lemma 7 supplies a predictable valid $\bar{\psi}_t$ from the optimizer residual, Welford path moments, and observed residual energy. We do not prove that this conservative quantity is small outside a restricted path. It enters the exploration term through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ (Equation* (59)*), not as an additive $O(T)$ term.*

The next lemma gives the primitive geometric bound underlying Assumption 4, expressed through the certifiable quantity ψ_t.

Lemma 6 (Primitive upper bound for prediction discrepancy)**.** *Let $z_s = (\mathbf{x}_s, a_s)$, $\mathbf{g}_s = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(z_s)$, $\mathbf{g}_{s,t} = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(z_s)$, and $\delta_{s,t} = \mathbf{g}_{s,t} - \mathbf{g}_s$. Define*

$$\bar{\mu}_{s,t} := \mu_{\boldsymbol{\theta}_s}(z_s) + \mathbf{g}_s^\top (\boldsymbol{\theta}_t - \boldsymbol{\theta}_s), \qquad e_{s,t} := \mu_{\boldsymbol{\theta}_t}(z_s) - \bar{\mu}_{s,t}, \qquad \bar{r}_{s,t} := \bar{\mu}_{s,t} - r_s.$$

Let $\bar{\mathcal{L}}_t$ be the frozen-feature linearized ridge objective whose minimizer is $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$, and define

$$M_t := \frac{1}{\sigma^2} \sum_{s<t} \left[\bar{r}_{s,t}\delta_{s,t} + e_{s,t}\mathbf{g}_s + e_{s,t}\delta_{s,t}\right].$$

Assumption	Status	Use	Checkability	Experimental instantiation
Conditional sub-Gaussian noise	Standard	Self-normalized confidence	Distributional	Gaussian generator
Realizability, $\|\theta^*\| \leq S$	Restrictive	Bias and Taylor schedules	Not generally testable	Known synthetic teacher
Bounded gradients	Standard on compact sets	Width and condition bounds	Analytic or monitored	Analytic tanh bound
L_μ, L_g on convex Θ	Restrictive	Linearization, transfer, centering	Analytic for tanh; otherwise hard	Global tanh Hessian bound and projection
Optimizer residual ζ_t	Algorithmic	Centering certificate	Pre-action gradient norm	Exact full-objective gradient norm
Operator and CG certificates	Algorithmic	Optimism and solve error	Residual/Loewner proof required	Analytic condition bound; float64 audit
Residual-energy envelope	Rate-only	Near-linear scaling	Observable after collection	Reported, not assumed by policy
Fixed active rank r	Restrictive	Closes γ_T	Supplied subspace, not terminal audit	Constructed active embedding
Window excitation κ_w	Restrictive	Closes $\Lambda_T^{\mathcal{C}}$	Analytic construction or assumed	Cyclic construction after burn-in

Table 2: Assumption ledger. Analytic and pre-action quantities are policy available; teacher comparisons and exact whitened quantities are post-hoc only.

If the nonlinear optimizer satisfies $\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \leq \zeta_t$, then for every candidate action $a \in \mathcal{A}_t$,

$$\left|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})\right| \;\leq\; \bar{s}_t(a)\left(\frac{\zeta_t}{\sqrt{\lambda}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\right), \qquad \bar{s}_t(a) := \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a)}.$$

The left-hand side is exactly the geometric bound $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \leq \bar{s}_t(a)\,\psi_t$ of Assumption 4 with $\psi_t = \zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}$ (Equation (31)). Consequently, since $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \leq \|M_t\|_2/\sqrt{\lambda}$, a valid centering certificate is

$$\bar{\psi}_t \;:=\; \frac{\zeta_t + \|M_t\|_2}{\sqrt{\lambda}} \;\geq\; \psi_t.$$

If, in addition, on a fixed event, $\boldsymbol{\theta} \mapsto \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}}(z)$ is L_g-Lipschitz, $|\bar{r}_{s,t}| \leq B_{t,\delta}$ for all $s < t$, and $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$, then

$$\|M_t\|_2 \;\leq\; \frac{1}{\sigma^2}\sum_{s<t}\left[B_{t,\delta}L_g\Delta_{s,t} + \frac{GL_g}{2}\Delta_{s,t}^2 + \frac{L_g^2}{2}\Delta_{s,t}^3\right].$$

Proof. The linearized ridge objective satisfies $\nabla\bar{\mathcal{L}}_t(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) = 0$ and, by its quadratic form,

$$\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \bar{\mathbf{C}}_t(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}).$$

Thus $\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} = \bar{\mathbf{C}}_t^{-1}\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)$. Using $\mu_{\boldsymbol{\theta}_t}(z_s) - r_s = \bar{r}_{s,t} + e_{s,t}$ and $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, the gradients of the nonlinear loss and the frozen-feature linearized loss satisfy

$$\nabla\mathcal{L}_t(\boldsymbol{\theta}_t) = \nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) + M_t, \qquad \text{so} \qquad \nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \nabla\mathcal{L}_t(\boldsymbol{\theta}_t) - M_t.$$

Therefore, by Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm,

$$|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \leq \bar{s}_t(a)\|\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \leq \bar{s}_t(a)\left(\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\right).$$

Since $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$, $\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \leq \zeta_t/\sqrt{\lambda}$, which gives the geometric inequality $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \leq \bar{s}_t(a)\big(\zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\big) = \bar{s}_t(a)\,\psi_t$, i.e. the bound of (31). For the certificate, $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$ also gives $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \leq \|M_t\|_2/\sqrt{\lambda}$, so $\psi_t \leq (\zeta_t + \|M_t\|_2)/\sqrt{\lambda} =: \bar{\psi}_t \geq \psi_t$, the proposed certificate. For the final display, Lipschitzness gives $\|\delta_{s,t}\| \leq L_g\Delta_{s,t}$ and the Taylor remainder bound $|e_{s,t}| \leq (L_g/2)\Delta_{s,t}^2$; applying these together with $\|\mathbf{g}_s\| \leq G$ (Assumption 2) and $|\bar{r}_{s,t}| \leq B_{t,\delta}$ term by term in the definition of M_t gives the claimed bound on $\|M_t\|_2$. □

Lemma 7 ($O(d)$-state path certificates)**.** *Suppose $\nabla_\theta \mu_\theta(z)$ is L_g-Lipschitz and bounded by G on a convex region containing the realized iterates. Before round t, let $n_t = t-1$, let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ summarize past iterates, and let $R_t^{\mathrm{col}} = \sum_{s<t} c_s^2$ with $c_s = \mu_{\boldsymbol{\theta}_s}(z_s) - r_s$. At $t=1$ use the initialization $\bar{\boldsymbol{\theta}}_{<1} = \mathbf{0}$, $J_{<1} = Q_1 = R_1^{\mathrm{col}} = 0$. Then, for all t,*

$$Q_t := J_{<t} + n_t \|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2 = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$$

and the following $\mathcal{H}_t$-measurable quantities are valid:

$$\bar{\chi}_t = \frac{L_g}{\sigma\sqrt{\lambda}} \sqrt{Q_t} \geq \chi_t, \tag{33}$$

$$\bar{M}_t = \frac{1}{\sigma^2} \left[L_g \sqrt{R_t^{\mathrm{col}} Q_t} + \frac{3GL_g}{2} Q_t + \frac{L_g^2}{2} Q_t^{3/2} \right] \geq \|M_t\|_2. \tag{34}$$

If $\|\boldsymbol{\theta}_s\|, \|\boldsymbol{\theta}_t\| \leq R$, the final term in (34) may be replaced by $L_g^2 R Q_t$. If a nonnegative $\mathcal{H}_t$-measurable optimizer certificate satisfies $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\| \leq \zeta_t$, then

$$\bar{\psi}_t = (\zeta_t + \bar{M}_t)/\sqrt{\lambda} \geq \psi_t.$$

If prediction gradients are L_μ-Lipschitz on a convex region containing $\boldsymbol{\theta}_t$ and $\boldsymbol{\theta}^*$ and $\|\boldsymbol{\theta}^*\| \leq S$, then

$$\bar{\epsilon}_t^{\mathrm{lin}} := \frac{L_\mu}{2} (S + \|\boldsymbol{\theta}_t\|)^2 \geq \varepsilon_{\mathrm{lin}}(t).$$

Consequently $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\mathrm{lin}})^2 \geq F_t$ and $\bar{E}_T = \sum_{t \leq T} \bar{\epsilon}_t^{\mathrm{lin}} \geq E_T$.

Proof. The parallel-axis identity follows after writing $\boldsymbol{\theta}_t - \boldsymbol{\theta}_s = (\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}) + (\bar{\boldsymbol{\theta}}_{<t} - \boldsymbol{\theta}_s)$; the cross term sums to zero. Lemma 11 and $\|\nabla \mu_{\boldsymbol{\theta}_t}(z_s) - \nabla \mu_{\boldsymbol{\theta}_s}(z_s)\| \leq L_g \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$ give (33). Let $d_{s,t} = \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. In Lemma 6,

$$|\bar{r}_{s,t}| \leq |c_s| + G d_{s,t}, \qquad \|\delta_{s,t}\| \leq L_g d_{s,t}, \qquad |e_{s,t}| \leq (L_g/2) d_{s,t}^2.$$

Thus each summand defining M_t has norm at most

$$L_g |c_s| d_{s,t} + \frac{3GL_g}{2} d_{s,t}^2 + \frac{L_g^2}{2} d_{s,t}^3.$$

Cauchy–Schwarz gives $\sum |c_s| d_{s,t} \leq \sqrt{R_t^{\mathrm{col}} Q_t}$, while $\sum d_{s,t}^2 = Q_t$ and $\sum d_{s,t}^3 \leq Q_t^{3/2}$, proving (34). Under the trust region, $d_{s,t} \leq 2R$, so $\sum d_{s,t}^3 \leq 2RQ_t$. The centering claim follows from Lemma 6 and $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$. Finally Taylor's theorem and $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| \leq S + \|\boldsymbol{\theta}_t\|$ prove the remainder, F_t, and E_T claims. ☐

Filtration-order implementation. At round t, the summaries through $t-1$, $\boldsymbol{\theta}_t$, and the certificate ζ_t are available before selection. The learner computes $Q_t, \bar{\chi}_t, u_t, \bar{M}_t, \bar{\psi}_t, \bar{\epsilon}_t^{\mathrm{lin}}$ and $\bar{\beta}_t$, then solves every candidate system to the uniform tolerance and selects a_t. After observing r_t, it adds c_t^2 to R_{t+1}^{col}, adds $\boldsymbol{\theta}_t$ with the Welford update

$$d = \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}, \quad \bar{\boldsymbol{\theta}}_{\leq t} = \bar{\boldsymbol{\theta}}_{<t} + d/t, \quad J_{\leq t} = J_{<t} + d^\top (\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{\leq t}),$$

and updates $\bar{F}_{t+1}$ and $\hat{\gamma}_t$. It then forms $\boldsymbol{\theta}_{t+1}$ and certifies the nonnegative bound ζ_{t+1} before the next action. No current reward enters the current score.

In frozen linear-head models Q_t may be nonzero while $L_g = L_\mu = 0$; then $\bar{\chi}_t = \bar{M}_t = \bar{\epsilon}_t^{\mathrm{lin}} = 0$, and exact optimization also gives $\bar{\psi}_t = 0$.

Corollary 8 (Bounded-path near-linear rate)**.** *Fix T and suppose for every $t \leq T$*

$$\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \leq R, \quad L_\mu \leq c_\mu/\sqrt{W}, \quad L_g \leq c_g/\sqrt{W}, \quad \zeta_t \leq \zeta_0/\sqrt{W}, \quad R_t^{\mathrm{col}} \leq C_R t P_T,$$

where $P_T \geq 1$ is an explicitly supplied residual-energy factor. For exact current-parameter curvature and a uniform CG tolerance $\bar{\varepsilon}_t \leq \bar{\varepsilon} < 1$, set

$$x_T = \frac{2c_g R\sqrt{T-1}}{\sigma\sqrt{\lambda W}}, \quad \epsilon_{\mathrm{drift,T}} = 2x_T + x_T^2, \quad \rho_\pm = 1 \pm \epsilon_{\mathrm{drift,T}}.$$

If $\rho_- > 0$, then $\rho_-\bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq \rho_+\bar{\mathbf{C}}_t$ for all $t \le T$. Writing

$$\alpha_I = \sqrt{\frac{1+\bar\varepsilon}{1-\bar\varepsilon}}, \qquad A_{\inf} = \alpha_I^2\frac{\rho_+}{\rho_-},$$

the observable information schedule satisfies $\gamma_T \le \hat\gamma_T \le A_{\inf}\gamma_T$, and

$$R_T \le 2\alpha_I\sqrt{\rho_+/\rho_-}\,\omega^{\mathrm{op}}_{\max,T}\sqrt{(\sigma^2+G^2/\lambda)T\gamma_T} + 2\bar E_T \tag{35}$$

holds with

$$\omega^{\mathrm{op}}_{\max,T} := \max_{t\le T}(\bar\beta_t + \bar\psi_t)$$

and

$$\begin{aligned}\omega^{\mathrm{op}}_{\max,T} \le{}& \sqrt{A_{\inf}\gamma_T + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} \\ &+ \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda}W}, \\ A_T ={}& \zeta_0 + \sigma^{-2}\Big[2c_gR\sqrt{C_RT(T-1)P_T} + 6Gc_gR^2(T-1)\Big], \\ B_T ={}& 4c_g^2R^3(T-1)/\sigma^2.\end{aligned}$$

In addition, (20) *holds and*

$$\bar\psi_t \le \frac{1}{\sqrt{\lambda}}\left\{\frac{\zeta_0}{\sqrt{W}} + \frac{1}{\sigma^2}\left[\frac{2c_gR\sqrt{C_Rt(t-1)P_T} + 6Gc_gR^2(t-1)}{\sqrt{W}} + \frac{4c_g^2R^3(t-1)}{W}\right]\right\}.$$

Proof. The trust region gives $Q_t \le 4R^2(t-1)$; substitution in Lemma 7 proves all displayed path bounds. Corollary 12 gives the two-sided operator relation. The CG sandwich and $u_t \le (1+x_T)^2 = \rho_+$ imply that every logarithm in $\hat\gamma_T$ is at most $\log(1+A_{\inf}\sigma^{-2}\bar s_t^2(a_t))$. For $A \ge 1$, $\log(1+Ax) \le A\log(1+x)$, proving the information comparison. Substitution into Corollary 29 gives the regret display. □

Corollary 9 (Conditional fixed-subspace adaptation rate)**.** *Under Corollary 8, additionally suppose a subspace $U \subseteq \mathbb{R}^d$ with $\dim U \le r$ is fixed and supplied before interaction, and every candidate or replayed tangent feature through round T lies in U. Hence* $\operatorname{rank}(\bar{\mathbf{C}}_t - \lambda\mathbf{I}) \le r$ *for every $t \le T+1$. Define $\Gamma_{T,r}$ by* (21)*. Then* (22) *holds with*

$$\begin{aligned}\Omega_{T,r} :={}& \sqrt{A_{\inf}\Gamma_{T,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda}W}, \\ A_T ={}& \zeta_0 + \sigma^{-2}\Big[2c_gR\sqrt{C_RT(T-1)P_T} + 6Gc_gR^2(T-1)\Big], \\ B_T ={}& 4c_g^2R^3(T-1)/\sigma^2,\end{aligned}$$

and $A_{\inf}, \rho_\pm$ are exactly those in Corollary 8. In particular, for fixed r and fixed problem constants, if $W \ge CT^2P_T$ for a constant C satisfying the explicit thresholds in (36)*, then*

$$R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W}).$$

This rate is conditional on the ex ante subspace U and every bounded-path, width, residual, optimizer, and CG premise of Corollary 8. It neither learns U nor covers unrestricted representation learning or end-to-end neural-network training.

Proof. Lemma 4 gives $\gamma_T \le \Gamma_{T,r}$. Every occurrence of γ_T in Corollary 8 is monotone, so direct substitution gives the display. Under the stated width scaling, x_T, the centering terms, and the linearization terms are bounded, while $\Gamma_{T,r} = O(r\log T)$. Hence $\Omega_{T,r} = O(\sqrt{r\log T} + 1)$ and the leading product is $O(r\sqrt{T}\log T)$. □

For explicit targets $\bar{\epsilon}_{\text{drift}} \in (0,1)$, $\Psi, b_F, \eta_E > 0$, put $x_* = \sqrt{1+\bar{\epsilon}_{\text{drift}}} - 1$. The sufficient conditions

$$\begin{aligned} &W \geq \frac{4c_g^2 R^2 (T-1)}{\lambda \sigma^2 x_*^2}, \qquad W \geq \frac{4c_\mu^2 R^4 T}{\sigma^2 b_F^2}, \quad W \geq \frac{4c_\mu^2 R^4}{\eta_E^2}, \\ &W \geq \max\left\{\frac{4A_T^2}{\lambda \Psi^2}, \frac{2B_T}{\sqrt{\lambda}\Psi}\right\} \end{aligned} \tag{36}$$

respectively ensure bounded drift, $\sqrt{\bar{F}_{T+1}}/\sigma \leq b_F$, $\bar{E}_T/T \leq \eta_E$, and $\bar{\psi}_T \leq \Psi$. Thus the centering requirement has leading sufficient scaling $W = \Omega(T^2 P_T)$; these conditions do not bound γ_T.

Corollary 10 (Nonlinear tanh-link Gaussian bandit)**.** *For $\mu_\theta(x,a) = \tanh(\phi(x,a)^\top \theta)$ with $\|\phi(x,a)\| \leq B_\phi$, one may take*

$$G = B_\phi, \qquad L_\mu = L_g = \frac{4}{3\sqrt{3}} B_\phi^2.$$

On any certified trust region, Corollary 2 therefore gives an executable nonlinear Gaussian-reward CC-UCB policy using only the online path statistics, a pre-action optimizer-residual certificate, and the stated operator/CG certificates.

Proof. $\nabla_\theta \mu = \operatorname{sech}^2(\phi^\top \theta)\phi$, so its norm is at most B_ϕ. The Hessian is $-2\tanh(z)\operatorname{sech}^2(z)\phi\phi^\top$. For $y = |\tanh z|$, $2y(1-y^2)$ is maximized at $y = 1/\sqrt{3}$ with value $4/(3\sqrt{3})$, proving both Lipschitz constants. Apply Lemma 7 and Corollary 2. $\square$

Generic algorithmic curvature. The analysis is conducted for a *generic predictable SPD algorithmic curvature sequence* $\{\mathcal{C}_t\}_{t \geq 1}$ satisfying

$$\mathcal{C}_t \text{ is symmetric}, \qquad \mathcal{C}_t \succeq \lambda \mathbf{I} \succ 0, \qquad \mathcal{C}_t \text{ is } \mathcal{H}_t\text{-measurable.} \tag{37}$$

This is the operator whose inverse the algorithm actually applies (via CG) to form widths. The exact full-history result is the special case $\mathcal{C}_t = \mathbf{C}_t$ (relinearized curvature (5)); approximate, subsampled, windowed, refreshed, or sketched operators $\hat{\mathbf{C}}_t$ are also represented by $\mathcal{C}_t$, but are covered only when the required transfer certificate below is available. Write the algorithmic width

$$s_{\mathcal{C},t}^2(a) \;:=\; \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a). \tag{38}$$

Assumption 5 (One-sided optimism transfer)**.** *For every round $t \leq T$ there is an $\mathcal{H}_t$-measurable factor $u_t(a) \geq 1$ such that, for every candidate action $a \in \mathcal{A}_t$,*

$$\bar{s}_t^2(a) \;\leq\; u_t(a)\, s_{\mathcal{C},t}^2(a). \tag{39}$$

This is the *only* spectral relation the main theorem needs, and it is one-sided: it dominates the predictable frozen-feature width $\bar{s}_t(a)$ (the object of the self-normalized confidence bound) by an inflated algorithmic width. No *lower* Loewner factor on $\mathcal{C}_t$ is required for optimism. The factor may be action-dependent; when a uniform per-round u_t satisfies (39) for every action we write u_t and state the action-dependent theorem with $u_t(a)$. For the exact-curvature instantiation $\mathcal{C}_t = \mathbf{C}_t$, the whitened relative-drift lemma below certifies (39) with the uniform factor $u_t = (1+\bar{\chi}_t)^2$; for an approximate operator, the composition of Lemma 11 with a one-sided spectral bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ gives $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ (Theorem 24).

The following lemma supplies the exact-curvature transfer factor through a *whitened, relative* feature-drift bound: the drift is measured in the $\bar{\mathbf{C}}_t^{-1}$ metric and accumulated in ℓ_2, and the resulting one-sided bound needs *no* smallness condition.

Lemma 11 (Whitened relative feature-drift bound)**.** *Let $\mathbf{g}_s := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$, $\mathbf{g}_{s,t} := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$, $\delta_{s,t} := \mathbf{g}_{s,t} - \mathbf{g}_s$, and $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. Using the symmetric (principal) square root $\bar{\mathbf{C}}_t^{1/2}$, define the $d \times (t-1)$ whitened column matrices*

$$\mathsf{G}_t := \Big[\bar{\mathbf{C}}_t^{-1/2}\mathbf{g}_s/\sigma\Big]_{s<t}, \qquad \mathsf{D}_t := \Big[\bar{\mathbf{C}}_t^{-1/2}\delta_{s,t}/\sigma\Big]_{s<t}, \qquad \chi_t := \|\mathsf{D}_t\|_{\text{op}}. \tag{40}$$

Then:

(a) $\mathsf{G}_t\mathsf{G}_t^\top = \mathbf{I} - \lambda\,\bar{\mathbf{C}}_t^{-1} \preceq \mathbf{I}$*, hence* $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$*.*

(b) $\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2} = \mathsf{G}_t\mathsf{D}_t^\top + \mathsf{D}_t\mathsf{G}_t^\top + \mathsf{D}_t\mathsf{D}_t^\top$*, so* $\big\|\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}\big\|_{\mathrm{op}} \le 2\chi_t + \chi_t^2$*.*

(c) ***(One-sided, no smallness condition.)*** *For every* $\chi_t \ge 0$*,*

$$\mathbf{C}_t \;\preceq\; (1+\chi_t)^2\,\bar{\mathbf{C}}_t, \qquad \text{hence} \qquad \bar{s}_t^2(a) \;\le\; (1+\chi_t)^2\,s_t^2(a) \;\; \forall a. \tag{41}$$

A predictable certificate $\bar{\chi}_t \ge \chi_t$ *therefore satisfies Assumption 5 in the exact case* $\mathcal{C}_t = \mathbf{C}_t$ *with the uniform factor* $u_t = (1+\bar{\chi}_t)^2$*.*

(d) ***(Whitened*** ℓ_2 ***primitive bound.)***

$$\chi_t \;\le\; \|\mathsf{D}_t\|_F = \left[\sigma^{-2}\sum_{s<t}\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}}\right]^{1/2} \;\le\; \frac{L_g}{\sigma\sqrt{\lambda}}\left[\sum_{s<t}\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2\right]^{1/2}, \tag{42}$$

the last step under the L_g*-Lipschitz-Jacobian assumption* $\|\delta_{s,t}\| \le L_g\Delta_{s,t}$*.*

Proof. (a) $\mathsf{G}_t\mathsf{G}_t^\top = \sigma^{-2}\sum_{s<t}\bar{\mathbf{C}}_t^{-1/2}\mathbf{g}_s\mathbf{g}_s^\top\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}\big(\sigma^{-2}\sum_{s<t}\mathbf{g}_s\mathbf{g}_s^\top\big)\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}(\bar{\mathbf{C}}_t - \lambda\mathbf{I})\bar{\mathbf{C}}_t^{-1/2} = \mathbf{I} - \lambda\bar{\mathbf{C}}_t^{-1}$. Since $\lambda\bar{\mathbf{C}}_t^{-1} \succeq 0$, $\mathsf{G}_t\mathsf{G}_t^\top \preceq \mathbf{I}$, and $\|\mathsf{G}_t\|_{\mathrm{op}}^2 = \|\mathsf{G}_t\mathsf{G}_t^\top\|_{\mathrm{op}} \le 1$.

(b) From $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, $\mathbf{g}_{s,t}\mathbf{g}_{s,t}^\top - \mathbf{g}_s\mathbf{g}_s^\top = \mathbf{g}_s\delta_{s,t}^\top + \delta_{s,t}\mathbf{g}_s^\top + \delta_{s,t}\delta_{s,t}^\top$. Conjugating each summand by $\bar{\mathbf{C}}_t^{-1/2}$ and weighting by σ^{-2}, the three groups sum exactly to $\mathsf{G}_t\mathsf{D}_t^\top$, $\mathsf{D}_t\mathsf{G}_t^\top$, and $\mathsf{D}_t\mathsf{D}_t^\top$. By the triangle inequality and $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$, $\|\mathsf{G}_t\mathsf{D}_t^\top + \mathsf{D}_t\mathsf{G}_t^\top + \mathsf{D}_t\mathsf{D}_t^\top\|_{\mathrm{op}} \le 2\|\mathsf{G}_t\|_{\mathrm{op}}\|\mathsf{D}_t\|_{\mathrm{op}} + \|\mathsf{D}_t\|_{\mathrm{op}}^2 \le 2\chi_t + \chi_t^2$.

(c) The matrix $W_t := \bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}$ is symmetric with $\|W_t\|_{\mathrm{op}} \le 2\chi_t + \chi_t^2$, hence $W_t \preceq (2\chi_t + \chi_t^2)\mathbf{I}$. Congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t - \bar{\mathbf{C}}_t \preceq (2\chi_t + \chi_t^2)\bar{\mathbf{C}}_t$, i.e. $\mathbf{C}_t \preceq (1+\chi_t)^2\bar{\mathbf{C}}_t$; no bound on χ_t is used. Both $\mathbf{C}_t, \bar{\mathbf{C}}_t \succ 0$, so inversion reverses the order: $\mathbf{C}_t^{-1} \succeq (1+\chi_t)^{-2}\bar{\mathbf{C}}_t^{-1}$. The quadratic form at $\mathbf{g}_t(a)$ gives $s_t^2(a) \ge (1+\chi_t)^{-2}\bar{s}_t^2(a)$, i.e. $\bar{s}_t^2(a) \le (1+\chi_t)^2 s_t^2(a)$. Since $x \mapsto (1+x)^2$ is increasing on $[0,\infty)$, replacing χ_t by any $\bar{\chi}_t \ge \chi_t$ preserves the inequality, giving $u_t = (1+\bar{\chi}_t)^2$.

(d) $\chi_t = \|\mathsf{D}_t\|_{\mathrm{op}} \le \|\mathsf{D}_t\|_F$, and $\|\mathsf{D}_t\|_F^2 = \sum_{s<t}\|\bar{\mathbf{C}}_t^{-1/2}\delta_{s,t}/\sigma\|^2 = \sigma^{-2}\sum_{s<t}\delta_{s,t}^\top\bar{\mathbf{C}}_t^{-1}\delta_{s,t} = \sigma^{-2}\sum_{s<t}\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}}$. Since $\bar{\mathbf{C}}_t^{-1} \preceq \lambda^{-1}\mathbf{I}$, $\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}} \le \|\delta_{s,t}\|^2/\lambda \le L_g^2\Delta_{s,t}^2/\lambda$; summing and taking square roots gives (42). ☐

Corollary 12 (Two-sided feature sandwich)**.** *If additionally* $\varepsilon_t := 2\chi_t + \chi_t^2 < 1$*, then*

$$(1-\varepsilon_t)\,\bar{\mathbf{C}}_t \;\preceq\; \mathbf{C}_t \;\preceq\; (1+\varepsilon_t)\,\bar{\mathbf{C}}_t. \tag{43}$$

Proof. From part (b), $W_t \succeq -(2\chi_t + \chi_t^2)\mathbf{I} = -\varepsilon_t\mathbf{I}$; congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t \succeq (1-\varepsilon_t)\bar{\mathbf{C}}_t$, and the upper bound is (41) with $(1+\chi_t)^2 \le 1+\varepsilon_t$ when $\varepsilon_t < 1$ (indeed $(1+\chi_t)^2 = 1+\varepsilon_t$ exactly). ☐

The main theorem uses only the one-sided bound (41), which holds for *any* $\chi_t \ge 0$; the smallness condition $\varepsilon_t < 1$ is needed only for the two-sided sandwich (43), which in turn is used by the explicit near-linear corollary and by any optional analysis that explicitly assumes a two-sided sandwich. An unrescaled current-parameter window is handled by Proposition 23 through a one-sided relation and does not require this condition. The binding quantity is the *whitened* ℓ_2 *accumulation* $\chi_t \le \|\mathsf{D}_t\|_F$ of replayed-Jacobian drift, measured in the $\bar{\mathbf{C}}_t^{-1}$ metric—a relative, whitened ℓ_2 bound that can be sharper than an ℓ_1 sum normalized only by λ when the drift occurs in well-observed directions (it credits directions the design has already explored), though it does not dominate uniformly. Together, Lemmas 6 and 11 identify exact whitened diagnostic targets. The operational upper bounds in Lemma 7 replace those targets by the predictable path statistics Q_t and R_t, the certified optimizer residual ζ_t, and analytic smoothness constants. Thus $\bar{\psi}_t$ and $u_t = (1+\bar{\chi}_t)^2$ require only $O(d)$ additional state; the exact whitened quantities remain post-hoc diagnostics used to measure the conservatism of these certificates.

Positive definiteness of both $\mathbf{C}_t$ and $\bar{\mathbf{C}}_t$ follows from $\lambda > 0$ and the outer-product form; it is not an independent assumption. All theoretical results hold under these explicit assumptions; we do *not* claim a regret guarantee for the full nonlinear model outside the local-linearization regime.

A.2 Novelty Relative to Prior Neural Bandits

Prior neural bandit algorithms use several gradient and last-layer feature conventions for uncertainty. NeuralUCB and NeuralTS use *full* neural-gradient Gram matrices in their stated algorithms, accumulating gradient features evaluated at the evolving trained parameters available when samples are processed (Zhou et al., 2020; Zhang et al., 2021); their experiments use diagonal approximations. Thus these algorithms are neither fixed-initialization Grams nor current-round replay relinearizations of every historical sample. NeuralLinear and last-layer Laplace instead restrict uncertainty to a learned representation or final layer (Riquelme et al., 2018; Kristiadi et al., 2020). We do *not* claim prior work uniformly replaces the full metric. Against this backdrop CC-UCB's contribution is narrower and specific: (i) it recomputes replayed historical Jacobians at the current parameter to define the *current-parameter relinearized* GGN/Fisher reference, rather than using historical-time, fixed-initialization, or last-layer features; (ii) it uses matrix-free CG/CVP solves for online per-action widths, forming and storing no $d \times d$ matrix and no per-sample gradient features; (iii) it separates the predictable frozen-feature confidence metric $\bar{\mathbf{C}}_t$ from the algorithmic curvature $\mathcal{C}_t$; and (iv) it prices, through a conditional transfer analysis, the costs of CG truncation, centering, feature drift, and a changing (nonmonotone) operator.

The main algorithmic contribution is adapting matrix-free full-network predictive-variance solves (Maddox et al., 2021) to current-relinearized online exploration with separate residual-checked per-action solves; we do not claim the underlying CG/CVP primitive as new. The main theoretical contribution is twofold. First, the regret bound is a *realized-width-complexity reduction*: it controls regret by the realized complexity $\Lambda_T^{\mathcal{C}} = \sum_t \log(1+\sigma^{-2}s^2_{\mathcal{C},t}(a_t))$ of the widths actually played, whose width-sum inequality follows directly from the definition and the bounded-width cap. Second, a *pathwise decomposition for that complexity* (Lemma 14) writes $\Lambda_T^{\mathcal{C}}$ as an endpoint log-determinant minus a determinant-variation sum, generalizing the standard monotone-Gram/elliptic-potential argument to a nonmonotone predictable SPD sequence. It does *not* by itself show $\Lambda_T^{\mathcal{C}}$ is small: the resulting complexity remains an assumed or observed trajectory quantity, and bounding its endpoint and variation terms under primitive assumptions is left open. These are paired with a one-sided optimism-transfer reduction: rather than a two-sided spectral sandwich, a single whitened, relative feature-drift factor (Lemma 11) transfers optimism, and the CG-accuracy and transfer costs enter through a sum of squares. Efficient estimators for nonlinear contextual bandits were proposed by Kassraie and Krause (2022) using an NTK-based uncertainty route; we compare approaches in Appendix B.

A.3 Computable Curvature Operator

Applying $\mathbf{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s<t}\mathbf{g}_t(\mathbf{x}_s,a_s)\big[\mathbf{g}_t(\mathbf{x}_s,a_s)^\top\mathbf{v}\big]$ *naively* requires one curvature–vector product (CVP) per historical sample per CG step—cost $O(It)$ CVPs per round. Each CVP evaluates one Jacobian–vector product (forward mode) followed by one vector–Jacobian product (reverse mode) through the network, at roughly the cost of one gradient evaluation (one JVP plus one VJP)[1]; we do not store per-sample gradient features $\mathbf{g}_s$, only a constant number of $O(d)$ CG vectors for CG, with CVPs computed on the fly. We describe three strategies that limit the per-step cost to $|\mathcal{S}_t|$ CVPs:

(i) Subsampled GGN (worst-case spectral baseline). At each round draw a uniform subsample $\mathcal{S}_t \subset \{1,\dots,t-1\}$ of size n (without replacement) and set $w_s = (t-1)/n$ to obtain the rescaled operator. The following lemma is a worst-case uniform spectral guarantee.

Lemma 13 (Worst-case uniform spectral guarantee for subsampled GGN)**.** *Fix round t and set $m = t-1$. Let $Y_s = \sigma^{-2}\mathbf{g}_t(\mathbf{x}_s,a_s)\mathbf{g}_t(\mathbf{x}_s,a_s)^\top$ for $s = 1,\dots,m$, so that $\mathbf{C}_t = \lambda\mathbf{I} + V$ with $V = \sum_{s=1}^m Y_s$. Each Y_s is PSD with $\|Y_s\|_{\mathrm{op}} \le G^2/\sigma^2 =: B$. For $\delta \in (0,1)$ and $1 \le n \le m$, draw $\mathcal{S}_t$ of size n uniformly without replacement from $\{1,\dots,m\}$ and define $\hat{V} = \frac{m}{n}\sum_{s\in\mathcal{S}_t} Y_s$ and $\hat{\mathbf{C}}_t = \lambda\mathbf{I} + \hat{V}$. Write $L := \log(2d/\delta)$.*

(i) Deviation bound. *With probability $\ge 1-\delta$,*

$$\|\hat{V} - V\|_{\mathrm{op}} \;\le\; mB\sqrt{\frac{2L}{n}} \;+\; \frac{4\,mB\,L}{3\,n}. \tag{44}$$

[1] Implementations based on double-backward Hessian–vector products cost closer to two gradient-equivalents per CVP; the constant does not affect the $O(\cdot)$ accounting.

(ii) Sufficient sample size. *For any* $\varepsilon \in (0,1)$*, if*

$$n \;\geq\; 8L\left[\frac{m^2\,B^2}{\varepsilon^2\,\lambda^2} \;+\; \frac{m\,B}{3\,\varepsilon\,\lambda}\right], \tag{45}$$

then $\Pr[\|\hat{V} - V\|_{\mathrm{op}} > \varepsilon\lambda] \leq \delta$ *and consequently* $(1{-}\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1{+}\varepsilon)\mathbf{C}_t$.

Proof. The $\{Y_s\}_{s=1}^m$ are fixed PSD self-adjoint matrices (deterministic conditional on the history $\mathcal{F}_{t-1}$ and the current parameter $\boldsymbol{\theta}_t$). The displayed proof is for *with-replacement* (i.i.d.) sampling. The without-replacement version follows from the domination argument of Gross and Nesme (2010), building on Hoeffding's reduction (Hoeffding, 1963): the matrix moment-generating function of a sum sampled without replacement from a fixed finite collection of self-adjoint matrices is dominated by that of the corresponding i.i.d. sum, so every tail bound derived from the matrix Laplace-transform method, including the matrix Bernstein inequality of Tropp (2015, Thm. 6.1.1), carries over with the same constants. In our setting the $\{Y_s\}$ are *fixed* PSD matrices (deterministic conditional on $\mathcal{F}_{t-1}$ and $\boldsymbol{\theta}_t$) and only the index set $\mathcal{S}_t$ is random, so the domination argument applies directly. We therefore state and prove the bound for i.i.d. draws $s_1, \ldots, s_n$ uniform on $\{1, \ldots, m\}$; the without-replacement guarantee is at least as strong. Define the centered random matrices $Z_k = (m/n)Y_{s_k} - V/n$ (mean zero, independent). The range satisfies $\|Z_k\|_{\mathrm{op}} \leq (m/n)B + (1/n)\|V\| \leq 2mB/n =: R$ (using $\|V\| \leq mB$). For the variance parameter, since $Y_s \preceq B\mathbf{I}$,

$$\mathbb{E}[Z_k Z_k^\top] \;\preceq\; \tfrac{m^2}{n^2}\mathbb{E}[Y_{s_1}^2] \;=\; \tfrac{m}{n^2}\sum_{s=1}^{m} Y_s^2 \;\preceq\; \tfrac{m}{n^2}B\,V \;\preceq\; \tfrac{m^2B^2}{n^2}\mathbf{I},$$

so $v := \big\|\sum_{k=1}^n \mathbb{E}[Z_k Z_k^\top]\big\| \leq m^2B^2/n$. By the matrix Bernstein inequality (Tropp, 2015, Thm. 6.1.1), $\Pr[\|\hat{V}{-}V\| \geq t] \leq 2d\exp\!\big({-t^2}/(2v+2Rt/3)\big)$. Choosing $t_0 = \sqrt{2vL} + \frac{2}{3}RL$ makes the matrix-Bernstein tail probability at most δ. Substituting $v \leq m^2B^2/n$ and $R = 2mB/n$ gives $t_0 \leq mB\sqrt{2L/n} + \frac{4mBL}{3n}$, yielding (44). For (45), require each summand in (44) to be $\leq \varepsilon\lambda/2$: $mB\sqrt{2L/n} \leq \varepsilon\lambda/2$ gives $n \geq 8m^2B^2L/(\varepsilon^2\lambda^2)$, and $4mBL/(3n) \leq \varepsilon\lambda/2$ gives $n \geq 8mBL/(3\varepsilon\lambda)$. Summing the two conditions yields (45); this is a conservative sufficient condition (using $a+b$ rather than $\max\{a,b\}$), not a sharp threshold. The multiplicative sandwich follows: $\|\hat{V}{-}V\| \leq \varepsilon\lambda$ gives $\mathbf{C}_t{-}\varepsilon\lambda\mathbf{I} \preceq \hat{\mathbf{C}}_t \preceq \mathbf{C}_t{+}\varepsilon\lambda\mathbf{I}$. Since $\mathbf{C}_t \succeq \lambda\mathbf{I}$, we have $\varepsilon\lambda\mathbf{I} \preceq \varepsilon\mathbf{C}_t$, so $\mathbf{C}_t{-}\varepsilon\lambda\mathbf{I} \succeq (1{-}\varepsilon)\mathbf{C}_t$ and $\mathbf{C}_t{+}\varepsilon\lambda\mathbf{I} \preceq (1{+}\varepsilon)\mathbf{C}_t$. □

With $B = G^2/\sigma^2$, the right-hand side of the sufficient condition (45) is $O\big(L \cdot [m^2G^4/(\varepsilon^2\lambda^2\sigma^4) + mG^2/(\varepsilon\lambda\sigma^2)]\big)$. The first (variance) term dominates unless $\varepsilon\lambda\sigma^2 \gtrsim mG^2$, so this worst-case sufficient threshold has a *quadratic leading dependence* on $m = t{-}1$ (it is a sufficient upper bound on the required sample size, not an exact or necessary sample complexity). The condition (45) is conservative and sufficient, not necessary, with worst-case leading dependence quadratic in m. If its threshold exceeds the population size $m = t{-}1$, it provides no nontrivial subsampling guarantee for $n < m$. Taking $n = m$ recovers the full population exactly for the separate reason that every summand is included.

This is a worst-case baseline guarantee, too conservative for practical use at large t. In particular, the bound above does not by itself justify computational savings at large t; it should be read as a worst-case spectral guarantee rather than as a practical scaling result. Subsampling reduces computation ($n \ll m$) only when m is moderate or $\varepsilon\lambda$ is large relative to mB; for large t the full history may be needed under the uniform Bernstein bound. Under additional spectral structure, leverage-score sampling or Frequent Directions sketches may yield useful approximations with smaller state. For linear bandits, Kuzborskij et al. (2019) give regret bounds in terms of the Frequent Directions spectral-tail error. Translating such structure into a uniform GGN sandwich and a concrete buffer or sketch size is left open. The Bernstein guarantee is not the mechanism by which we expect practical savings at scale. Practical savings require small buffers treated heuristically, spectral decay/effective dimension, ridge-leverage sampling, Frequent Directions, or empirical verification of the spectral sandwich. We leave structure-aware sampling and empirical spectral certification as future work. To obtain a spectral sandwich that holds simultaneously for all $t \leq T$, union-bound over t (replacing δ by δ/T). Theorem 24 then exposes the one-sided transfer factor and realized width complexity as operator-dependent terms along the realized path; it does not identify a standalone causal regret inflation.

(ii) Windowed curvature (unrescaled, current parameter). Maintain a sliding window $\mathcal{S}_t \subseteq \{1, \ldots, t{-}1\}$ of the most recent n interactions with unit weights ($w_s{=}1$), *re-evaluating the Jacobians at the current parameter* $\boldsymbol{\theta}_t$. This is a nonnegative-weighted *subset* of the same current-parameter outer products, so $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (Proposition 23);

Theorem 24 then applies with the *deterministic* approximate-to-full factor $\kappa_{+,t}$=1 and the full-to-frozen factor $u_t = (1 + \bar{\chi}_t)^2$—no two-sided sandwich is required, though the overall guarantee remains conditional on that transfer certificate. The cost of the smaller operator is not hidden: it surfaces as larger realized widths $s^2_{\mathcal{C},t}(a_t)$ and hence a larger dynamic width complexity $\Lambda^{\mathcal{C}}_T$. The same holds for any current-parameter downweighted subset with $w_s \in [0, 1]$ (including an unrescaled subsample). A *stale* window that reuses Jacobians from an older parameter is *not* covered; see (iii).

(iii) Periodic refresh (heuristic; stale operator). Recompute the curvature operator every M rounds on a cached buffer and reuse it for M consecutive actions. Because the reused operator evaluates Jacobians at an *older* parameter, it is no longer a subset of the current-parameter outer products, so the $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ argument of Proposition 23 does not apply and the stale operator can both over- and under-estimate $\mathbf{C}_t$. This remains a heuristic: Theorem 24 covers it only when a one-sided factor $\kappa_{+,t}$ (an explicit bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$) is separately verified, which we evaluate empirically. The rescaled subsample of (i), with weights $w_s = (t-1)/n > 1$, is likewise not covered by Proposition 23; it instead uses the two-sided matrix Bernstein bound of Lemma 13.

A.4 Computational Cost Model

Primitive assumptions. CC-UCB requires a single computational primitive: the *curvature–vector product* (CVP) $\mathbf{v} \mapsto \mathbf{C}_t\mathbf{v}$, which decomposes into per-sample operations $\mathbf{v} \mapsto \mathbf{g}_t(\mathbf{x}_s, a_s)\big[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \mathbf{v}\big]$. Each such operation consists of:

1. a Jacobian–vector product (JVP) $\mathbf{v} \mapsto \mathbf{g}_t^\top \mathbf{v}$ (forward-mode AD, cost $\approx 1\times$ forward pass);
2. a vector–Jacobian product (VJP) $\alpha \mapsto \alpha\,\mathbf{g}_t$ (reverse-mode AD, cost $\approx 1\times$ backward pass).

Sharded autodiff can distribute JVP/VJP evaluation, but its communication pattern and sample parallelism are implementation-dependent and are not benchmarked here. A full round of CC-UCB with action screening costs $K \times I$ CG iterations, each involving $|\mathcal{S}_t|$ forward–backward passes. Additional curvature-state memory is $O(d)$ for a constant number of CG vectors ($\tilde{\mathbf{u}}$, residual, search direction); no $d \times d$ matrix is formed or stored.

Krylov methods for Hessian spectral analysis have been run on networks with $> 10^8$ parameters under distributed training (Granziol and Juarev, 2026), which indicates that curvature–vector primitives of this general form are implementable at scale. We do *not* claim foundation-model feasibility from this: that work times Hessian–vector products, not the exact per-example GGN JVP/VJP used here, and we report no wall-clock benchmark of our primitive. Warm-starting CG from the previous round's solution can reduce the iteration count in practice.

Cost accounting. Table 3 summarizes the per-round FLOP budget, where one "gradient-equivalent" (GE) denotes the cost of a single forward-plus-backward pass through the network.

Table 3: Per-round cost of CC-UCB in gradient-equivalent (GE) units. K: actions scored; I: CG iterations; n=$|\mathcal{S}_t|$: buffer size.

Component	Cost (GE)	Notes
Mean predictions	K	fwd. only
Gradient features	K	one bwd. each
CG solves	KIn	JVP+VJP per CVP
$\boldsymbol{\theta}$-update	$O(n)$	warm-started opt.
Total	$O(K(1{+}In){+}n)$	CG-dominated

The following is an operation *count*, not a wall-clock benchmark: this paper reports no timing measurement of the per-example GGN JVP/VJP primitive. For a concrete configuration (GPT-2 124M, K=10 screened actions, I=25 CG steps, n=256 buffer) the CG cost is $10 \times 25 \times 256 = 64{,}000$ CVPs, each $\approx$ one gradient-equivalent. The total KIn is a *sample-equivalent FLOP count*, not a wall-clock figure: batching over the buffer and parallelism across actions and shards change the realized time substantially, and we do not measure it here. Reducing K via action screening (§3) is the main practical lever.

Cost of the analyzed configuration. The full-history instantiation $\mathcal{C}_t = \mathbf{C}_t$, $\mathcal{S}_t = \{1, \ldots, t-1\}$ with adaptive CG stopping has the certificate $\bar{\kappa}_t = 1 + (t-1)G^2/(\lambda\sigma^2)$, so maintaining $\max_a \varepsilon_{t,a} \le \bar{\varepsilon}_t$ is certified by

$I_t = O\big(\sqrt{t}\, G/(\sigma\sqrt{\lambda})\, \log(1/\bar{\varepsilon}_t)\big)$ iterations per solve (Eq. (75)), so the per-round CG cost is $K\, I_t\, (t-1) = O(K\, t^{3/2})$ sample-CVPs, i.e. $O(K\, T^{5/2})$ cumulatively—the worst-case cost of the exact analyzed method under the stated CG certificate. Table 3 and the numerical example describe fixed-buffer operators ($n \ll t$), which are computationally practical but are covered by Theorem 1 only under the corrected window/subsample conditions of §A.3: an unrescaled current-parameter buffer has the deterministic approximate-to-full factor $\kappa_{+,t}$=1 (Proposition 23)—the overall guarantee remaining conditional on the full-to-frozen transfer and the other certificates of Theorem 1—while a rescaled or stale buffer additionally requires the verified transfer factor of Theorem 24.

For the growing window of Theorem 3, one solve costs $O(m_t^{3/2} \log(1/\bar{\varepsilon}))$ sample-CVPs under the analytic condition-number bound, so the following exponents are worst-case certified counts (logarithmic factors suppressed):

q	regret exponent	cumulative sample-CVP exponent
$\frac{1}{2}$	$T^{\frac{3}{4}}$	$KT^{\frac{7}{4}}$
$\frac{2}{3}$	$T^{\frac{2}{3}}$	KT^{2}
1	$T^{\frac{1}{2}}$	$KT^{\frac{5}{2}}$

These regret exponents additionally require polylogarithmic predictable certificate factors and $E_T = o(T)$; the operation counts are not wall-clock measurements.

A.5 Dynamic Log-Determinant Potential

The regret bound is driven not by γ_T but by the realized *width complexity* of the widths the algorithm actually uses,

$$\Lambda_T^{\mathcal{C}} \;:=\; \sum_{t=1}^{T} \log\!\big(1 + \sigma^{-2}\, s_{\mathcal{C},t}^2(a_t)\big). \tag{46}$$

Unlike $\bar{\mathbf{C}}_t$, the algorithmic curvature $\mathcal{C}_t$ need *not* be monotone in t: relinearization, subsampling, windowing, and periodic refresh can all *shrink* it, so the elementary determinant telescoping of (25) no longer applies. The following lemma prices this nonmonotonicity exactly, through a pathwise log-determinant identity. This *pathwise determinant identity* (a dynamic potential decomposition) is what lets the analysis handle a general predictable SPD curvature sequence rather than a monotone Gram matrix.

Lemma 14 (Dynamic log-determinant potential)**.** *Let $\{\mathcal{C}_t\}_{t\geq 1}$ satisfy* (37) *with $\mathcal{C}_1 = \lambda\mathbf{I}$. For the played feature $\mathbf{h}_t := \mathbf{g}_t(\mathbf{x}_t, a_t)$ define the rank-one update $\mathcal{C}_t^+ := \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$ and, using the symmetric (principal) root $(\mathcal{C}_t^+)^{-1/2}$, the normalized perturbation*

$$\Xi_t \;:=\; (\mathcal{C}_t^+)^{-1/2}\, (\mathcal{C}_{t+1} - \mathcal{C}_t^+)\, (\mathcal{C}_t^+)^{-1/2}, \tag{47}$$

so that $\mathbf{I} + \Xi_t \;=\; (\mathcal{C}_t^+)^{-1/2}\mathcal{C}_{t+1}(\mathcal{C}_t^+)^{-1/2}$ is SPD; equivalently, every eigenvalue of Ξ_t exceeds -1 and every eigenvalue of $\mathbf{I} + \Xi_t$ is positive (with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$). Fix any end-of-round-$T$ *terminal operator $\mathcal{C}_{T+1}$ constructed only from data and randomness available through round T ($\mathcal{F}_T$-measurable, hence not a function of $\mathbf{x}_{T+1}$, $\mathcal{A}_{T+1}$, or future rewards; e.g. the canonical rank-one update $\mathcal{C}_{T+1} = \mathcal{C}_T^+$, or the operator the algorithm commits for round T+1). The identity below is then an a posteriori statement about the completed sequence $\mathcal{C}_1, \ldots, \mathcal{C}_{T+1}$; the quantities Ξ_T, $V_T^{\mathcal{C}}$, and Γ_T^{dyn} are defined relative to this choice, and the bound $\Lambda_T^{\mathcal{C}} \leq \Gamma_T^{\mathrm{dyn}}$ holds for* every *such valid choice. Then*

$$\Lambda_T^{\mathcal{C}} \;=\; \log \frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} \;-\; \sum_{t=1}^{T} \log\det(\mathbf{I} + \Xi_t). \tag{48}$$

With the determinant-destroying variation charge and the dynamic potential

$$V_T^{\mathcal{C}} := \sum_{t=1}^{T} \big[-\log\det(\mathbf{I} + \Xi_t)\big]_+, \qquad \Gamma_T^{\mathrm{dyn}} := \log \frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} + V_T^{\mathcal{C}}, \tag{49}$$

we have $\Lambda_T^{\mathcal{C}} \leq \Gamma_T^{\mathrm{dyn}}$, and, using $s_{\mathcal{C},t}^2(a_t) \leq G^2/\lambda$ and $\log(1+x) \geq x/(1+x)$,

$$\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \;\leq\; (\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}} \;\leq\; (\sigma^2 + G^2/\lambda)\, \Gamma_T^{\mathrm{dyn}}. \tag{50}$$

Proof. By the matrix determinant lemma applied to $\mathcal{C}_t^+ = \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$,

$$\frac{\det(\mathcal{C}_t^+)}{\det(\mathcal{C}_t)} = 1 + \sigma^{-2}\mathbf{h}_t^\top \mathcal{C}_t^{-1}\mathbf{h}_t = 1 + \sigma^{-2}s_{\mathcal{C},t}^2(a_t). \tag{51}$$

Writing $B_t := (\mathcal{C}_t^+)^{-1/2}$, we have $\mathbf{I} + \Xi_t = B_t\mathcal{C}_t^+ B_t + B_t(\mathcal{C}_{t+1} - \mathcal{C}_t^+)B_t = B_t\mathcal{C}_{t+1}B_t$, a congruence of the SPD matrix $\mathcal{C}_{t+1}$ by the invertible symmetric B_t, hence SPD (in particular its eigenvalues exceed -1 with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$), and

$$\frac{\det(\mathcal{C}_{t+1})}{\det(\mathcal{C}_t^+)} = \det\bigl(B_t^{-1}(\mathbf{I} + \Xi_t)B_t^{-1}\bigr)\big/\det(\mathcal{C}_t^+) = \det(\mathbf{I} + \Xi_t), \tag{52}$$

using $\det(\mathcal{C}_{t+1}) = \det(\mathcal{C}_t^+)\det(\mathbf{I} + \Xi_t)$. Taking logs of (51)–(52) and adding,

$$\log\bigl(1 + \sigma^{-2}s_{\mathcal{C},t}^2(a_t)\bigr) = \log\det(\mathcal{C}_{t+1}) - \log\det(\mathcal{C}_t) - \log\det(\mathbf{I} + \Xi_t).$$

Summing over $t = 1, \dots, T$ telescopes the first difference to $\log\det(\mathcal{C}_{T+1}) - \log\det(\mathcal{C}_1)$; with $\mathcal{C}_1 = \lambda\mathbf{I}$ this is $\log[\det(\mathcal{C}_{T+1})/\det(\lambda\mathbf{I})]$, proving (48). Since $-\log\det(\mathbf{I}+\Xi_t) \le [-\log\det(\mathbf{I}+\Xi_t)]_+$ term by term, $-\sum_t \log\det(\mathbf{I}+\Xi_t) \le V_T^{\mathcal{C}}$, so $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. For (50): $\mathcal{C}_t \succeq \lambda\mathbf{I}$ gives $s_{\mathcal{C},t}^2(a_t) = \mathbf{h}_t^\top\mathcal{C}_t^{-1}\mathbf{h}_t \le \|\mathbf{h}_t\|^2/\lambda \le G^2/\lambda =: c_{\max}\sigma^2$. With $x_t := \sigma^{-2}s_{\mathcal{C},t}^2(a_t) \in [0, c_{\max}]$ and $\log(1+x) \ge x/(1+x) \ge x/(1+c_{\max})$, $\Lambda_T^{\mathcal{C}} \ge (1+c_{\max})^{-1}\sum_t x_t$, i.e. $\sum_t s_{\mathcal{C},t}^2(a_t) \le (\sigma^2 + G^2/\lambda)\Lambda_T^{\mathcal{C}}$; the second inequality is $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. ☐

Remark (Interpretation and scope of the dynamic potential)**.** *Identity* (48) *is* pathwise*: it uses no martingale or probabilistic argument.* $\Lambda_T^{\mathcal{C}}$ *is the exact accumulated width complexity of the actions actually played.* $V_T^{\mathcal{C}}$ *charges only* determinant-destroying *changes in the nonmonotone curvature sequence—rounds where* $\det(\mathbf{I} + \Xi_t) < 1$*, i.e. where* $\mathcal{C}_{t+1}$ *loses volume relative to the rank-one-updated* $\mathcal{C}_t^+$*; changes that* increase *determinant volume contribute zero. If* $\mathcal{C}_1 \ne \lambda\mathbf{I}$ *but still* $\mathcal{C}_1 \succeq \lambda\mathbf{I}$*, the denominator in* (48)*–*(49) *is* $\det(\mathcal{C}_1)$ *rather than* $\det(\lambda\mathbf{I})$*. The charge* $V_T^{\mathcal{C}}$ *can be large for unstable curvature sequences: we do* not *claim the identity itself shows the dynamic complexity is small, nor that the dynamic theorem automatically yields sublinear regret. Note that* $V_T^{\mathcal{C}}$ *alone does not control* $\Lambda_T^{\mathcal{C}}$ *or* Γ_T^{dyn}*: by* (49) *the potential* $\Gamma_T^{\mathrm{dyn}} = \log\det(\mathcal{C}_{T+1}/\lambda\mathbf{I}) + V_T^{\mathcal{C}}$ *also carries the endpoint log-determinant term. Bounding* both *the variation charge* $V_T^{\mathcal{C}}$ *and the endpoint* $\log\det(\mathcal{C}_{T+1}/\lambda\mathbf{I})$ *is* one sufficient route *to controlling* $\Lambda_T^{\mathcal{C}}$*, through* $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$*; it is not equivalent to it, since the exact identity* (48) *can contain cancellation that makes* $\Lambda_T^{\mathcal{C}}$ *small even when* Γ_T^{dyn} *is not. Outside the fixed-rank and persistently excited regimes of Corollary 9 and Theorem 3, controlling* $\Lambda_T^{\mathcal{C}}$ *for a genuinely adapting backbone remains open.*

Lemma 15 (Rank-sensitive refresh bound)**.** *Let* $\mathcal{R} = \{t : \mathcal{C}_{t+1} \ne \mathcal{C}_t^+\}$*. Suppose that for every* $t \in \mathcal{R}$*, the negative part* $(\Xi_t)_- = (-\Xi_t)_+$ *has rank at most* r_t *and* $\lambda_{\min}(\mathbf{I} + \Xi_t) \ge 1 - \nu_t$ *for* $0 \le \nu_t < 1$*. Then*

$$V_T^{\mathcal{C}} \le \sum_{t\in\mathcal{R}} r_t \log\frac{1}{1-\nu_t}. \tag{53}$$

If additionally $\mathcal{C}_{T+1} = \lambda\mathbf{I} + A$*, where* $A \succeq 0$*,* $\operatorname{rank}(A) \le r_{\mathrm{end}}$*, and* $\operatorname{tr}(A) \le W_{\mathrm{end}}G^2/\sigma^2$ *for a specified total endpoint weight* $W_{\mathrm{end}} \ge 0$*, then, for* $r_{\mathrm{end}} > 0$*,*

$$\Lambda_T^{\mathcal{C}} \le r_{\mathrm{end}}\log\left(1 + \frac{W_{\mathrm{end}}G^2}{r_{\mathrm{end}}\lambda\sigma^2}\right) + \sum_{t\in\mathcal{R}} r_t\log\frac{1}{1-\nu_t}. \tag{54}$$

For $r_{\mathrm{end}} = 0$*, the first term is defined as zero.*

Proof. Let $\xi_{t,i}$ be the eigenvalues of Ξ_t. At most r_t are negative, each is at least $-\nu_t$, and every nonnegative eigenvalue makes a nonpositive contribution to $-\sum_i \log(1+\xi_{t,i})$. Hence $[-\log\det(\mathbf{I}+\Xi_t)]_+ \le r_t\log(1/(1-\nu_t))$, proving (53). List the nonzero eigenvalues of A and pad with zeros to obtain $a_1, \dots, a_{r_{\mathrm{end}}}$. Concavity gives

$$\sum_i \log(1 + a_i/\lambda) \le r_{\mathrm{end}}\log\left(1 + \frac{\sum_i a_i}{r_{\mathrm{end}}\lambda}\right),$$

and the trace premise plus Lemma 14 proves (54). ☐

Consequences. Sequential Gram updates have $\Xi_t = 0$. If canonical rank-one updates are used outside J_T refresh rounds, and each refresh has $r_t \le r$, $\nu_t \le \nu < 1$, then $V_T^{\mathcal{C}} \le J_T r \log(1/(1-\nu))$; a geometric schedule is therefore polylogarithmic only when these rank and spectral-floor bounds hold. For a replacement $\mathcal{C}_{t+1} - \mathcal{C}_t^+ = A_t^{\text{new}} - A_t^{\text{old}}$ with both terms PSD and $\operatorname{rank}(A_t^{\text{old}}) \le r$, Sylvester inertia gives $r_t \le r$, but a spectral floor is still required.

For a fixed-feature window of length m, after warm-up each deletion has rank one. Writing $N_{\text{drop}} = \max\{0, T-m\}$, Sherman–Morrison gives the following exact charge for each $t > m$. Let $v_t^{\text{drop}} = \sigma^{-1}\mathbf{g}_{t-m}(\mathbf{x}_{t-m}, a_{t-m})$ and let $D_t = \mathcal{C}_t^+ - v_t^{\text{drop}}(v_t^{\text{drop}})^\top = \mathcal{C}_{t+1}$ be the remaining-window matrix. Then $-\log\det(\mathbf{I} + \Xi_t) = \log[1 + (v_t^{\text{drop}})^\top D_t^{-1} v_t^{\text{drop}}]$, and therefore

$$V_T^{\mathcal{C}} \le N_{\text{drop}} \log\left(1 + \frac{G^2}{\lambda\sigma^2}\right).$$

This can be linear in T. A relinearized window need not be a rank-one replacement. Likewise, merely holding a stale operator fixed does not make non-refresh transitions canonical rank-one updates, so geometric refresh timing alone does not imply a logarithmic charge.

Corollary 16 (Observable CG upper bound on the width complexity)**.** *If the played-action CG solve satisfies $\varepsilon_{t,a_t} \le \bar{\varepsilon}_t < 1$ so that $\tilde{s}_t^2(a_t) \ge (1-\bar{\varepsilon}_t)\, s_{\mathcal{C},t}^2(a_t)$ (Lemma 17), then*

$$\Lambda_T^{\mathcal{C}} \;\le\; \widehat{\Lambda^{\mathcal{C}}}_T \;:=\; \sum_{t=1}^{T} \log\!\Big(1 + \frac{\tilde{s}_t^2(a_t)}{\sigma^2\,(1-\bar{\varepsilon}_t)}\Big). \tag{55}$$

Proof. $s_{\mathcal{C},t}^2(a_t) \le \tilde{s}_t^2(a_t)/(1-\bar{\varepsilon}_t)$ and $x \mapsto \log(1+\sigma^{-2}x)$ is increasing, so each summand of $\Lambda_T^{\mathcal{C}}$ is dominated term by term. □

This gives an observable upper bound on the dynamic width-complexity term, conditional on the certified CG tolerances; $V_T^{\mathcal{C}}$ itself is *not* observable in general (it requires determinant estimation of the curvature sequence).

A.6 CG Approximation Lemma

Lemma 17 (CG multiplicative accuracy)**.** *Fix a round t and an action a with right-hand side $\mathbf{g} := \mathbf{g}_t(a) \ne \mathbf{0}$. Let $\mathbf{u} = \mathcal{C}_t^{-1}\mathbf{g}$ and let $\tilde{\mathbf{u}}$ be the iterate after $I_{t,a}$ steps of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$, with $\mathcal{C}_t$ held* fixed *across all iterations. The relative energy-norm error is the per-action quantity $\varepsilon_{t,a}$ of* (9)*, and*

$$\varepsilon_{t,a} \;=\; \frac{\|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} \;\le\; 2\left(\frac{\sqrt{\bar{\kappa}_t} - 1}{\sqrt{\bar{\kappa}_t} + 1}\right)^{I_{t,a}}, \tag{56}$$

where $\kappa_t := \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number and $\bar{\kappa}_t \ge \kappa_t$ is a certified $\mathcal{H}_t$-measurable upper bound. The convergence rate in (56) *is right-hand-side independent, so it holds for every action with the same $\bar{\kappa}_t$. For a* nonnegative weighted outer-product *operator $\mathcal{C}_t = \lambda\mathbf{I} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ ($w_s \ge 0$, so $\mathcal{C}_t \succeq \lambda\mathbf{I} \succ 0$ and CG is well-posed),*

$$\kappa_t \le \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \le 1 + \frac{W_t G^2}{\lambda\sigma^2} =: \bar{\kappa}_t,$$

with $W_t = \sum_{s\in\mathcal{S}_t} w_s$ and $\|\mathbf{g}_t(\mathbf{x}_s, a_s)\| \le G$ from Assumption 2; here $W_t = n$ for unrescaled windowed/refresh buffers, giving $\bar{\kappa}_t$ and α_t independent of T, and $W_t = t-1$ for full history and for rescaled uniform subsampling with $w_s = (t-1)/n$. For a general SPD operator that is not such an outer product, $\bar{\kappa}_t$ must be certified separately. Write $\tilde{s}_t^2(a) = \mathbf{g}^\top\tilde{\mathbf{u}}$ and $s_{\mathcal{C},t}^2(a) = \mathbf{g}^\top\mathcal{C}_t^{-1}\mathbf{g}$.

Assume *$\varepsilon_{t,a} < 1$ (see Appendix E for the minimum $I_{t,a}$ ensuring this as a function of $\bar{\kappa}_t$; Algorithm 1 enforces a known* per-round *tolerance $\bar{\varepsilon}_t$ uniformly over the scored actions, $\max_a \varepsilon_{t,a} \le \bar{\varepsilon}_t < 1$, via adaptive residual stopping or a fixed budget under a certified condition-number bound; see §3). Then*

$$(1-\varepsilon_{t,a})\, s_{\mathcal{C},t}^2(a) \;\le\; \tilde{s}_t^2(a) \;\le\; (1+\varepsilon_{t,a})\, s_{\mathcal{C},t}^2(a). \tag{57}$$

For $\mathbf{g}_t(a) = \mathbf{0}$ the algorithm sets $\tilde{s}_t^2(a) = 0 = s_{\mathcal{C},t}^2(a)$ and (57) *holds trivially.*

Proof. The CG minimax property gives $\|e_I\|_{\mathcal{C}_t}/\|e_0\|_{\mathcal{C}_t} \le \min_{p\in\mathcal{P}_I,\, p(0)=1} \max_{x\in[\lambda_{\min},\lambda_{\max}]} |p(x)|$. Choosing the scaled Chebyshev polynomial bounds this maximum by $2[(\sqrt{\kappa_t}-1)/(\sqrt{\kappa_t}+1)]^I$; this ratio is increasing in κ_t, so $\bar\kappa_t \ge \kappa_t$ proves (56). $\tilde s_t^2(a) - s^2_{\mathcal{C},t}(a) = \mathbf{g}^\top(\tilde{\mathbf{u}}-\mathbf{u}) = (\mathcal{C}_t\mathbf{u})^\top(\tilde{\mathbf{u}}-\mathbf{u}) = \mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}}-\mathbf{u})$. By Cauchy–Schwarz in the $\mathcal{C}_t$-norm, $|\mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}}-\mathbf{u})| \le \|\mathbf{u}\|_{\mathcal{C}_t}\|\tilde{\mathbf{u}}-\mathbf{u}\|_{\mathcal{C}_t} \le \varepsilon_{t,a}\|\mathbf{u}\|^2_{\mathcal{C}_t} = \varepsilon_{t,a}\, s^2_{\mathcal{C},t}(a)$. For the outer-product special case the condition-number bound follows from $\lambda_{\max}(\mathcal{C}_t) \le \lambda + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\|\mathbf{g}_t(\mathbf{x}_s,a_s)\|^2 \le \lambda + W_tG^2/\sigma^2$. □

The algorithmic bonus. With the *per-round* certified tolerance $\bar\varepsilon_t \ge \max_a \varepsilon_{t,a}$ and the one-sided transfer factor $u_t(a)\ge 1$ of Assumption 5, Algorithm 1 plays the bonus

$$w_t(a) \;:=\; \omega_t\sqrt{\frac{u_t(a)}{1-\bar\varepsilon_t}}\,\sqrt{\tilde s_t^2(a)}, \tag{58}$$

where $\bar\beta_t \ge \beta_t$ is the predictable confidence upper bound, $\bar\psi_t$ the centering certificate (Assumption 4), and

$$\omega_t := \bar\beta_t + \bar\psi_t \tag{59}$$

is the combined confidence-plus-centering width factor. Define the *per-round* CG inflation factor $\alpha_t := \sqrt{(1+\bar\varepsilon_t)/(1-\bar\varepsilon_t)}$.

Lemma 18 (Bonus sandwiching)**.** *Under Assumption 5 and* $\max_{a\in\mathcal{A}_t}\varepsilon_{t,a} \le \bar\varepsilon_t < 1$*, the bonus* (58) *satisfies, for every* $a\in\mathcal{A}_t$*,*

$$\omega_t\,\bar s_t(a) \;\le\; w_t(a) \;\le\; \alpha_t\,\omega_t\,\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a). \tag{60}$$

Proof. For $\mathbf{g}_t(a)=\mathbf{0}$ all three terms vanish and the bound is trivial. Otherwise Lemma 17 with $\varepsilon_{t,a}\le\bar\varepsilon_t$ gives $(1-\bar\varepsilon_t)s^2_{\mathcal{C},t}(a) \le \tilde s_t^2(a) \le (1+\bar\varepsilon_t)s^2_{\mathcal{C},t}(a)$. The lower bound and $\omega_t\ge 0$ give $\sqrt{\tilde s_t^2(a)} \ge \sqrt{1-\bar\varepsilon_t}\,s_{\mathcal{C},t}(a)$, so $w_t(a) \ge \omega_t\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a)$; then (39) gives $\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a) \ge \bar s_t(a)$, i.e. $w_t(a)\ge\omega_t\bar s_t(a)$. The upper bound gives $\sqrt{\tilde s_t^2(a)} \le \sqrt{1+\bar\varepsilon_t}\,s_{\mathcal{C},t}(a)$, whence $w_t(a) \le \omega_t\sqrt{u_t(a)/(1-\bar\varepsilon_t)}\sqrt{1+\bar\varepsilon_t}\,s_{\mathcal{C},t}(a) = \alpha_t\omega_t\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a)$. Both bounds are per-round; no horizon maximum is taken before summing. □

A.7 Confidence and Discrepancy Lemmas

Lemma 19 (Confidence for the linearized ridge estimator)**.** *Suppose Assumptions 1 to 3 hold and* $\mu^* = \mu_{\boldsymbol\theta^*}$ *with* $\|\boldsymbol\theta^*\|\le S$*. Define the linearized ridge estimator* $\hat{\boldsymbol\theta}_t^{\mathrm{lin}}$ *by* (70)*, the frozen-feature variance* $\bar s_t(a) = \sqrt{\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a)}$*, and the cumulative squared linearization error* $F_t := \sum_{s=1}^{t-1}\varepsilon_{\mathrm{lin}}(s)^2$*. Then with probability at least* $1-\delta$*, for all* $t\ge 1$ *and all* $a\in\mathcal{A}_t$ *simultaneously,*

$$\left|\mathbf{g}_t(a)^\top\big(\hat{\boldsymbol\theta}_t^{\mathrm{lin}} - \boldsymbol\theta^*\big)\right| \;\le\; \bar s_t(a)\,\beta_t, \tag{61}$$

where

$$\beta_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\,S \;+\; \frac{1}{\sigma}\sqrt{F_t}. \tag{62}$$

(Here $\gamma_{t-1} = \log\det(\bar{\mathbf{C}}_t)/\det(\lambda\mathbf{I})$ *is the information gain through round* $t-1$*.)*

The quantity β_t *in* (62) *is a theoretical radius. The algorithm may use any predictable upper bound* $\bar\beta_t\ge\beta_t$*; all results below remain valid with* $\bar\beta_t$ *in place of* β_t*. This is necessary because* β_t *depends on the unknown cumulative linearization term* F_t*.*

Proof. From (69), $y_s = \mathbf{g}_s^\top\boldsymbol\theta^* + \rho_s + \eta_s$ with $|\rho_s|\le\varepsilon_{\mathrm{lin}}(s)$. The closed form of $\hat{\boldsymbol\theta}_t^{\mathrm{lin}}$ gives

$$\hat{\boldsymbol\theta}_t^{\mathrm{lin}} - \boldsymbol\theta^* = -\lambda\bar{\mathbf{C}}_t^{-1}\boldsymbol\theta^* + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\rho_s + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\eta_s.$$

Taking the inner product with $\mathbf{g}_t(a)$ and bounding in absolute value:

(i) *Prior bias.* $|\lambda\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\boldsymbol\theta^*| \le \lambda\,\bar s_t(a)\,\|\boldsymbol\theta^*\|_{\bar{\mathbf{C}}_t^{-1}} \le \bar s_t(a)\sqrt{\lambda}\,S$. (Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm; $\lambda\|\boldsymbol\theta^*\|_{\bar{\mathbf{C}}_t^{-1}} \le \lambda\|\boldsymbol\theta^*\|/\sqrt{\lambda_{\min}(\bar{\mathbf{C}}_t)} \le \sqrt{\lambda}\,S$ since $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$.)

(ii) *Linearization bias.* Define $\Phi = [\mathbf{g}_1/\sigma, \dots, \mathbf{g}_{t-1}/\sigma]$ (the $d \times (t-1)$ design matrix) and $\hat{\rho} = (\rho_1, \dots, \rho_{t-1})^\top$. Then $\sigma^{-2} \sum \mathbf{g}_s \rho_s = \sigma^{-1} \Phi \hat{\rho}$. Since $\bar{\mathbf{C}}_t \succeq \Phi\Phi^\top$, we have $\Phi^\top \bar{\mathbf{C}}_t^{-1} \Phi \preceq \mathbf{I}$, so $\|\Phi\hat{\rho}\|_{\bar{\mathbf{C}}_t^{-1}} = \|\hat{\rho}\|_{\Phi^\top \bar{\mathbf{C}}_t^{-1} \Phi} \le \|\hat{\rho}\| = \sqrt{F_t}$. Therefore $|\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \sigma^{-1} \Phi \hat{\rho}| \le \bar{s}_t(a) \sigma^{-1} \sqrt{F_t}$.

(iii) *Noise.*

$$\left\| \sum_{s<t} \frac{\mathbf{g}_s}{\sigma} \frac{\eta_s}{\sigma} \right\|_{\bar{\mathbf{C}}_t^{-1}} \le \sqrt{\gamma_{t-1} + 2\log(1/\delta)}$$

holds for all $t \ge 1$ with probability at least $1 - \delta$ by the time-uniform vector self-normalized martingale bound (Abbasi-Yadkori et al., 2011). Here $x_s = \mathbf{g}_s/\sigma$ is predictable with respect to the pre-reward sigma-algebra $\mathcal{G}_s$, and $\xi_s = \eta_s/\sigma$ is conditionally 1-sub-Gaussian given $\mathcal{G}_s$. For any candidate action a, Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm gives

$$\left| \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \sum_{s<t} \frac{\mathbf{g}_s \eta_s}{\sigma^2} \right| \le \bar{s}_t(a) \sqrt{\gamma_{t-1} + 2\log(1/\delta)}.$$

Summing gives (61)–(62). □

Corollary 20 (Predictable confidence schedule)**.** *Suppose a nonnegative $\mathcal{H}_t$-measurable sequence $\{\bar{F}_t\}_{t\ge1}$ satisfies $F_t \le \bar{F}_t$ for all t. Then Algorithm 1 may set*

$$\bar{\beta}_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\, S \;+\; \sigma^{-1}\sqrt{\bar{F}_t},$$

which is predictable and satisfies $\bar{\beta}_t \ge \beta_t$.

Proof. $\bar{F}_t \ge F_t$ term-by-term implies each summand in (62) is dominated, so $\bar{\beta}_t \ge \beta_t$. Predictability holds because γ_{t-1}, S, $\bar{F}_t$, λ, σ, and δ are all determined before round t. □

Lemma 7 supplies the computable choice $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\mathrm{lin}})^2$; a tighter externally certified predictable sequence may replace it.

Corollary 21 (Observable information-gain surrogate via the one-sided transfer factor)**.** *Run Algorithm 1 with a predictable algorithmic curvature sequence $\mathcal{C}_s$, per-round CG tolerance with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s < 1$ for the played action, and an $\mathcal{H}_s$-measurable one-sided optimism-transfer factor $u_s(a) \ge 1$ satisfying $\bar{s}_s^2(a) \le u_s(a)\, s_{\mathcal{C},s}^2(a)$ (Assumption 5). Suppose also that the nonnegative predictable sequence $\bar{F}_t$ satisfies $\bar{F}_t \ge F_t$. Define the observable information-gain surrogate*

$$\hat{\gamma}_{t-1} \;:=\; \sum_{s=1}^{t-1} \log\!\Big(1 + \sigma^{-2} \frac{u_s(a_s)}{1 - \bar{\varepsilon}_s}\, \tilde{s}_s^2(a_s)\Big),$$

where $\tilde{s}_s^2(a_s) = \mathbf{g}_s(a_s)^\top \tilde{\mathbf{u}}_s$ is the CG variance proxy computed at round s for the played action. Then $\hat{\gamma}_{t-1}$ is $\mathcal{H}_t$-measurable and $\hat{\gamma}_{t-1} \ge \gamma_{t-1}$, so

$$\hat{\beta}_t \;:=\; \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\, S \;+\; \sigma^{-1}\sqrt{\bar{F}_t}$$

is a predictable schedule with $\hat{\beta}_t \ge \beta_t$. Played-action CG accuracy is sufficient for this information comparison. If the same tolerance also holds uniformly for every candidate action before selection, then Theorem 1 holds with $\bar{\beta}_t = \hat{\beta}_t$.

Proof. The transfer factor gives $\bar{s}_s^2(a_s) \le u_s(a_s)\, s_{\mathcal{C},s}^2(a_s)$, and Lemma 17 with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s$ gives $s_{\mathcal{C},s}^2(a_s) \le \tilde{s}_s^2(a_s)/(1 - \bar{\varepsilon}_s)$; hence $\bar{s}_s^2(a_s) \le u_s(a_s)\, \tilde{s}_s^2(a_s)/(1 - \bar{\varepsilon}_s)$. By monotonicity of $x \mapsto \log(1 + \sigma^{-2}x)$, each summand of γ_{t-1} in (25) is dominated by the corresponding summand of $\hat{\gamma}_{t-1}$. Measurability holds because $\tilde{s}_s^2(a_s)$, $u_s(a_s)$, and $\bar{\varepsilon}_s$ are all $\mathcal{H}_s$-measurable. The schedule property then follows term by term as in Corollary 20. □

This makes the information-gain component observable when $u_s(a_s)$ is certified. Together with Lemma 7, it renders the full confidence and centering schedules operational; it does not make them tight.

Remark (Computability and dimension dependence of β_t)**.** *Two caveats about β_t deserve emphasis. First, γ_{t-1} is defined through the frozen-feature Gram matrix $\bar{\mathbf{C}}_t$, whose per-sample Jacobians the matrix-free implementation deliberately does not store; γ_{t-1} is therefore not directly computable in the stated memory model. Second, the worst-case substitute $\gamma_{t-1} \le d \log\big(1 + (t-1)G^2/(d\lambda\sigma^2)\big)$ does not exploit low realized information gain and can be order t when $d \gg t$, rendering the resulting bonus and theorem bound vacuous. Corollary 21 addresses both issues with an observable surrogate built from quantities the algorithm already computes; the resulting guarantee is data-dependent, and its tightness still depends on the realized surrogate together with the other theorem factors.*

Lemma 22 (Action-scaled centering discrepancy)**.** *Under Assumption 4, $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{\psi}_t\, \bar{s}_t(a)$ for all $a \in \mathcal{A}_t$.*

Proof. Immediate from (32). □

Remark (status of the centering certificate $\bar{\psi}_t$). Lemma 7 computes a predictable $\bar{\psi}_t \ge \psi_t$ from the certified optimizer residual, Welford path moments, observed collection residuals, and analytic smoothness constants. The schedule is operational but can be loose; the corrected-center variant (Corollary 5) removes it at the cost of frozen-feature access.

In the convex last-layer regime, under exact optimization, $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\text{lin}}$ and $\bar{\psi}_t = 0$. The full prediction error at $(\mathbf{x}_t, a)$ is bounded by the triangle inequality, with the centering discrepancy *action-scaled*:

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)| \;\le\; (\beta_t + \bar{\psi}_t)\, \bar{s}_t(a) \;+\; \varepsilon_{\text{lin}}(t) \;\le\; \omega_t\, \bar{s}_t(a) + \varepsilon_{\text{lin}}(t), \tag{63}$$

using (61), Lemma 22, the per-round linearization remainder $|\rho_t| \le \varepsilon_{\text{lin}}(t)$, and $\bar{\beta}_t \ge \beta_t$.

A.8 Cost of Approximate Curvature

Proposition 23 (Unrescaled current-parameter subset)**.** *Let $\mathcal{S}_t \subseteq \{1, \ldots, t-1\}$ and $0 \le w_s \le 1$, and define the current-parameter operator*

$$\hat{\mathbf{C}}_t \;=\; \lambda\mathbf{I} + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top,$$

using the same relinearized Jacobians $\mathbf{g}_t(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$ that define $\mathbf{C}_t$ (5). Then $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$, so Theorem 24 applies with $\kappa_{+,t} = 1$ and, in the exact-transfer case, $u_t = (1 + \bar{\chi}_t)^2$ (Lemma 11).

Proof. $\mathbf{C}_t - \hat{\mathbf{C}}_t = \sigma^{-2} \sum_{s \in \mathcal{S}_t} (1 - w_s)\mathbf{g}_t\mathbf{g}_t^\top + \sigma^{-2} \sum_{s \notin \mathcal{S}_t} \mathbf{g}_t\mathbf{g}_t^\top$, a sum of PSD outer products (each $1 - w_s \ge 0$), hence $\mathbf{C}_t - \hat{\mathbf{C}}_t \succeq 0$. With $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (so $\kappa_{+,t} = 1$ in (64)) and $\hat{\mathbf{C}}_t \succeq \lambda\mathbf{I}$, the hypotheses of Theorem 24 hold. □

This covers an unrescaled current-parameter sliding window, an unrescaled current-parameter subsample, and any current-parameter downweighted subset with weights in $[0, 1]$. It does *not* cover rescaled subsampling with weights $(t-1)/n > 1$ (Lemma 13), a stale periodic-refresh operator evaluated at an older parameter, or an arbitrary randomized sketch (e.g. an eigenspace Lanczos surrogate), for which a factor $\kappa_{+,t}$ must be certified separately.

Theorem 24 (Approximate curvature via one-sided transfer)**.** *Adopt the hypotheses of Theorem 1 with the algorithmic curvature $\mathcal{C}_t = \hat{\mathbf{C}}_t$, an approximate operator that is symmetric, $\mathcal{H}_t$-measurable, and satisfies $\hat{\mathbf{C}}_t \succeq \lambda\mathbf{I}$. Suppose the two* one-sided *bounds*

$$\hat{\mathbf{C}}_t \;\preceq\; \kappa_{+,t}\, \mathbf{C}_t, \qquad \mathbf{C}_t \;\preceq\; (1 + \bar{\chi}_t)^2\, \bar{\mathbf{C}}_t \tag{64}$$

hold for every $t \le T$, where $\kappa_{+,t} \ge 1$ is $\mathcal{H}_t$-measurable and $\bar{\chi}_t \ge \chi_t$ is the whitened-drift certificate of Lemma 11. Then, for every action a,

$$\bar{s}_t^2(a) \;\le\; \kappa_{+,t}\, (1 + \bar{\chi}_t)^2\, \hat{s}_t^2(a), \tag{65}$$

so Assumption 5 holds with the uniform factor $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ and Theorem 1 applies verbatim with $\mathcal{C}_t = \hat{\mathbf{C}}_t$ and $s_{\mathcal{C},t} = \hat{s}_t$:

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}} \sum_{t=1}^{T} \alpha_t^2\, u_t(a_t)\, \omega_t^2} \;+\; 2E_T.$$

No lower *spectral factor on* $\hat{\mathbf{C}}_t$ *is required for optimism; the cost of an excessively small, stale, or nonmonotone approximate operator is charged through its realized width complexity* $\Lambda_T^{\mathcal{C}}$ *(and its variation* $V_T^{\mathcal{C}}$ *inside* Γ_T^{dyn}*), not through a lower Loewner bound.*

Proof. $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ with both sides SPD inverts to $\hat{\mathbf{C}}_t^{-1} \succeq \kappa_{+,t}^{-1}\mathbf{C}_t^{-1}$, so $\hat{s}_t^2(a) \geq \kappa_{+,t}^{-1}s_t^2(a)$; and (41) gives $\bar{s}_t^2(a) \leq (1+\bar{\chi}_t)^2 s_t^2(a)$. Chaining, $\bar{s}_t^2(a) \leq (1+\bar{\chi}_t)^2\kappa_{+,t}\hat{s}_t^2(a)$, which is (65) with $s_{\mathcal{C},t} = \hat{s}_t$. Theorem 1 applies with $\mathcal{C}_t = \hat{\mathbf{C}}_t$. The two-sided $\sqrt{\rho_+/\rho_-}$ inflation—which instead assumes a lower spectral factor—is retained for comparison in Appendix F; the two results trade different assumptions (a lower spectral bound versus dynamic width complexity), and neither numerically dominates the other in general. □

Proposition 25 (Roundwise scalar-width invariance)**.** *If two width functions satisfy* $\hat{s}_t(a) = c_t s_t(a)$ *for every candidate action, where* $c_t > 0$ *is action independent, then replacing a scalar bonus coefficient* b_t *by* b_t/c_t *gives identical UCB scores and therefore the same selected action under the same tie rule.*

Proof. $m_t(a) + (b_t/c_t)\hat{s}_t(a) = m_t(a) + b_t s_t(a)$ for every a. □

This is a roundwise statement. If two policies also have identical centers, histories, optimization, and coupled exogenous randomness, if c_t is predictable, and if proportionality holds at every subsequent common history, induction gives identical actions and rewards after the coefficient adjustment. Absent those conditions the proposition makes no regret claim. Accordingly, regret differences can reflect anisotropic width distortion or a difference in damping, center, adaptive path, optimization, calibration, or solve accuracy.

Proposition 26 (Off-diagonal linear-Gram witness)**.** *Let* $\sigma = 1$, $\mathbf{u} = (1,1)/\sqrt{2}$, $\mathbf{v} = (1,-1)/\sqrt{2}$, *the action set be* $\{\mathbf{u}, \mathbf{v}\}$, *and* $\boldsymbol{\theta}^* = \epsilon\mathbf{u} + \Delta\mathbf{v}$ *with* $\Delta > \epsilon > 0$. *Rewards are noiseless, the ridge is* $\lambda\mathbf{I}$, *all policies use the same full-ridge mean center, and ties select* $\mathbf{u}$. *If*

$$\max\left\{\sqrt{\lambda(\epsilon^2+\Delta^2)}, \epsilon\sqrt{\lambda}\left(1+\sqrt{\frac{\lambda}{\lambda+1}}\right)\right\} < b \leq \left(\Delta - \frac{\epsilon}{\lambda+1}\right)\sqrt{\lambda+1}, \tag{66}$$

then full-Gram UCB plays $\mathbf{u}$ *once and* $\mathbf{v}$ *thereafter, so* $R_T^{\mathrm{full}} = \Delta - \epsilon$. *The raw diagonal policy and the policy inflated by the uniform analytic factor* $\kappa_n = (\lambda + n/2)/\lambda$ *play* $\mathbf{u}$ *forever and have* $R_T^{\mathrm{diag}} = T(\Delta - \epsilon)$.

Proof. After n pulls of $\mathbf{u}$ and no pull of $\mathbf{v}$,

$$C_n = \lambda\mathbf{I} + n\mathbf{u}\mathbf{u}^\top, \qquad \hat{\boldsymbol{\theta}}_n = \frac{n\epsilon}{\lambda+n}\mathbf{u}.$$

Thus the full scores are $n\epsilon/(\lambda+n) + b/\sqrt{\lambda+n}$ and $b/\sqrt{\lambda}$. The strict lower inequality in (66) makes the second score larger at $n = 1$. The confidence lower bound also implies $b > \Delta\sqrt{\lambda}$; after any number of $\mathbf{v}$ pulls its score is strictly larger than Δ, whereas the upper inequality makes the fixed $\mathbf{u}$ score at most Δ. Hence there is no return. In contrast, $D_n = \mathrm{diag}(C_n) = (\lambda + n/2)\mathbf{I}$, so both diagonal widths are equal and the center gives $\mathbf{u}$ a strict advantage after round one. Moreover $D_n \preceq \kappa_n C_n$ analytically, but multiplying both diagonal widths by $\sqrt{\kappa_n}$ leaves them equal; uniform certified inflation therefore does not change the diagonal action. For example, $(\lambda, \epsilon, \Delta, b) = (1, 0.1, 1, 1.1)$ satisfies the interval. □

This is a two-dimensional linear-Gram existence witness. For a linear predictor with Gaussian squared loss, this Gram is exactly the GGN/Fisher operator, so the construction isolates a GGN-relevant geometric mechanism. It is neither a nonlinear-network result nor a uniform dominance result. A dense action-dependent factor recovers the full widths, but it uses the reference geometry that the diagonal approximation was meant to avoid.

Composing subsampling with feature drift. Lemma 13 guarantees a sandwich for the *relinearized* $\mathbf{C}_t$: $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$. Only the *upper* half is needed here: $\hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$ gives $\kappa_{+,t} = 1+\varepsilon$, and composing with the one-sided whitened bound $\mathbf{C}_t \preceq (1+\bar{\chi}_t)^2\bar{\mathbf{C}}_t$ (Lemma 11) yields $u_t = (1+\varepsilon)(1+\bar{\chi}_t)^2$ in Theorem 24. A union bound over $t \leq T$ (replacing δ' by δ'/T in Lemma 13) makes the subsampling event time-uniform and contributes to δ_{cert}.

For an *unrescaled current-parameter window* (or any current-parameter downweighted subset with weights in $[0,1]$), Proposition 23 gives the *deterministic* upper factor $\kappa_{+,t} = 1$ with *no* distributional assumption, so Theorem 24 applies directly with $u_t = (1 + \bar{\chi}_t)^2$. Only the *stale periodic-refresh* operator (§A.3(iii)), which reuses Jacobians from an older parameter, lacks a deterministic $\kappa_{+,t}$; there Theorem 24 provides a guarantee only when a one-sided bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ is separately verified, which we evaluate empirically.

What remains open. The path schedules close the computability gap, but their constants can be extremely loose. Beyond the bounded low-rank and window-excitation regimes, useful control of $\Lambda_T^{\mathcal{C}}$ for a genuinely adapting backbone, tighter smoothness/residual certificates, and verified numerical enclosures for floating-point solver and spectral calculations remain open.

B Related Work

Neural and linear contextual bandits. LinUCB (Abbasi-Yadkori et al., 2011) and Thompson sampling (Thompson, 1933; Agrawal and Goyal, 2013) provide the analytical backbone for contextual bandit exploration. Neural extensions—NeuralUCB (Zhou et al., 2020), NeuralTS (Zhang et al., 2021), neural-linear models (Riquelme et al., 2018)—use different neural feature covariances. NeuralUCB and NeuralTS accumulate full neural-gradient Grams in their stated algorithms (and diagonalize them in experiments), while NeuralLinear maintains Bayesian linear uncertainty over learned last-layer features. CC-UCB instead re-evaluates every replayed Jacobian at the current parameter and accesses the resulting full operator implicitly. Neural-linear designs that train a deep representation but explore only in the last layer (Xu et al., 2022a) occupy the opposite design point. Recent alternatives sharpen different parts of this landscape. Salgia (2023) analyze overparameterized smooth-activation networks through non-asymptotic network–NTK approximation and give an efficient sublinear-regret algorithm. Deb et al. (2024) instead reduce contextual bandits to online neural regression, using perturbed predictions to obtain their regression guarantees. Bui et al. (2025) combine a learned deep representation with a variance-aware linear-UCB exploration layer. These are, respectively, an NTK analysis, a regression-oracle route, and a learned last-layer geometry; none analyzes the dynamically relinearized full GGN and its CG error. Our low-rank/excitation result is correspondingly narrower than generic neural regression, but controls that current-parameter operator.

Online Bayesian and EKF-style neural bandits. Durán-Martín et al. (2022) apply extended Kalman filter (EKF) updates in a learned or random low-dimensional affine parameter subspace. They maintain a dense or diagonal covariance in that subspace, inducing a low-rank covariance over parameters spanning all network layers, with memory independent of the number of rounds. This differs from applying exact curvature–vector products in the full parameter space. CC-UCB differs by using CG with the full GGN operator and providing a regret analysis that explicitly quantifies the CG truncation error.

Last-layer and structured Laplace approximations. Last-layer Laplace fixes the backbone and restricts the approximate posterior to the final layer (Kristiadi et al., 2020), omitting backbone-weight uncertainty by construction. This is only one scalable option: all-layer block-diagonal Kronecker curvature has also been used (Ritter et al., 2018), and Daxberger et al. (2021) systematize all-weight, subnetwork, and last-layer variants with full, low-rank, KFAC, or diagonal curvature. CC-UCB uses the uncompressed current-parameter GGN as its reference operator without materializing the covariance matrix.

Sampling-based full-parameter exploration. Langevin Monte Carlo Thompson sampling (LMC-TS) (Xu et al., 2022b) explores with approximate samples of a loss-defined parameter posterior, obtained by noisy gradient steps on the regularized loss; like CC-UCB it never materializes a covariance matrix. Its regret guarantee is for linear contextual bandits; its neural-bandit evidence is empirical. The two routes pay for exploration differently: LMC-TS through the mixing of a per-round sampling chain, CC-UCB through CG truncation and spectral distortion of an explicit curvature metric, which yields deterministic widths, an optimism-style analysis, and a fixed curvature operator reused across separate per-action CG solves.

Efficient nonlinear UCB. Kassraie and Krause (2022) analyze NTK-UCB and propose NN-UCB and its convolutional variant CNN-UCB. NN-UCB combines a trained-network mean with confidence widths from the empirical NTK feature map at initialization. They prove sublinear regret for the SupNN-UCB variant under NTK-RKHS, sufficient-exploration, and network-width assumptions. Their fixed-feature uncertainty differs from CC-UCB's replayed GGN/Fisher curvature at the current parameter $\boldsymbol{\theta}_t$, accessed matrix-free via CG. CC-UCB thus targets the regime where the practitioner has a specific pretrained model and wants to use its curvature as-is, rather than constructing a surrogate NTK-based estimator.

Spectral sketching for linear bandits. Kuzborskij et al. (2019) analyze linear bandits where the Gram matrix is replaced by a Frequent Directions sketch, deriving regret bounds that depend on the spectral approximation quality. Wen et al. (2026) show that fixed-size sketch regret bounds can degenerate to the trivial linear rate under heavy spectral tails, and propose an adaptive dyadic block sketch. Our spectral-distortion theorem (Theorem 24) instantiates the same conceptual lever in the matrix-free GGN/CVP setting by exposing transfer and realized-width terms along the realized trajectory. We do not claim novelty for the sandwich-to-regret reduction itself; the contribution is adapting it to the CG/GGN context with explicit CG truncation and feature-drift factors.

Laplace approximation in deep learning. The Laplace approximation (MacKay, 1992; Ritter et al., 2018) provides a Gaussian posterior centered at a MAP estimate with covariance given by the inverse curvature. Scalable variants use diagonal, KFAC (Martens and Grosse, 2015), or subnetwork approximations (Daxberger et al., 2021). Matrix-free full-Jacobian/Fisher predictive variance using implicit products and CG predates this work (Maddox et al., 2021); CC-UCB adapts that computational route to an online, current-relinearized GGN operator and supplies bandit-specific transfer and solver-error accounting.

GGN–vector products, Hessian spectral analysis, and influence functions. Pearlmutter's algorithm (Pearlmutter, 1994) computes exact Hessian–vector products; the analogous GGN–vector product has the same Jacobian-transpose-times-Jacobian-times-vector structure (Schraudolph, 2002). Recent work on Krylov methods for Hessian spectral analysis (Granziol and Juarev, 2026) demonstrates related distributed HVP/Krylov infrastructure at $> 10^8$-parameter scale, but does not benchmark the exact per-example GGN operator or end-to-end CC-UCB latency. Influence functions (Koh and Liang, 2017) use inverse-curvature–vector products to attribute predictions to training points; the CG-based implicit solve we employ is the same computational primitive.

Optimal experimental design. Information-theoretic exploration objectives—such as maximizing $\log\det(\bar{\mathbf{C}}_t)$ or minimizing the trace of $\bar{\mathbf{C}}_t^{-1}$—connect bandit exploration to the classical D-optimal and A-optimal design criteria (Valko et al., 2013). CC-UCB's use of the full GGN/Fisher curvature enables stochastic estimates of these quantities via Hutchinson-type trace estimators (Ubaru et al., 2017), without forming the matrix.

Filtration. We separate *pre-action* randomness, revealed before the agent commits to a_t, from *post-reward* randomness, drawn only after r_t is observed. Let Ω_t collect all pre-action algorithmic randomness used at round t: curvature subsampling indices, randomized sketches, and stochastic solver seeds. The randomized sketch or subsample that defines the curvature operator is drawn into Ω_t and *fixed before action selection*; it is not re-randomized during any conjugate-gradient (CG) solve at round t. (Standard CG requires a fixed symmetric positive definite matrix–vector oracle across its iterations and does not apply to a stochastic oracle that changes between iterations.) Let U_{t+1} collect the *post-reward* randomness used to form the next parameter estimate $\boldsymbol{\theta}_{t+1}$ (minibatch order, dropout, perturbed warm starts). Define $\mathcal{F}_0 = \sigma(\boldsymbol{\theta}_1)$ and, for $t \geq 1$, the recursively built σ-algebra

$$\mathcal{F}_t \ := \ \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t, \Omega_t, a_t, r_t, U_{t+1}\big), \tag{67}$$

so that $\boldsymbol{\theta}_{t+1}$—computed from the data through round t together with the post-reward randomness U_{t+1}—is $\mathcal{F}_t$-measurable. Define the *pre-action history* σ-algebra and the pre-reward σ-algebra

$$\mathcal{H}_t^- \ := \ \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t\big), \qquad \mathcal{H}_t \ := \ \mathcal{H}_t^- \vee \sigma(\Omega_t), \qquad \mathcal{G}_t \ := \ \mathcal{H}_t \vee \sigma(a_t). \tag{68}$$

At round t the context $\mathbf{x}_t$, the finite action set $\mathcal{A}_t$, and the pre-action randomness Ω_t are revealed; the agent selects a_t; then the reward $r_t = \mu^*(\mathbf{x}_t, a_t) + \eta_t$ is observed; finally U_{t+1} is drawn and $\boldsymbol{\theta}_{t+1}$ is computed. $\mathcal{H}_t^-$ is the history immediately before the randomized certificate draw; after that draw, its realized operator and schedules are $\mathcal{H}_t$-measurable. By construction of Algorithm 1 the selection rule is a deterministic function of $\mathcal{H}_t$-measurable quantities, so the chosen action a_t is itself $\mathcal{H}_t$-measurable and hence $\mathcal{G}_t = \mathcal{H}_t$; we retain the symbol $\mathcal{G}_t$ where the played action is used. The following are all $\mathcal{H}_t$-measurable (i.e. determined before the reward r_t): the candidate action set $\mathcal{A}_t$ and every candidate feature $\mathbf{g}_t(\mathbf{x}_t, a)$, $a \in \mathcal{A}_t$; the chosen action a_t; the algorithmic curvature operator $\mathcal{C}_t$ (Section 3); the confidence schedule $\bar{\beta}_t$; the centering certificate $\bar{\psi}_t$ (Assumption 4); the per-round CG tolerance $\bar{\varepsilon}_t$ together with any condition-number certificate used by the CG stopping rule; and the one-sided optimism-transfer factor $u_t(a)$ (Assumption 5). Since $\boldsymbol{\theta}_t$ is $\mathcal{F}_{t-1} \subseteq \mathcal{H}_t$-measurable, each candidate design vector $\mathbf{g}_t(\mathbf{x}_t, a)$ is $\mathcal{H}_t$-measurable and the realized design vector $\mathbf{g}_t(\mathbf{x}_t, a_t)$ is $\mathcal{G}_t$-measurable. We assume $\{U_s\}_{s\geq 1}$ is independent of the reward noise $\{\eta_s\}_{s\geq 1}$, so that revealing U_{t+1} and forming $\boldsymbol{\theta}_{t+1}$ after r_t preserves the conditional sub-Gaussian property of Assumption 1 for the future noise $\{\eta_s\}_{s>t}$. Index every randomized certificate source (operator sketch, transfer bound, condition bound, or other randomized oracle) by $j \in \mathcal{J}$. For its round-t validity event $\mathcal{E}_{j,t}$ choose in advance $\delta_{j,t} \geq 0$ such that its stated conditional guarantee gives $\Pr(\mathcal{E}_{j,t}^c \mid \mathcal{H}_t^-) \leq \delta_{j,t}$ and

$$\sum_{j\in\mathcal{J}} \sum_{t=1}^{T} \delta_{j,t} \leq \delta_{\mathrm{cert}}, \qquad \mathcal{E}_{\mathrm{cert}} := \bigcap_{j\in\mathcal{J}} \bigcap_{t=1}^{T} \mathcal{E}_{j,t}.$$

The tower property and union bound yield $\Pr(\mathcal{E}_{\mathrm{cert}}) \geq 1 - \delta_{\mathrm{cert}}$. Deterministic or almost-sure certificates use $\delta_{j,t} = 0$. The self-normalized confidence event is separate and receives probability δ.

Auxiliary linearized ridge estimator. For the analysis we define a *linearized ridge estimator* based on frozen features. Set the predictable intercept $b_s := \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s) - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}_s$ and the pseudo-response $y_s := r_s - b_s$. Under Assumption 3 and realizability ($\mu^* = \mu_{\boldsymbol{\theta}^*}$),

$$y_s \;=\; \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}^* \;+\; \rho_s(\mathbf{x}_s, a_s, \boldsymbol{\theta}^*) \;+\; \eta_s, \tag{69}$$

where $|\rho_s| \leq \varepsilon_{\mathrm{lin}}(s)$. The linearized estimator is

$$\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} \;:=\; \arg\min_{\boldsymbol{\theta}} \left[\frac{1}{2\sigma^2} \sum_{s=1}^{t-1} \big(y_s - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}\big)^2 \;+\; \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2 \right] \;=\; \bar{\mathbf{C}}_t^{-1}\, \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s)\, y_s. \tag{70}$$

This is a standard linear ridge regression with predictable design vectors $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$, so the self-normalized martingale bound (Abbasi-Yadkori et al., 2011) applies to it directly (Lemma 19).

GGN versus Hessian (context only). The regret theorem uses the GGN/Fisher curvature as the exploration metric. It does not require the GGN to be an accurate Hessian approximation. For squared loss (3) the full Hessian of the data-fit term is

$$\mathbf{H}_t \;=\; \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \Big[\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \;-\; \big(r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\big) \nabla^2 \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s) \Big] \;+\; \lambda \mathbf{I}.$$

The curvature matrix $\mathbf{C}_t$ drops the residual-times-second-derivative term, retaining only the outer-product part that is guaranteed PSD. The dropped residual-times-second-derivative term $D_t = \sigma^{-2} \sum_{s<t} (r_s - \mu_{\boldsymbol{\theta}_t}) \nabla^2 \mu_{\boldsymbol{\theta}_t}$ has operator norm at most $(t-1)\epsilon_r H/\sigma^2$, where ϵ_r bounds the residuals and H the Hessian norms. When this quantity is small relative to λ, $\mathbf{C}_t$ approximates the full Hessian in the Loewner sense (a formal statement with proof is given in Appendix G). This comparison is *not* invoked in the regret analysis (Theorem 1 uses $\mathbf{C}_t$ directly as the exploration metric); it serves only to contextualize $\mathbf{C}_t$ as a proxy for the full Hessian. Outside the small-residual regime, the GGN is not an exact Hessian approximation; it is the PSD Fisher/GGN metric used by the algorithm.

Remark (Fisher, empirical Fisher, and GGN). Three curvature objects appear in the literature and must be distinguished. The *empirical Fisher* is the realized score outer product $\sum_s \nabla_{\boldsymbol{\theta}} \log p(r_s \mid \mathbf{x}_s, a_s; \boldsymbol{\theta})\, \nabla_{\boldsymbol{\theta}} \log p(r_s \mid \mathbf{x}_s, a_s; \boldsymbol{\theta})^\top$; it depends on observed rewards and is *not* the operator used by CC-UCB. The *true (model) Fisher* is the model expectation of the score outer product, $F(\boldsymbol{\theta}) = \mathbb{E}_{r\sim p(\cdot|\mathbf{x},a;\boldsymbol{\theta})}[\nabla_{\boldsymbol{\theta}} \log p\; \nabla_{\boldsymbol{\theta}} \log p^\top]$. The *generalized Gauss–Newton* (GGN) drops the residual-times-Hessian term from the full Hessian, retaining only the outer-product Jacobian part that is guaranteed PSD (Martens, 2020; Schraudolph, 2002). For Gaussian regression with fixed variance (the squared-loss setting of this paper), the true Fisher and GGN are identical, and both reduce to the outer-product curvature $\sigma^{-2} \sum_s \mathbf{g}_s \mathbf{g}_s^\top$ used in (5). More generally, for exponential-family negative log-likelihoods parameterized by natural parameters $z_{\boldsymbol{\theta}}(\mathbf{x}, a)$, the true Fisher equals the GGN: $J_{\boldsymbol{\theta}}^\top \nabla^2 A(z)\, J_{\boldsymbol{\theta}}$, where $J_{\boldsymbol{\theta}} = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}}$ and $\nabla^2 A$ is the Hessian of the log-partition function (Martens, 2020; Pascanu and Bengio, 2013). Bernoulli rewards yield scalar weight $p(1-p)$; categorical rewards yield $\mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$. See Appendix H for the extension.

We access $\mathbf{C}_t$ only through curvature–vector products $\mathbf{v} \mapsto \mathbf{C}_t\, \mathbf{v}$, computed via forward-mode Jacobian–vector products composed with reverse-mode vector–Jacobian products (Pearlmutter, 1994; Schraudolph, 2002).

Extension to exponential-family likelihoods. The same matrix-free CG/CVP machinery extends algorithmically to exponential-family likelihoods (e.g. Bernoulli or categorical rewards) via the Fisher/GGN curvature $J_t^\top (\nabla^2 A)\, J_t$, but no regret guarantee beyond Gaussian squared loss is claimed in this draft. Details of the curvature definitions are in Appendix H.

Per-round CG tolerance $\bar{\varepsilon}_t$. The algorithm uses a *known* per-round tolerance $\bar{\varepsilon}_t \in (0, 1)$ in the UCB bonus (line 8), not the true per-action energy-norm errors $\varepsilon_{t,a}$ (which are unobservable without the exact solves). It suffices that $\max_{a\in\mathcal{A}_t} \varepsilon_{t,a} \le \bar{\varepsilon}_t$ at every round ((10); zero-gradient actions have $\varepsilon_{t,a} = 0$ by convention and are trivially within tolerance); the bonus is then conservative. Two strategies ensure this:

1. *Adaptive residual stopping (recommended).* In exact arithmetic, for each action monitor the computable Euclidean relative residual $\|r_I\|/\|\mathbf{g}_t(a)\|$ at each CG step (skipping actions with $\mathbf{g}_t(a) = \mathbf{0}$). Since $\varepsilon_{t,a} \le \sqrt{\bar{\kappa}_t} \cdot \|r_I\|/\|\mathbf{g}_t(a)\|$ (standard CG bound, with $\bar{\kappa}_t$ a certified upper bound on the condition number of $\mathcal{C}_t$), stopping that solve when $\|r_I\|/\|\mathbf{g}_t(a)\| \le \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$ guarantees $\varepsilon_{t,a} \le \bar{\varepsilon}_t$, hence (10). This is conservative by the factor $\sqrt{\bar{\kappa}_t}$ but requires no access to $\mathcal{C}_t^{-1}$ and adapts automatically to the per-round condition number. A floating-point implementation needs a verified residual enclosure to inherit this certification literally.
2. *Fixed budget (requires a uniform condition-number bound).* If an a priori bound $\bar{\kappa}_t \le \kappa_{\max}$ holds for *all rounds* $t \le T$, choose I via Eq. (75) at $\kappa_{\max}$ and a fixed target $\bar{\varepsilon}_t \equiv \bar{\varepsilon}$. For a *nonnegative weighted outer-product* operator $\mathcal{C}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ the condition number is bounded in closed form, $\bar{\kappa}_t \le 1 + W_t G^2/(\lambda\sigma^2)$ with $W_t = \sum_{s\in\mathcal{S}_t} w_s$; a fixed buffer of size n ($W_t = n$, unrescaled) makes this independent of T. For the full-history operator ($W_t = t{-}1$), and for rescaled uniform subsampling with $w_s = (t{-}1)/n$ (so $W_t = \sum_{s\in\mathcal{S}_t} w_s = t{-}1$), the bound grows with t. For a known finite horizon, a fixed budget at $\kappa_{\max} = 1 + (T-1)G^2/(\lambda\sigma^2)$ is valid but scales as $O(\sqrt{T}\log(1/\bar{\varepsilon}))$; a horizon-independent fixed budget is not uniform, so adaptive stopping or an increasing I_t is preferable (see Appendix E). For a *general* SPD operator that is not a nonnegative weighted outer product (e.g. a randomized sketch), the closed-form bound $1 + W_t G^2/(\lambda\sigma^2)$ does not apply, and a separately certified $\mathcal{H}_t$-measurable condition-number upper bound $\bar{\kappa}_t$ must be supplied.

In the regret bound (Theorem 1), the CG inflation enters through the per-round factor $\alpha_t = \sqrt{(1{+}\bar{\varepsilon}_t)/(1{-}\bar{\varepsilon}_t)}$, inside the sum of squares $\sum_t \alpha_t^2 u_t(a_t)\omega_t^2$; no horizon maximum over $\bar{\varepsilon}_t$ is taken.

Action screening (practical heuristic). When the action set $\mathcal{A}_t$ is large, computing the CG solve for every action is prohibitive ($|\mathcal{A}_t| \times I$ CG steps per round). A practical heuristic is to screen the top-K actions by mean prediction $m_t(a)$ (forward passes only), then compute UCB widths for the K candidates. This reduces cost to $K \times I$ CG solves. *The regret guarantee in Theorem 1 analyzes the full-enumeration case ($K = |\mathcal{A}_t|$); screening can drop the optimal arm and is not covered by the theory unless one adds an assumption that $a_t^* \in \mathcal{A}_t^K$ for all t.*

Algorithm 2 Operational CC-UCB with path certificates

Require: Reward predictor $\mu_{\boldsymbol{\theta}}$, damping $\lambda > 0$, noise proxy σ, confidence level δ, radius S, analytic constants G, L_g, L_μ; algorithmic curvature mode $\mathcal{C}_t$ (exact full-history $\mathbf{C}_t$ or approximate $\hat{\mathbf{C}}_t$) with certified one-sided factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; per-round CG tolerance $\bar{\varepsilon}_t < 1$

Initialize $\boldsymbol{\theta}_1$, certified $\zeta_1 \geq \|\nabla \mathcal{L}_1(\boldsymbol{\theta}_1)\|$, $n \leftarrow 0$, $\bar{\boldsymbol{\theta}} \leftarrow \mathbf{0}$, $J \leftarrow 0$, $R^{\mathrm{col}} \leftarrow 0$, $\bar{F} \leftarrow 0$, $\hat{\gamma} \leftarrow 0$, and $\mathcal{D}_0 \leftarrow \emptyset$
for $t = 1, 2, \ldots, T$ **do**
  Observe context $\mathbf{x}_t$ and action set $\mathcal{A}_t$; draw pre-action randomness Ω_t {$\mathcal{H}_t$ of (68)}
  $Q_t \leftarrow J + n\|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}\|^2$; $\bar{\chi}_t \leftarrow L_g\sqrt{Q_t}/(\sigma\sqrt{\lambda})$; $u_t \leftarrow \kappa_{+,t}(1 + \bar{\chi}_t)^2$
  $\bar{M}_t \leftarrow \sigma^{-2}[L_g\sqrt{R^{\mathrm{col}}Q_t} + (3GL_g/2)Q_t + (L_g^2/2)Q_t^{3/2}]$; $\bar{\psi}_t \leftarrow (\zeta_t + \bar{M}_t)/\sqrt{\lambda}$ {in a radius-R region use $L_g^2 R Q_t$ for the last term}
  $\bar{\epsilon}_t^{\mathrm{lin}} \leftarrow (L_\mu/2)(S + \|\boldsymbol{\theta}_t\|)^2$; $\bar{\beta}_t \leftarrow \sqrt{\hat{\gamma} + 2\log(1/\delta)} + \sqrt{\lambda}S + \sigma^{-1}\sqrt{\bar{F}}$; $\omega_t \leftarrow \bar{\beta}_t + \bar{\psi}_t$ {corrected center uses $\omega_t = \bar{\beta}_t$}
  Form or receive a fixed $\mathcal{H}_t$-measurable SPD matrix–vector oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$, $\mathcal{C}_t \succeq \lambda\mathbf{I}$, held fixed across all CG solves this round {exact/buffered special case $\mathcal{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}]$, weights w_s per §A.3}
  for each $a \in \mathcal{A}_t$ **do**
    $m_t(a) \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {mean prediction (or corrected center (28))}
    $\mathbf{g}_t(a) \leftarrow \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {gradient feature}
    if $\mathbf{g}_t(a) = \mathbf{0}$ **then**
      $\tilde{s}_t^2(a) \leftarrow 0$ {skip CG; no residual ratio}
    else
      $\tilde{\mathbf{u}} \leftarrow \mathrm{CG}(\mathcal{C}_t,\ \mathbf{g}_t(a))$; stop only once $\|r_I\|/\|\mathbf{g}_t(a)\| \leq \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$, or use a certified budget I_t from (75) {ensures $\varepsilon_{t,a} \leq \bar{\varepsilon}_t$}
      $\tilde{s}_t^2(a) \leftarrow \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}$ {approx. predictive variance}
    end if
  end for
  $a_t \leftarrow \arg\max_{a\in\mathcal{A}_t}\left[m_t(a) + \omega_t\sqrt{\dfrac{u_t(a)}{1 - \bar{\varepsilon}_t}}\,\sqrt{\tilde{s}_t^2(a)}\right]$ {bonus (58); ties broken by a fixed rule (e.g. lowest index) or by Ω_t}
  Play a_t, observe r_t
  $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{x}_t, a_t, r_t)\}$
  $c_t \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a_t) - r_t$; $R^{\mathrm{col}} \leftarrow R^{\mathrm{col}} + c_t^2$; $\bar{F} \leftarrow \bar{F} + (\bar{\epsilon}_t^{\mathrm{lin}})^2$; $\hat{\gamma} \leftarrow \hat{\gamma} + log[1 + \sigma^{-2}u_t\tilde{s}_t^2(a_t)/(1 - \bar{\varepsilon}_t)]$
  $n \leftarrow n + 1$; $d \leftarrow \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}$; $\bar{\boldsymbol{\theta}} \leftarrow \bar{\boldsymbol{\theta}} + d/n$; $J \leftarrow J + d^\top(\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}})$ {vector Welford update}
  Optimize on $\mathcal{D}_t$ to obtain $\boldsymbol{\theta}_{t+1}$, project to the certified region if required, and certify $\zeta_{t+1} \geq \|\nabla\mathcal{L}_{t+1}(\boldsymbol{\theta}_{t+1})\|$
end for

For the exact-curvature theorem, Algorithm 1 is run with $\mathcal{S}_t = \{1, \ldots, t-1\}$, $w_s = 1$, full action enumeration, $\mathcal{C}_t = \mathbf{C}_t$, and the transfer factor $u_t = (1 + \bar{\chi}_t)^2$ from Lemma 11. For an approximate operator $\mathcal{C}_t = \hat{\mathbf{C}}_t$, use $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ from Theorem 24. The selection rule needs only the *one-sided* upper transfer factor $u_t(a)$ for optimism; no lower spectral factor is required (the cost of the operator is charged through the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$). Algorithm 2 computes u_t, $\bar{\psi}_t$, and $\bar{\beta}_t$ from pre-action state. The operator-specific inputs are the proved factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; exact full and unrescaled-subset operators use $\kappa_{+,t} = 1$. Windowed/refresh curvature and action screening are practical heuristics unless the corresponding one-sided transfer factor or optimal-arm-survival condition is separately verified.

Corrected-center remark. Replacing the center $m_t(a)$ by the corrected center $m_t^{\mathrm{corr}}(a)$ of (28) sets $\bar{\psi}_t = 0$ (so $\omega_t = \bar{\beta}_t$) and removes the centering-stability assumption exactly (Corollary 5), at the cost of frozen-feature access for $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$; this departs from the $O(d)$ additional curvature-state-memory, no-stored-feature design unless a frozen-feature oracle, replay mechanism, or certified sketch is supplied (see the remark after Corollary 5).

CC-UCB is the analyzed method throughout.

How to read the theorem. The theorem is a transfer result for a general predictable SPD curvature sequence. Lemma 7 and Corollary 2 instantiate its nonlinear confidence, centering, and feature-transfer inputs from $O(d)$ path state, observed collection residuals, analytic smoothness constants, and a certified optimizer residual. These schedules are executable but can be very loose. A sublinear guarantee from this bound still requires control of the realized dynamic width complexity and a restricted parameter path; unrestricted end-to-end fine-tuning is not covered. No horizon-uniform lower transfer is required by Theorem 1; the lower sandwich appears only in the explicit near-linear corollary.

Discussion of terms.

- *Exploration term.* Write $S_T := \sum_{t=1}^{T} \alpha_t^2\, u_t(a_t)\, \omega_t^2$ for the sum of squares of the per-round factors. The exploration term is $2\sqrt{(\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}}\, S_T}$. A sufficient condition for sublinear regret is
$$\Lambda_T^{\mathcal{C}}\, S_T = o(T^2), \qquad E_T = o(T).$$
Concretely this holds whenever the per-round factors are polylogarithmically bounded ($\sup_t \alpha_t^2 u_t(a_t)\omega_t^2 = O(\operatorname{polylog} T)$, so $S_T = O(T \operatorname{polylog} T)$) and the realized dynamic complexity is small ($\Lambda_T^{\mathcal{C}} = O(\operatorname{polylog} T)$, e.g. via $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ with both the variation charge $V_T^{\mathcal{C}}$ and the endpoint $\log\det(\mathcal{C}_{T+1}/\lambda\mathbf{I})$ polylogarithmically bounded). Note ω_t^2 contains the cumulative squared linearization error through $\bar{F}_t$ (via β_t) and the centering certificate $\bar{\psi}_t$, so a small $\Lambda_T^{\mathcal{C}}$ alone is insufficient. α_t is bounded whenever the CG tolerance $\bar{\varepsilon}_t$ is bounded away from 1; in the exact-curvature case $u_t = (1 + \bar{\chi}_t)^2$ is bounded whenever the whitened-drift certificate $\bar{\chi}_t$ is bounded (Lemma 11); the whitened ℓ_2 accumulation $\chi_t \le \|\mathsf{D}_t\|_F$ makes this a near-stationarity condition on replayed-Jacobian drift. The unstable-curvature cost is isolated in $\Lambda_T^{\mathcal{C}}$ (through $V_T^{\mathcal{C}}$), not hidden in a spectral constant. The factor $\sqrt{\sigma^2 + G^2/\lambda}$ arises from the width-sum bound of the dynamic width-complexity lemma (Lemma 14, Eq. (50)); when $G^2 \le \lambda\sigma^2$ it is at most $\sigma\sqrt{2}$.
- *Linearization remainder.* The bound involves $E_T = \sum_t \varepsilon_{\mathrm{lin}}(t)$, not $T\varepsilon_{\mathrm{lin}}$. Near-linear regimes also require the squared-error sum $F_T = \sum_t \varepsilon_{\mathrm{lin}}(t)^2$ (through $\bar{F}_t$, hence ω_t) to be controlled. $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ gives $E_T = O(1)$; $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$ gives $E_T = O(\sqrt{T})$. The additive role of E_T mirrors misspecified linear bandits, where a uniformly bounded per-round model error commonly contributes an additive regret term that scales linearly in the cumulative approximation error (Ghosh et al., 2017; Lattimore et al., 2020). The displayed bound requires $E_T/T \to 0$; a nonvanishing uniform per-round error bound alone yields an $O(T)$ additive term.
- *Centering.* The centering certificate enters *multiplicatively*, through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ inside the sum of squares S_T, rather than as a separate additive term. It vanishes ($\bar{\psi}_t = 0$) when $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ (convex last-layer models with exact optimization), and the corrected-center variant (Corollary 5) removes it in general at the cost of a frozen-feature solve.
- *Regret improvement over diagonal/last-layer surrogates.* The bound itself does not prove that full curvature yields strictly lower regret than diagonal or last-layer methods in any specific regime; it provides an upper bound with explicit per-round factors. *Demonstrating regret improvement is an empirical hypothesis* (Section 5). The theorem exposes the transfer factor u_t and realized width complexity $\Lambda_T^{\mathcal{C}}$ as operator-dependent terms along the realized trajectory; it does not identify a causal regret cost because the adaptive path and other factors may also change.
- *Scope.* The bound is proved for Gaussian/squared-loss rewards. Extending it to exponential-family likelihoods (Bernoulli, categorical) would require a weighted frozen-feature analysis accounting for the data-dependent Fisher weights $\nabla^2 A(z)$ and is left as future work (Appendix H).

Corollary 27 (Sanity-check: frozen features recover linear UCB)**.** *Suppose the backbone is frozen and the reward head is linear in its parameters, the ridge objective is optimized exactly, and take the algorithmic curvature to be the sequential frozen Gram matrix, $\mathcal{C}_t = \bar{\mathbf{C}}_t$. Then: $\varepsilon_{\mathrm{lin}}(t) = 0$ for all t ($L = 0$ in Assumption 3), so $E_T = F_T = 0$; $\bar{\psi}_t = 0$ (under exact optimization the MAP and linearized estimators coincide, so Assumption 4 holds with equality), giving $\omega_t = \bar{\beta}_t$; $u_t(a) = 1$ for all a (indeed $\bar{s}_t^2(a) = s_{\mathcal{C},t}^2(a)$, so Assumption 5 holds with equality); and, since $\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top$ by (7), the update is exactly the rank-one step, i.e. $\mathcal{C}_{t+1} = \mathcal{C}_t^+$, so $\Xi_t = 0$, $V_T^{\mathcal{C}} = 0$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$. Then Theorem 1 gives*
$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\gamma_T \sum_{t=1}^{T} \alpha_t^2\, \bar{\beta}_t^2}.$$
If in addition $\bar{\varepsilon}_t \le \bar{\varepsilon}$ so that $\alpha_t \le \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$, the looser Corollary 28 recovers the familiar LinUCB form $R_T \le 2\,\alpha_I\, \bar{\beta}_{\max,T} \sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T}$, where $\bar{\beta}_{\max,T} := \max_{t \le T} \bar{\beta}_t$.

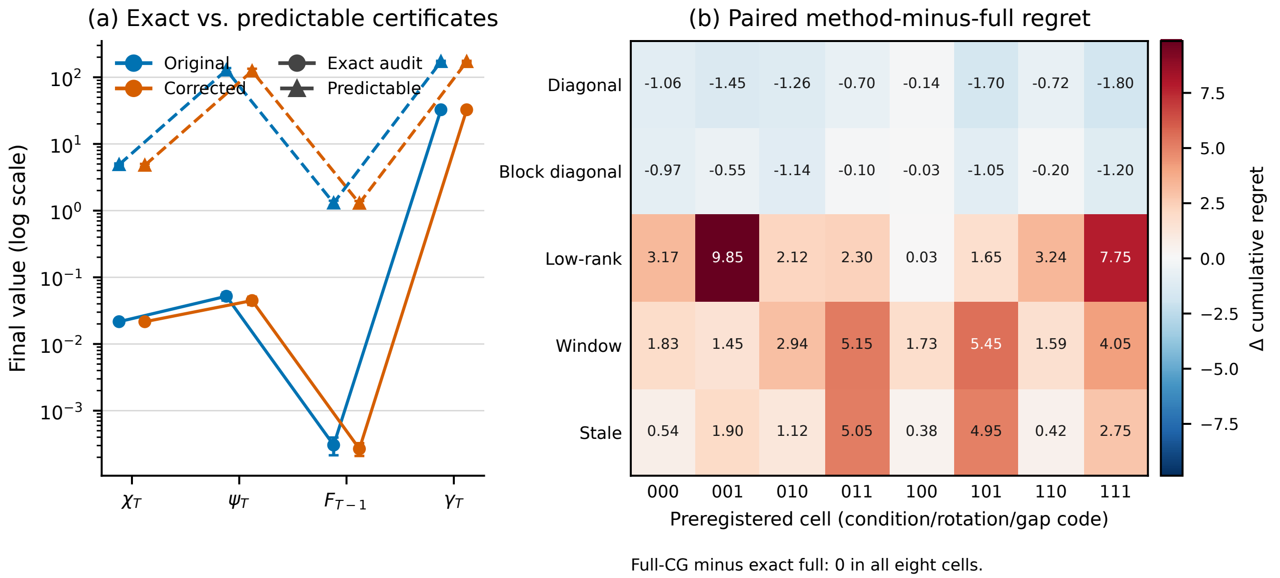

Figure 2: Supplementary certificate-tightness and curvature-mechanism overview. Panel (a) compares final-round float64 audit quantities with predictable schedules in the 50-seed tanh execution. Panel (b) gives paired regret differences in the preregistered curvature phase map. These are descriptive point audits.

Proof. Substitute $\varepsilon_{\text{lin}}(t) = 0$, $\bar{\psi}_t = 0$, $u_t(a) = 1$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$, $V_T^{\mathcal{C}} = 0$ into (15); the frozen Gram is monotone, so $\Xi_t = 0$ and the dynamic potential collapses to the information gain γ_T. The LinUCB form is Corollary 28 with $\alpha_t \le \alpha_I$, $u_t = 1$, $\omega_t = \bar{\beta}_t \le \bar{\beta}_{\max,T}$. □

C Executed-Policy Audit Protocols

Interpretation and statistics. Primary regret results are generated by online policies executed from scratch. Synthetic evaluators know the teacher only to compute pseudo-regret, optimism failures, and post-hoc theorem-event checks; the action rule never receives it. Hyperparameter selection, where used, is performed on the declared tuning seeds before evaluation. Common random numbers pair methods within an evaluation seed. We report arithmetic means, marginal 95% Student-t intervals, and paired Student-t intervals for method differences. No multiplicity adjustment is applied, so the intervals are descriptive. Scaling-curve bands instead use pointwise percentile bootstrap intervals. Their slope intervals resample whole seed trajectories, reaverage the same five nested prefixes, and refit the specified unweighted OLS model; they measure seed sensitivity conditional on that finite grid, not whether a power law is appropriate. An offline common-trajectory diagnostic uses one logged action and checkpoint sequence for all operators and is never mixed with online regret.

Predictable tanh execution. The analytically bounded tanh instance is $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $d = 6$, four actions, $\|\phi\| \le 0.75$, Gaussian noise $\sigma = 0.20$, teacher radius $S = 0.35$, and an optimizer projected to radius 0.50. We use the global constants $G = 0.75$ and $L_g = L_\mu = 4(0.75)^2/(3\sqrt{3})$, $\lambda = 1$, $\delta = 0.05$, eight clipped gradient steps per round, and a CG energy target 0.05 with the analytic condition bound $1 + (t-1)G^2/(\lambda\sigma^2)$. At every round the policy records Q_t, R_t^{col}, $\bar{F}_t$, $\hat{\gamma}_t$, $\bar{\chi}_t$, $\bar{M}_t$, $\bar{\psi}_t$, $\bar{\beta}_t$, the optimizer gradient norm, all candidate residuals, and whether any post-hoc exact check fails. No post-hoc field is read by action selection. Seeds 60–69 run a preregistered eight-cell one-factor sensitivity grid; the fixed reference configuration is rerun from scratch on evaluation seeds 160–209 for both centers. The 50-seed means and intervals in Section 5.1 come only from those evaluation runs.

Linear theorem-event audit. The context is $x \in \{-1, +1\}^8/\sqrt{8}$, there are five actions, and $\phi(x, a) = [x, e_a, x \otimes e_a]$ has dimension 53 and norm $\sqrt{3}$. The fixed teacher contains shared, arm-intercept, and context–arm interaction blocks; exact enumeration of all 2^8 contexts confirms that multiple arms are optimal. Noise is

r	Method	Regret slope [95% interval]	R^2	Max. $\lvert \log R - \widehat{\log R} \rvert$	Λ slope [95% interval]
4	Exact current	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Full CG	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.105 [0.104, 0.106]
4	Window $q = 1/2$	0.851 [0.843, 0.859]	0.9937	8.445e-02	0.465 [0.464, 0.465]
4	Window $q = 2/3$	0.309 [0.292, 0.326]	0.8529	1.550e-01	0.334 [0.333, 0.334]
4	Window $q = 1$	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Frozen	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.106 [0.106, 0.106]
4	Diagonal	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.673 [0.673, 0.673]
4	Greedy	0.000 [-0.000, 0.000]	1.0000	5.551e-17	0.115 [0.114, 0.115]
8	Exact current	0.134 [0.115, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Full CG	0.141 [0.122, 0.161]	0.6969	1.247e-01	0.099 [0.098, 0.099]
8	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.383 [0.383, 0.383]
8	Window $q = 2/3$	0.994 [0.993, 0.995]	1.0000	7.659e-03	0.304 [0.304, 0.304]
8	Window $q = 1$	0.134 [0.116, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Frozen	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.115 [0.115, 0.115]
8	Diagonal	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.671 [0.671, 0.671]
8	Greedy	0.037 [0.018, 0.057]	0.5542	4.606e-02	0.125 [0.124, 0.126]
16	Exact current	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Full CG	0.829 [0.815, 0.842]	0.9841	1.418e-01	0.118 [0.117, 0.118]
16	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.406 [0.406, 0.406]
16	Window $q = 2/3$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.278 [0.278, 0.278]
16	Window $q = 1$	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Frozen	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.125 [0.125, 0.125]
16	Diagonal	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.669 [0.669, 0.669]
16	Greedy	0.273 [0.229, 0.315]	0.9024	1.086e-01	0.140 [0.138, 0.141]

Table 4: Five-point OLS diagnostics for every method at $d = 2048$. Slope brackets are obtained from 2,000 paired resamples of whole seed trajectories. R^2 and maximum absolute log residual describe the fit to the five selected prefix means; they are not tests of a power-law model or asymptotic exponents.

$\mathcal{N}(0, 0.25^2)$. Evaluation uses seeds 100–119 after tuning on seeds 0–9. The tuning grid is $\lambda \in \{0.5, 1, 2\}$ and $c_{\text{bonus}} \in \{1, 1.25\}$; a candidate is eligible only if every tuning run passes the recorded confidence and small-scale numerical checks. Fresh evaluation policies use dense full Gram, CG full Gram, diagonal, window 64, rescaled subsample 64, rank-16 Lanczos–Ritz, and refresh period 20. The CG target is 0.05 with at most 106 iterations. Fixed-reference and independently tuned comparisons are both saved.

Linear policy checks and bound scale. The implemented base radius is

$$\beta_t^{\text{base}} = \sqrt{d \log\left(1 + \frac{(t-1)G^2}{d\lambda\sigma^2}\right) + 2\log(1/\delta)} + \sqrt{\lambda}\, S, \qquad \bar{\beta}_t = c_{\text{bonus}} \beta_t^{\text{base}}, \quad c_{\text{bonus}} \geq 1,$$

with $d = 53$, $G = \sqrt{3}$, $S = \|\theta^*\|_2$, and $\delta = 0.05$; hence tuning $c_{\text{bonus}} \in \{1, 1.25\}$ can only inflate the valid radius. Because $\bar{\psi}_t = 0$, the policy uses $\omega_t = \bar{\beta}_t$. Dense, CG, unrescaled window, and stale-refresh policies use the analytic factor $u_t = 1$. Diagonal, rescaled-subsample, and Lanczos policies materialize dense float64 small matrices before action selection and use $u_t = \max\{1, (1 + 32\epsilon_{\text{mach}})\lambda_{\max}(\bar{C}_t^{-1/2} \mathcal{C}_t \bar{C}_t^{-1/2})\}$. The multiplicative $1 + 32\epsilon_{\text{mach}}$ is a heuristic inflation, not a verified rounding-error enclosure. The CG policy likewise sets $\bar{\kappa}_t$ to a float64 point estimate of $\text{cond}(\mathcal{C}_t)$ before action selection and enforces $\bar{\varepsilon}_t = 0.05$ using relative residual target $\bar{\varepsilon}_t / \sqrt{\bar{\kappa}_t}$; the other policies have $\bar{\varepsilon}_t = 0$. These pre-action computations are deterministic conditional on each seeded operator draw, but predictability does not turn an unenclosed point estimate into an upper certificate. The analytic $u_t = 1$ transfer statements for dense full Gram, unrescaled window, and stale-refresh are unaffected; diagonal, rescaled-subsample, Lanczos, and full-Gram CG remain numerical audits of theorem semantics rather than certified executions. Tuning-run event checks are a selection gate, not the mathematical basis of the evaluation checks. The machine-readable per-policy trace is `results/derived/certification_audit.json`.

Nonlinear drift stress test. The teacher and student are one-hidden-layer width-four tanh networks with four inputs, five outputs, and 45 displacement parameters. They share the initialized backbone, making the exact frozen-head ridge solve a 25-parameter linear regime. Full-network regimes use respectively $(\text{learning rate}, \text{updates}, \text{step cap}) = (0.002, 2, 0.005), (0.004, 3, 0.020), (0.008, 4, 0.080)$. Each original- or corrected-center policy executes for 100 rounds on seeds 110–129 with a fixed, time-only schedule; no evaluation-seed tuning occurs. Dense float64 replay computes E_T, F_T, the dense-reference centering ratio and primitive ψ_t, χ_t, generalized transfer factors, condition numbers, dense-reference and residual-based CG errors, $\Lambda_T^{\mathcal{C}}$, endpoint log

Study	Policy	Regret [95% CI]	Measured run time (s)	Event fail.	Status
Tanh fixed	Original center	7.86 [7.66, 8.07]	0.094	0/50	Post-hoc verified
Tanh fixed	Corrected center	5.69 [5.46, 5.92]	0.095	0/50	Post-hoc verified
Balanced tuned	CC-UCB (full GGN-CG)	21.94 [20.64, 23.23]	0.966	–	Uncertified
Balanced tuned	Diagonal full network	16.11 [14.62, 17.61]	0.024	–	Uncertified
Balanced tuned	LinUCB	10.58 [9.82, 11.33]	0.015	–	Uncertified
Balanced tuned	LinearTS	12.67 [12.02, 13.31]	0.017	–	Uncertified
Balanced tuned	NeuralLinear	10.01 [9.33, 10.68]	0.014	–	Uncertified
Balanced tuned	NeuralUCB	23.04 [21.81, 24.27]	0.027	–	Uncertified
Balanced tuned	NeuralTS	25.07 [23.40, 26.75]	0.032	–	Uncertified
Balanced tuned	Frozen last layer	7.44 [6.90, 7.99]	0.012	–	Uncertified
Balanced tuned	Greedy	16.26 [14.44, 18.09]	0.017	–	Uncertified
Balanced tuned	UCB1	90.35 [87.95, 92.75]	0.003	–	Uncertified
Balanced tuned	Context-free TS	89.40 [86.53, 92.26]	0.004	–	Uncertified

Table 5: Supplementary executed-policy summary. Tanh rows use one fixed theoretical configuration; balanced rows are selected only on tuning seeds and rerun on disjoint evaluation seeds. Float64 event checks are not verified enclosures, and balanced rows are uncertified performance comparisons. Measured run times are resource accounting under each executed implementation, not a controlled speed or matched-wall-clock comparison.

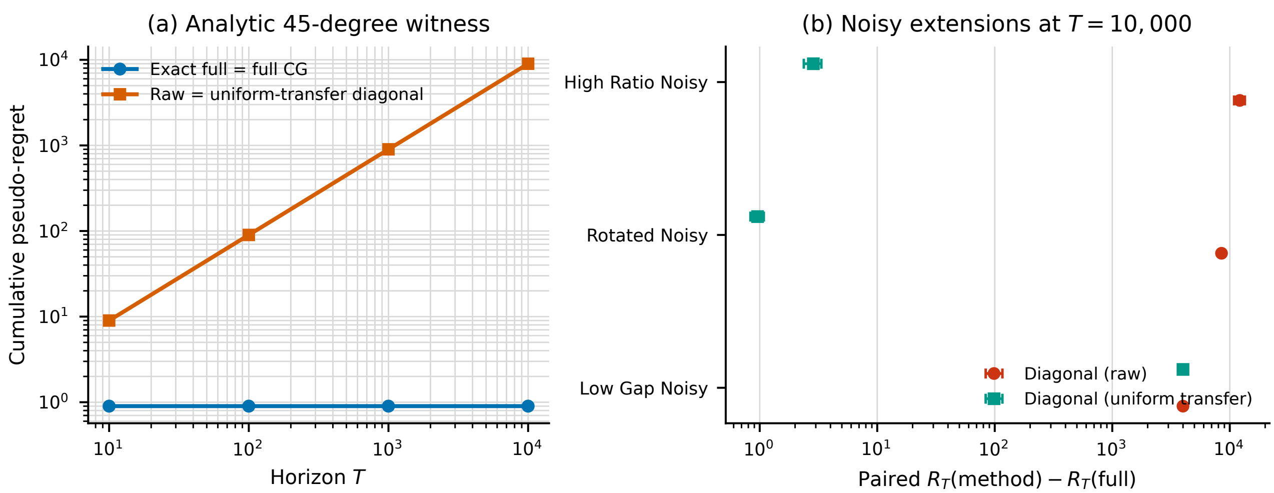

Figure 3: Off-diagonal linear-Gram witness. The analytic cell is constructive; the noisy cells are uncertified extensions. Paired differences are descriptive and establish no uniform ordering of curvature surrogates.

determinant, $V_T^{\mathcal{C}}$, Γ_T^{dyn}, and all-action optimism failures. Policy-available and post-hoc fields are separated in every JSONL record. The corrected-center identity is checked directly.

A secondary post-hoc analysis across the eight dependent regime–center cell means gives descriptive Spearman correlation 0.83 between the post-hoc decomposition score and regret, versus 0.57 for relative curvature change. With $n = 8$, reused seeds, and no seed-level sensitivity or inferential analysis, this is neither a primary result nor a causal or inferential claim. This stress test is secondary to the predictable tanh execution because its schedules fail.

Preregistered curvature-mechanism map. The fixed-gap bounded-linear family has $d = 10$, four actions, $T = 30$, and Gaussian noise 0.2. Its eight-cell fractional design crosses active-spectrum condition number $\{3, 50\}$, rotation $\{0, 60°\}$, action gap $\{0.15, 0.50\}$, nuisance strength $\{0.4, 1.2\}$, effective rank $\{3, 8\}$, and damping $\{0.5, 2\}$ according to the generators stored in the configuration; all cells are fixed before execution and no tuning is run. Exact full, residual-checked full CG, diagonal, block-diagonal, rank-three Lanczos, window-eight, and refresh-period-five policies execute independently on seeds 200–229. The exact-full trajectory is additionally replayed through every operator. These common-trajectory records contain per-action width ratios, cross-action coefficient of variation, Spearman and Kendall ranks, raw and scalar-rescaled top-score disagreement, normalized

Policy	R_{10}	R_{100}	R_{1000}	R_{10000}	Log–log slope
Exact full	0.90	0.90	0.90	0.90	0.000
Residual-checked full CG	0.90	0.90	0.90	0.90	0.000
Diagonal (raw)	9.00	90.00	900.00	9000.00	1.000
Diagonal (uniform transfer)	9.00	90.00	900.00	9000.00	1.000
Diagonal (actionwise reference)	0.90	0.90	0.90	0.90	0.000
Greedy	9.00	90.00	900.00	9000.00	1.000

Table 6: Executed analytic off-diagonal linear-Gram witness by horizon. Every value is generated from the provenance-validated derived artifact; fitted slopes are finite-horizon diagnostics.

Method	$T = 250$	$T = 500$	$T = 1000$
full CG	375.7	438.1	479.3
full dense	361.5	422.4	461.7
diagonal	4120.8	7710.5	13456.0
Lanczos–Ritz	495.7	496.4	509.3
subsample 64	339.7	373.1	396.6
refresh 20	370.3	465.1	526.7
window 64	252.0	255.3	260.2

Table 7: Fixed-reference linear RHS-to-regret ratio RHS/R_T. The JSON artifact also records R_T, RHS, R_T/T, and RHS/T at every horizon for both fixed and tuned configurations. Every displayed RHS exceeds T times the exact maximum one-round pseudo-regret 0.8118 on the finite context support, so the reported RHS values are numerically vacuous. For the four affected policies they are not rigorously certified bounds because their spectral point estimates lack verified upper enclosures.

decision-margin distortion, candidate-gradient alignment with retained and discarded eigenspaces, global and action-set transfer, effective rank, and condition number. Paired intervals are method-minus-exact-full; they are descriptive and unadjusted across cells.

Balanced contextual benchmark. Normalized Rademacher contexts have dimension four, five available actions, and Gaussian noise 0.10; exact enumeration of the 16 contexts verifies that every arm is optimal somewhere. Each method selects $(\lambda, c_{\text{bonus}})$ from $\{0.5, 1\} \times \{0.5, 1, 2\}$ by mean pseudo-regret over tuning seeds 70–79 at $T = 80$, then runs from scratch for $T = 200$ on seeds 260–289 with shared contexts and reward noise. Full-network methods use the same width-four tanh backbone, one clipped float64 update per round, learning rate 0.01, and model ridge 0.01. The CC-UCB row relinearizes all selected historical samples at the current parameter and solves one fixed full-GGN operator per action with explicit residual checks. NeuralLinear is a Bayesian Gaussian head on the frozen initialized representation; the frozen-last-layer row uses the same posterior with a UCB rule. NeuralUCB and NeuralTS are identified as local matched implementations rather than exact reproductions of every published training detail. LinUCB, linear Thompson sampling, diagonal full-network UCB, greedy, Gaussian UCB1, and Gaussian context-free Thompson sampling complete the comparison. Arithmetic means, marginal intervals, paired differences, selected hyperparameters, run times, and all seed-level records are retained.

Secondary operator and CG audits. The operator factorial varies windows 16/64, uniform resamples 16/64, Lanczos ranks 4/16, and refresh periods 5/20 in addition to full, frozen, and diagonal operators. On both the bounded linear task and the medium-drift, corrected-center tanh task, twenty online evaluation seeds (120–139) and matching offline common trajectories record exact global and action-set transfer, widths, disagreement, determinant terms, regret, runtime, and memory. The nonlinear policies use the fixed schedule and remain post-hoc audits rather than certified executions. The CG audit uses $d = 64, n = 256, K = 5$, condition numbers 10/100/1000, five target errors, two initializations, and two preconditioners on seeds 130–139. Its solver audit contains 600,000 dense-reference action solves; its separate policy audit executes 200 full-enumeration linear-bandit policies for 200 rounds with the same five tolerances and start/preconditioner factorial. Unlike the one-step scaling construction, the rotated-SPD right-hand sides exercise nontrivial Krylov convergence. At target energy error 0.1 with zero start and no preconditioner, mean iteration counts are 6.026, 22.643, and 55.043 for condition numbers 10, 10^2, and 10^3; mean measured energy errors are 2.23×10^{-2}, 6.87×10^{-3}, and 2.45×10^{-3}, and mean relative width errors are 2.54×10^{-4}, 2.43×10^{-5}, and 3.28×10^{-6}. The residual certificate is at most the

Policy	$T = 200$	$T = 500$	$T = 1000$	$T = 1500$
full GGN-CG	163.3 / 0.183	399.0 / 0.202	794.3 / 0.206	1189.7 / 0.207
frozen Gram	164.0 / 0.180	401.1 / 0.198	797.5 / 0.202	1193.0 / 0.205
diagonal full	166.4 / 0.168	407.7 / 0.185	807.2 / 0.193	1204.4 / 0.197
last-layer full	170.7 / 0.146	420.9 / 0.158	849.4 / 0.151	1273.1 / 0.151
last-layer diagonal	171.1 / 0.144	426.0 / 0.148	851.6 / 0.148	1274.9 / 0.150
greedy	162.3 / 0.189	393.9 / 0.212	781.9 / 0.218	1166.2 / 0.223
UCB1	145.4 / 0.273	332.5 / 0.335	616.2 / 0.384	895.6 / 0.403
Beta–Bernoulli TS	124.7 / 0.376	284.9 / 0.430	544.0 / 0.456	802.6 / 0.465

Table 8: Covertype evaluation-seed means, shown as cumulative pseudo-regret / accuracy. Contextual hyperparameters are selected only on validation seeds; UCB1 and Thompson sampling are parameter-free. Per-horizon marginal and paired 95% Student-t intervals, runtimes, policy types, and certification status are in `results/derived/covertype_horizon_results.json`.

prescribed 0.1 in every retained solve. These solver audits do not establish a method-speed advantage or a policy benefit.

Systems audit. Five evaluation seeds (140–144) cross $d \in \{32, 64, 128\}$, $n \in \{32, 128, 512\}$, $K \in \{5, 10\}$, and $I \in \{5, 15, 30\}$. We measure dense exact, separate full CG, diagonal, final-coordinate block, Lanczos–Ritz, row-batched independent CG, and row-batched Jacobi-PCG. Every action retains its own original-system residual; batched methods share only the operator evaluation. An additional matrix-free CPU grid uses $d \in \{512, 2048, 8192\}$, $n = 32$, $K = 4$, and $I = 16$ for the two batched solvers, with no dense allocation. Records include wall time, host memory, operator calls, equivalent sample CVPs, per-action iterations and residuals, and width error. These are synthetic float64 operator timings on one CPU host and use different computational primitives. Timings are descriptive resource accounting under the executed implementations, not matched-wall-clock comparisons; no trained-network or accelerator extrapolation is made. The configured 131,841-parameter `torch.func` JVP/VJP benchmark was not run because its declared Buck PyTorch dependency failed to configure.

C.1 Covertype real-data baseline audit

We use UCI Covertype DOI `10.24432/C50K5N` via scikit-learn 1.9.0; the cached dataset checksum is `eeab5db05bb3a9c00c62ced4c7d13112a00cd3487f59ac1943aaf5983d7c3ddf`. A PCG64 permutation with seed 20260717 creates non-stratified train/validation/test counts 348,607/116,202/116,203. Training means and scales standardize columns 0–9; binary columns 10–53 are unchanged. A width-four tanh network is initialized from seeded fan-in normals and optimized by one online SGD step (learning rate 0.01, ridge 1) using only selected-arm feedback. Full-GGN, frozen-Gram, diagonal, and greedy policies train all 255 parameters; last-layer policies train 35 head parameters. Damping $\{1, 10\}$ and bonus $\{1, 2, 4\}$ are selected on validation seeds 50–59; greedy fixes bonus zero. Evaluation seeds 150–159 use test contexts only after selection. CG uses zero initialization, relative residual 10^{-3}, at most 80 iterations, and one fixed operator per solve. The selection artifact and its validation evidence are SHA-256 committed inside every evaluation manifest.

The exact test-split label counts are $(42267, 56737, 7221, 536, 1902, 3444, 4096)$, giving a label-2 majority rate 0.488258. Uniform random has exact expected accuracy $1/7 = 0.142857$. The fixed majority arm is a whole-test-label oracle diagnostic, not a deployable policy. We add deployable context-free UCB1 (one forced pull per arm, then $\hat{\mu}_a + \sqrt{2 \log t / N_a}$) and independent Beta$(1, 1)$–Bernoulli Thompson sampling. All eight executed methods share evaluation seeds and context order.

Both non-contextual policies beat every contextual method at every horizon. At $T = 1500$, UCB1 and Thompson sampling have regret 895.6 and 802.6 (accuracy 0.403 and 0.465), while full GGN-CG has 1189.7 (0.207). Their paired method-minus-full differences are -294.1 $[-388.1, -200.1]$ and -387.1 $[-479.3, -294.9]$. Full and diagonal last-layer policies have regret 1273.1 and 1274.9 (accuracy 0.151 and 0.150). Mean complete-run times at $T = 1500$ are 5.665 seconds for full GGN-CG, 0.602 for diagonal full-network, 0.007 for UCB1, and 0.017 for Thompson sampling. No method is theorem-certified: binary rewards are routed through Gaussian squared-loss curvature. The full-GGN CG calls reach their relative-residual rule, but no energy-error certificate is supplied, so they are not called solver-certified. This study is a failed baseline check, not external validation of curvature-aware exploration.

Environment and artifacts. All reported new runs used CPython 3.14.0 on one arm64 host with 16 logical CPUs, 64 GiB physical memory, Darwin 25.5.0, and no accelerator. Core versions are NumPy 2.5.1, SciPy 1.18.0, matplotlib 3.11.0, psutil 7.2.2, and scikit-learn 1.9.0. Each run directory contains the resolved configuration, seed, UTC timestamp, Git revision and dirty flag, package and hardware record, raw per-round JSONL, and a summary. Derived JSON, CSV, table, and figure artifacts carry SHA-256 sidecars linking them to the aggregate and raw inputs.

Corollary 28 (Looser worst-case reduction)**.** *If $\alpha_t \le \alpha_{\max}$, $u_t(a_t) \le u_{\max}$, and $\omega_t \le \omega_{\max}$ for all $t \le T$, then $S_T \le \alpha_{\max}^2 u_{\max} \omega_{\max}^2 T$ and*

$$R_T \;\le\; 2\,\alpha_{\max}\,\omega_{\max}\,\sqrt{u_{\max}}\,\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\Lambda_T^{\mathcal{C}}} \;+\; 2E_T.$$

This $\sqrt{T\,\Lambda_T^{\mathcal{C}}}$ form recovers the familiar worst-case expression; it is weakly looser than (15) *(no tighter, with equality when the per-round factors are constant) and is* not *the main result.*

D Proof of Theorem 1

We recall the key objects: $\mathbf{g}_s(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}, a)$ (frozen features), $\bar{\mathbf{C}}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s<t} \mathbf{g}_s(\mathbf{x}_s, a_s)\mathbf{g}_s(\mathbf{x}_s, a_s)^\top$ (frozen-feature Gram matrix), $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}$, the algorithmic curvature $\mathcal{C}_t$ with width $s_{\mathcal{C},t}(a) = \sqrt{\mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a)}$, the linearized ridge estimator $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ (Eq. 70), the combined width factor $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, the per-round CG inflation $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, and the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$ (Eq. 46). Write $m_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.

Step 1: Confidence, centering, and prediction-error bound. By Lemma 19, with probability $\ge 1-\delta$,

$$\left|\mathbf{g}_t(a)^\top (\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}^*)\right| \;\le\; \bar{s}_t(a)\,\beta_t, \qquad \forall\, t,\, a, \tag{71}$$

where β_t is as in (62). The self-normalized martingale inequality (Abbasi-Yadkori et al., 2011) is applied to the 1-sub-Gaussian sequence $\xi_s = \eta_s/\sigma$ with the predictable design $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$. Each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is $\mathcal{G}_s$-measurable (where $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ is the pre-reward sigma-algebra of (68)), and the sequential update (7) ensures the hypotheses of the self-normalized bound are satisfied. By Lemma 22 (Assumption 4), $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \le \bar{\psi}_t\, \bar{s}_t(a)$. Combined with $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$ and $\bar{\beta}_t \ge \beta_t$, the triangle inequality gives the prediction-error bound (63)

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - m_t(a)| \;\le\; (\beta_t + \bar{\psi}_t)\,\bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t) \;\le\; \omega_t\,\bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t), \qquad \forall\, a. \tag{72}$$

(The corrected-center variant of Corollary 5 reaches (72) with $\omega_t = \bar{\beta}_t$ and without Assumption 4.)

Step 2: Bonus sandwiching. By Lemma 18 (which uses Lemma 17 with $\max_{a\in\mathcal{A}_t} \varepsilon_{t,a} \le \bar{\varepsilon}_t$ and the one-sided transfer (39)), for all a,

$$\omega_t\,\bar{s}_t(a) \;\le\; w_t(a) \;\le\; \alpha_t\,\omega_t\,\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a). \tag{73}$$

The lower bound and (72) give $\mu^*(\mathbf{x}_t, a) \le m_t(a) + w_t(a) + \varepsilon_{\mathrm{lin}}(t)$ for all a: the UCB covers μ^* up to the additive $\varepsilon_{\mathrm{lin}}(t)$ model-mismatch term.

Step 3: Instantaneous regret. From the UCB rule $m_t(a_t) + w_t(a_t) \ge m_t(a_t^*) + w_t(a_t^*)$, and using $\mu^*(a_t^*) \le m_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\mathrm{lin}}(t)$ (Step 2 at a_t^*, valid since $a_t^* \in \mathcal{A}_t$),

$$\begin{aligned}
\mu^*(a_t^*) - \mu^*(a_t) \;&\le\; [m_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\mathrm{lin}}(t)] - \mu^*(a_t) \\
&\le\; [m_t(a_t) + w_t(a_t) + \varepsilon_{\mathrm{lin}}(t)] - \mu^*(a_t) \quad \text{(UCB selection)} \\
&=\; w_t(a_t) + [m_t(a_t) - \mu^*(a_t)] + \varepsilon_{\mathrm{lin}}(t) \\
&\le\; w_t(a_t) + \omega_t\,\bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t) \quad \text{((72) at } a_t\text{)} \\
&\le\; 2\,w_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t) \quad \text{(lower bound in (73)).}
\end{aligned}$$

Step 4: Upper bonus bound. The upper bound in (73) gives

$$\mu^*(a_t^*) - \mu^*(a_t) \;\le\; 2\,\alpha_t\,\omega_t\,\sqrt{u_t(a_t)}\,s_{\mathcal{C},t}(a_t) \;+\; 2\varepsilon_{\mathrm{lin}}(t). \tag{74}$$

Step 5: Cauchy–Schwarz after summing. Summing (74) over $t = 1, \ldots, T$ and applying Cauchy–Schwarz to the sum (*not* per round),

$$R_T \;\le\; 2\sum_{t=1}^{T} \alpha_t\omega_t\sqrt{u_t(a_t)}\,s_{\mathcal{C},t}(a_t) + 2E_T \;\le\; 2\sqrt{\sum_{t=1}^{T}\alpha_t^2\omega_t^2 u_t(a_t)}\sqrt{\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t)} + 2E_T.$$

Step 6: Dynamic potential and collection. Lemma 14 gives $\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \le (\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}$. Substituting,

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}\sum_{t=1}^{T}\alpha_t^2\,u_t(a_t)\,\omega_t^2} \;+\; 2E_T,$$

which is (15). The interpretable and observable versions (27) follow from $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ (Lemma 14) and $\Lambda_T^{\mathcal{C}} \le \widehat{\Lambda^{\mathcal{C}}}_T$ (Corollary 16). The Cauchy–Schwarz step is applied only after summing, so the time-varying factors $\alpha_t, u_t(a_t), \omega_t$ enter through a sum of squares; no horizon maximum is taken before Cauchy–Schwarz, and the usual $\sqrt{T}$ scaling is recovered when the per-round factors are uniformly bounded (Corollary 28). The stated probability follows from a union bound over the self-normalized confidence event $\mathcal{E}_{\mathrm{conf}}$ ($\ge 1-\delta$) and the certificate event $\mathcal{E}_{\mathrm{cert}}$ ($\ge 1-\delta_{\mathrm{cert}}$), giving $\ge 1-\delta-\delta_{\mathrm{cert}}$. $\square$

E CG Convergence Analysis

Fix a round t and an action a with $\mathbf{g} := \mathbf{g}_t(a) \ne \mathbf{0}$. After $I_{t,a}$ iterations of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ (starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$), the standard energy-norm bound gives

$$\varepsilon_{t,a} \;=\; \frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\le\; 2\left(\frac{\sqrt{\kappa_t}-1}{\sqrt{\kappa_t}+1}\right)^{I_{t,a}},$$

where $\kappa_t = \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number. The bound is right-hand-side independent, so a common iteration budget bounds $\max_a \varepsilon_{t,a}$. This zero-start form is what the fixed-budget certificate (75) uses. For a *warm start* $\tilde{\mathbf{u}}_0 \ne \mathbf{0}$, energy-norm minimization over the shifted Krylov space instead gives

$$\frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\le\; 2\,q^{\,I_{t,a}}\,\frac{\|\tilde{\mathbf{u}}_0 - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}, \qquad q = \frac{\sqrt{\kappa_t}-1}{\sqrt{\kappa_t}+1},$$

i.e. the extra factor is the initial relative energy error $\|e_0\|_{\mathcal{C}_t}/\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}$ (equal to one at $\tilde{\mathbf{u}}_0 = \mathbf{0}$, recovering the zero-start bound). Under the same certified condition-number bound, adaptive residual stopping (line 5) remains valid with a warm start because it certifies the realized residual directly; but the fixed-budget count (75) is valid *as written* only for zero initialization, and a fixed-budget warm-start certificate additionally requires a bound on that initial relative energy error (the zero-start iteration count does not by itself certify a warm-started solve). Any certified upper bound $\bar{\kappa}_t \ge \kappa_t$ may replace κ_t; for a nonnegative weighted outer-product operator, $\lambda_{\min}(\mathcal{C}_t) \ge \lambda$ and $\lambda_{\max}(\mathcal{C}_t) \le \lambda + W_t G^2/\sigma^2$ (where $W_t = \sum_{s\in\mathcal{S}_t} w_s$) give

$$\kappa_t \le \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \le 1 + \frac{W_t G^2}{\lambda\sigma^2} =: \bar{\kappa}_t,$$

whereas a general SPD operator requires a separately certified $\bar{\kappa}_t$.

Two cases at value one must be distinguished. The true condition number $\kappa_t = 1$ forces all eigenvalues equal, i.e. $\mathcal{C}_t = c_t\mathbf{I}$ for some scalar $c_t > 0$, and one CG iteration from $\tilde{\mathbf{u}}_0 = \mathbf{0}$ is *exact* ($\varepsilon_{t,a} = 0$) for every $\mathbf{g} \ne \mathbf{0}$; this does *not* by itself imply $c_t = \lambda$. With the specific certificate used here, $\bar{\kappa}_t = 1$ additionally forces $\lambda_{\max}(\mathcal{C}_t) = \lambda$, which together with $\mathcal{C}_t \succeq \lambda\mathbf{I}$ gives $\mathcal{C}_t = \lambda\mathbf{I}$ exactly. In either reading the single-step solve is exact, so we keep

$\bar\kappa_t = 1$ as a separate branch and never evaluate the logarithmic budget below at its zero denominator. For $\bar\kappa_t > 1$, setting the right-hand side of the rate equal to $\bar\varepsilon_{\mathrm{CG}}$ and requiring $\bar\varepsilon_{\mathrm{CG}} < 1$ (where $\bar\varepsilon_{\mathrm{CG}}$ denotes the round-t target tolerance $\bar\varepsilon_t$) gives the per-solve budget

$$I_{\min}(\bar\kappa_t, \bar\varepsilon_{\mathrm{CG}}) \;=\; \left\lceil \frac{\log(2/\bar\varepsilon_{\mathrm{CG}})}{\log\!\big((\sqrt{\bar\kappa_t}+1)/(\sqrt{\bar\kappa_t}-1)\big)} \right\rceil \;=\; O\!\big(\sqrt{\bar\kappa_t}\,\log(1/\bar\varepsilon_{\mathrm{CG}})\big), \tag{75}$$

which, applied to every scored action, ensures $\max_a \varepsilon_{t,a} \le \bar\varepsilon_{\mathrm{CG}} = \bar\varepsilon_t$. For $W_t = n$ (unrescaled fixed buffer size), $\bar\kappa_t \le 1 + nG^2/(\lambda\sigma^2)$ is bounded independently of T, so $I_{\min}$ is constant. For full history, and for rescaled uniform subsampling with $w_s = (t-1)/n$, we have $W_t = t-1$, so the condition-number *bound* $\bar\kappa_t = O(tG^2/(\lambda\sigma^2))$ grows with t, and the worst-case certificate is then obtained with $I_{\min} = \Theta\big(G\sqrt{t/(\lambda\sigma^2)}\,\log(1/\bar\varepsilon_{\mathrm{CG}})\big)$ iterations per solve—i.e. $I = O(\sqrt{t}\,\log(1/\bar\varepsilon_{\mathrm{CG}}))$ suffices under this bound (this is an upper-bound certificate, not a lower bound on the number of iterations CG actually needs). CG may converge faster than this worst-case certificate suggests under favorable spectral clustering of $\mathcal{C}_t$; the realized iteration counts would be measured empirically.

F Two-Sided Spectral-Distortion Corollary

The main-text Theorem 24 uses only a one-sided upper transfer factor. For comparison we record the earlier two-sided result, which instead assumes both an upper *and* a lower spectral factor and yields the familiar $\sqrt{\rho_+/\rho_-}$ inflation. The two results trade different assumptions—the corollary below needs a lower spectral bound; the dynamic Theorem 1 needs the dynamic width complexity $\Lambda_T^{\mathcal{C}}$ instead—and neither numerically dominates the other in general.

Corollary 29 (Two-sided spectral-distortion inflation)**.** *Adopt the hypotheses of Theorem 1 with $\mathcal{C}_t = \hat{\mathbf{C}}_t$, retaining the* original nonlinear center *(so $\omega_t = \bar\beta_t + \bar\psi_t$; in the convex last-layer regime under exact optimization, $\bar\psi_t = 0$ and $\omega_t = \bar\beta_t$), and a uniform CG tolerance $\bar\varepsilon_t \le \bar\varepsilon$ so that $\alpha_t \le \alpha_I := \sqrt{(1+\bar\varepsilon)/(1-\bar\varepsilon)}$. Suppose predictable factors $\rho_{-,t}, \rho_{+,t}$ are certified before action selection and the* two-sided *sandwich $\rho_{-,t}\bar{\mathbf{C}}_t \preceq \hat{\mathbf{C}}_t \preceq \rho_{+,t}\bar{\mathbf{C}}_t$ holds for all $t \le T$ with $0 < \rho_{-,t} \le \rho_{+,t} < \infty$; set $\rho_- = \min_t \rho_{-,t}$, $\rho_+ = \max_t \rho_{+,t}$, and (to satisfy the convention $u_t \ge 1$ of Assumption 5) take the transfer factor $u_t := \max\{1, \rho_{+,t}\} \le \max\{1, \rho_+\}$. Write $\omega_{\max,T} := \max_{t\le T}(\bar\beta_t + \bar\psi_t)$ and $\rho_+^\star := \max\{1, \rho_+\}$. Then, on the intersection of the confidence event $\mathcal{E}_{\mathrm{conf}}$ ($\ge 1-\delta$) and the certificate event $\mathcal{E}_{\mathrm{cert}}$ (which now also contains the two-sided sandwich event; $\ge 1-\delta_{\mathrm{cert}}$),*

$$R_T \;\le\; 2\,\alpha_I\,\sqrt{\rho_+^\star/\rho_-}\;\omega_{\max,T}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} \;+\; 2E_T$$

with probability at least $1-\delta-\delta_{\mathrm{cert}}$, where γ_T is the frozen-feature information gain (25)*. Under the standard normalization $\rho_+ \ge 1$ (achievable by rescaling the sandwich so its upper factor is at least one), $\rho_+^\star = \rho_+$ and the inflation is the familiar $\sqrt{\rho_+/\rho_-}$.*

Proof. This is a *direct* argument through the monotone frozen Gram $\bar{\mathbf{C}}_t$; it does *not* substitute $\hat s_t$ for the dynamic width in Corollary 28. *Step 1 (variance sandwich).* Inverting the two-sided sandwich gives $\rho_{+,t}^{-1}\bar{\mathbf{C}}_t^{-1} \preceq \hat{\mathbf{C}}_t^{-1} \preceq \rho_{-,t}^{-1}\bar{\mathbf{C}}_t^{-1}$, so, evaluating the quadratic form at $\mathbf{g}_t(a)$, $\rho_{+,t}^{-1/2}\,\bar s_t(a) \le \hat s_t(a) \le \rho_{-,t}^{-1/2}\,\bar s_t(a)$. The lower half gives $\bar s_t^2(a) \le \rho_{+,t}\hat s_t^2(a) \le u_t \hat s_t^2(a)$, so Assumption 5 holds with $u_t = \max\{1, \rho_{+,t}\}$. *Step 2 (per-round regret).* With $s_{\mathcal{C},t} = \hat s_t$, Theorem 1 Steps 1–4 give $\mathrm{regret}_t \le 2\,\alpha_t\,\omega_t\,\sqrt{u_t}\,\hat s_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t)$; using $\hat s_t(a_t) \le \rho_{-,t}^{-1/2}\bar s_t(a_t)$, $u_t = \max\{1, \rho_{+,t}\} \le \rho_+^\star$, $\alpha_t \le \alpha_I$, and $\omega_t \le \omega_{\max,T}$,

$$\mathrm{regret}_t \;\le\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_{-,t}}\;\bar s_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t) \;\le\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\;\bar s_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

Step 3 (frozen-feature elliptic potential). Sum over t and apply Cauchy–Schwarz, then the classical elliptic-potential identity for the *monotone* Gram $\bar{\mathbf{C}}_t$ ((7)), $\sum_t \bar s_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\sum_t \log(1 + \sigma^{-2}\bar s_t^2(a_t)) = (\sigma^2 + G^2/\lambda)\,\gamma_T$:

$$R_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{T\sum_t \bar s_t^2(a_t)} + 2E_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} + 2E_T.$$

The probability statement is the union bound over $\mathcal{E}_{\mathrm{conf}}$ and $\mathcal{E}_{\mathrm{cert}}$. $\square$

The time-uniform requirement is essential: a curvature sandwich that holds only at a single t does not suffice for a regret bound that sums over all rounds. For the subsampled GGN in Lemma 13, a per-round union bound over $t \le T$ (allocating δ_{cert}/T per round within the certificate event $\mathcal{E}_{\mathrm{cert}}$) gives a time-uniform sandwich with the sample size in (45) evaluated at the worst-case $m = T{-}1$.

G GGN Approximation Error

Lemma 30 (GGN approximation error)**.** *Suppose $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a)$ is twice differentiable in $\boldsymbol{\theta}$. Suppose $|r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)| \le \epsilon_r$ and $\|\nabla^2 \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\|_{\mathrm{op}} \le H$ for all $s < t$. Define $\zeta := (t{-}1)\,\epsilon_r\, H/(\sigma^2\lambda)$. Then:*

(i) $\|\mathbf{H}_t - \mathbf{C}_t\|_{\mathrm{op}} \le (t{-}1)\,\epsilon_r\, H/\sigma^2 = \zeta\lambda$.
(ii) If $\zeta < 1$, then $(1{-}\zeta)\mathbf{C}_t \preceq \mathbf{H}_t \preceq (1{+}\zeta)\mathbf{C}_t$.

Proof. The dropped term is $D_t = \sigma^{-2}\sum_{s<t}(r_s - \mu_{\boldsymbol{\theta}_t})\nabla^2\mu_{\boldsymbol{\theta}_t}$. By the triangle inequality, $\|D_t\|_{\mathrm{op}} \le \sigma^{-2}\sum_{s<t}\epsilon_r H = \zeta\lambda$, proving (i). For (ii): since $-\|D_t\|\mathbf{I} \preceq -D_t \preceq \|D_t\|\mathbf{I}$, we have $\mathbf{C}_t - \zeta\lambda\mathbf{I} \preceq \mathbf{H}_t \preceq \mathbf{C}_t + \zeta\lambda\mathbf{I}$. Because $\mathbf{C}_t \succeq \lambda\mathbf{I}$, the lower bound gives $\mathbf{C}_t - \zeta\lambda\mathbf{I} \succeq \mathbf{C}_t - \zeta\mathbf{C}_t = (1{-}\zeta)\mathbf{C}_t$ (using $\zeta\lambda\mathbf{I} \preceq \zeta\mathbf{C}_t$), and similarly $\mathbf{C}_t + \zeta\lambda\mathbf{I} \preceq \mathbf{C}_t + \zeta\mathbf{C}_t = (1{+}\zeta)\mathbf{C}_t$. □

H Exponential-Family Curvature Definitions

For a generic negative log-likelihood $\ell\big(r,\, z_{\boldsymbol{\theta}}(\mathbf{x}, a)\big)$ with natural parameter $z_{\boldsymbol{\theta}} \in \mathbb{R}^k$ (scalar for Bernoulli, k-dimensional for categorical), the GGN/Fisher curvature matrices are

$$\mathbf{C}_t^{\exp} \;=\; \lambda\,\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_t(\mathbf{x}_s, a_s)^\top\, \nabla^2 A\big(z_t(\mathbf{x}_s, a_s)\big)\, J_t(\mathbf{x}_s, a_s), \tag{76}$$

$$\bar{\mathbf{C}}_t^{\exp} \;=\; \lambda\,\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_s(\mathbf{x}_s, a_s)^\top\, \nabla^2 A\big(z_s(\mathbf{x}_s, a_s)\big)\, J_s(\mathbf{x}_s, a_s), \tag{77}$$

where $J_t(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \in \mathbb{R}^{k\times d}$ is the Jacobian of the natural parameter and $\nabla^2 A(z)$ is the output-space Hessian of the log-partition function. For Bernoulli rewards $\nabla^2 A(z) = p(1{-}p)$ (scalar); for categorical rewards $\nabla^2 A(z) = \mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$). Setting $k = 1$ and $\nabla^2 A = \sigma^{-2}$ recovers the Gaussian curvature (5)–(6).

I Broader Impact

The motivating applications—recommendation and learned reward heads—carry deployment risks. Exploration deliberately selects uncertain actions, which can expose users to under-vetted content; a miscalibrated reward head can amplify feedback loops between the policy and its future data. Deployment should combine exploration with explicit safety constraints, action filters, offline evaluation, and online monitoring. This paper analyzes regret under stated certificates and does not establish safe exploration.

J Data and Code Availability

The reviewed repository contains the configuration-driven experiment pipeline, pinned Buck environment, unit tests, aggregation and figure scripts, validated derived artifacts, the retained $d = 128, r = 4$ scaling slice, and selected raw off-diagonal and blocker records. The 2.1 GB full scaling tree and several historical provenance inputs are stored outside Git and are not present in a clean compact checkout. Therefore the current checkout does not by itself regenerate every historical raw-derived chain. The anonymous submission bundle will contain only tiers whose provenance inputs have been restored and validated; until that final bundle is rebuilt, "will contain" rather than "contains" is the operative release claim. Internal run manifests carry input hashes and local execution provenance. The release builder writes into a fresh directory using `tools/build_anonymous_supplement.py`: it converts paths to repository-relative form, replaces the Git revision with an anonymous source-tree content hash, writes an artifact-preserving per-round JSONL projection compressed with zstd, regenerates sidecars and the top-level manifest, and fails on missing provenance inputs or identifying scans. The public UCI Covertype dataset (Blackard, 1998) is distributed under CC BY 4.0 (DOI `10.24432/C50K5N`); its exact cached-file and split hashes are in each Covertype manifest. The compact horizon report records all real-data means, intervals, resource-accounting times, policy types, and certification categories. The defining autodiff benchmark has status `not_run` and contributes no timing result. No identifying repository URL or local path will appear in the final manuscript bundle.